



Gen AI / LLM Training

1

Rules for Today

- Participate. Get engaged.
- Make mistakes.
- Learn from mistakes. Learn from each other.
- Raise your hands.
- Ask questions. There are no dumb questions.
- Bring your own questions.
- If you need a break, please take a break.
- Bring notepad so you can do exercise.
- Discuss.
- Laugh. Have a fun.

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2

2

Introduction

Name:

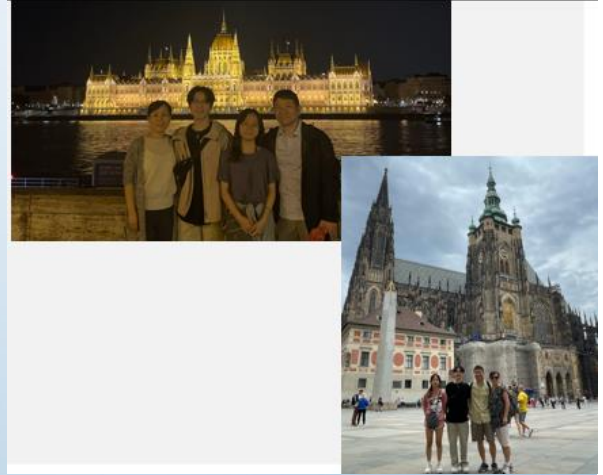
Kevin Lee

What you do:

Managing member and CTO
at ClinVia

Something special about you:

Love to travel with the family



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3

3

Agenda

- Introduction of Large Language Model (LLM), Generative AI (Gen AI)
- ChatGPT Introduction and Application
- ChatGPT Use Cases
- Prompts, not Search
- Prompt Engineering in Gen AI (ChatGPT)
- Gen AI functionality – ChatGPT, Copilot, Gemini, Sonnet, DeepSeek
- Application development using Gen AI API (SAS & Python)
- LangChain & RAG LLM application development
- AI Agent and MCP
- Enterprise level Gen AI (ChatGPT) Implementation
- Future of Gen AI (ChatGPT)
- Discussion

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4

Large Language Model

- Introduction
- LLM in AI Landscape
- LLM Architecture
- History of LLM Development

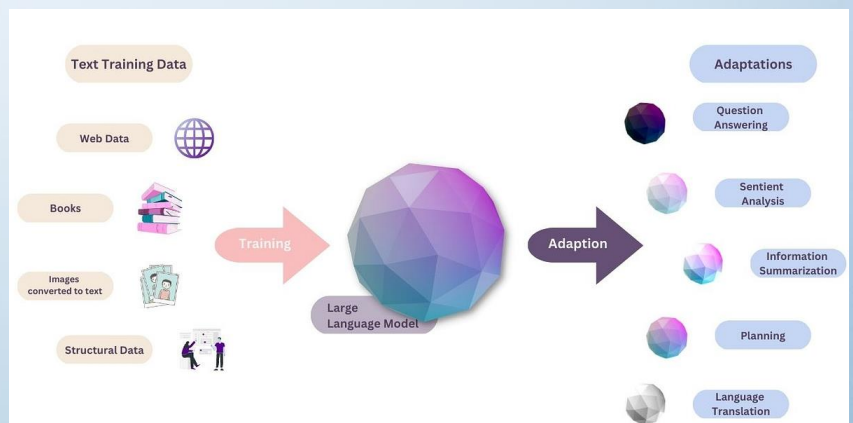
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5

What is LLM?

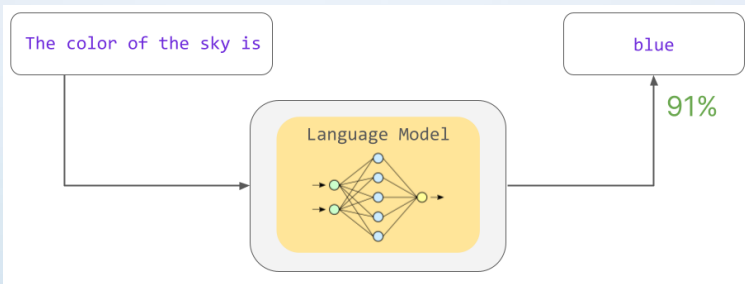
- Large Language Model (LLM) – a trained Machine Learning model that understands and generates text in a human-like fashion.
- LLM has been trained on a huge corpus of data (a large numbers of texts)
- Highly efficient in generating language-based outputs.
- Popular LLM
 - BERT
 - ChatGPT3.5
 - ChatGPT4
 - GPTs
 - Llama
 - Copilot
 - Bard / Gemini



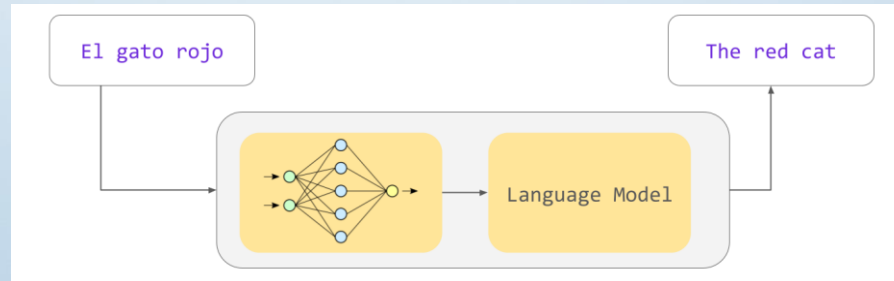
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How LLM Works



- Large Languages are used to train LLM
- So, when 'input texts' are prompted into LLM, LLM provide 'response', which show a highest probability.



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History of LLM Model Development

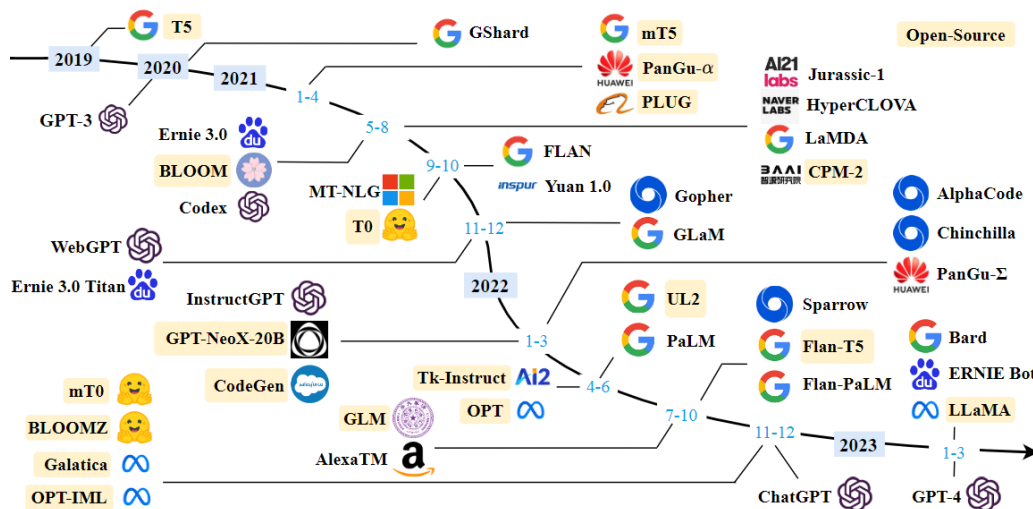


Fig. 1. A timeline of existing large language models (having a size larger than 10B) in recent years. We mark the open-source LLMs in yellow color.

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Gen AI (Generative AI)

- Introduction
- Gen AI Market Trend

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What is Gen AI?

Gen AI – a trained Machine Learning model that generate new contents with a simple prompt.

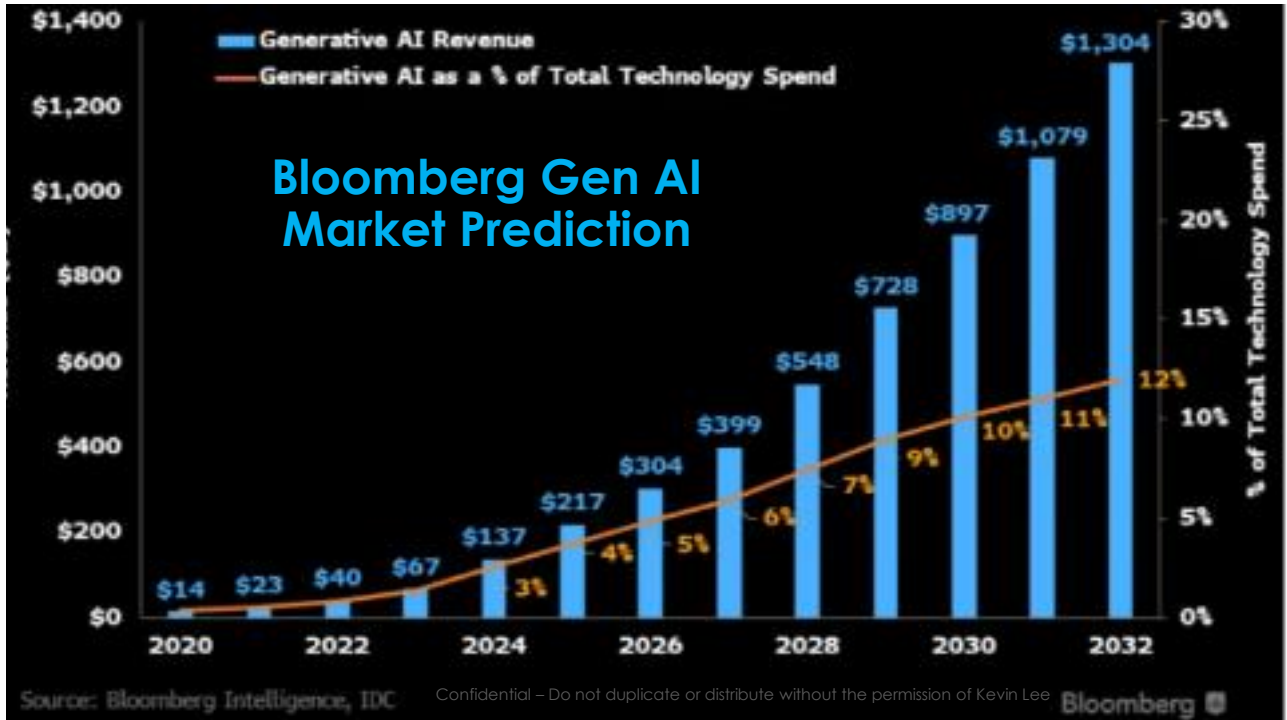
- Text : Content Writing, Chatbots, Assistants, Search
- Code : Code Generation, Data Set Generation
- Image : Image Generation, Image Edit
- Audio : Voice Generation/Edit, Sound creation, Audio Translation
- Video : Video Creation/Edit, Voice Translation, Deepfake



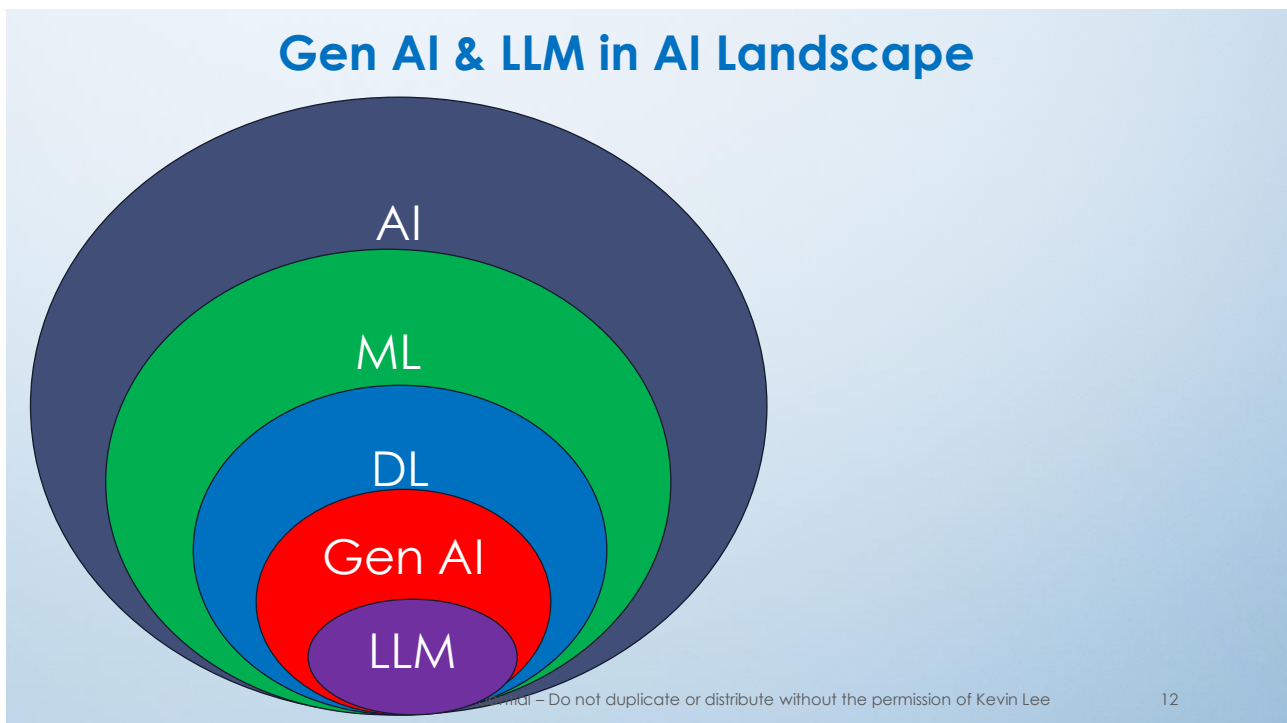
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10



11



12

ChatGPT

- Introduction
- Development
- Plan
- Popularity
- Why Should I use it & Why not

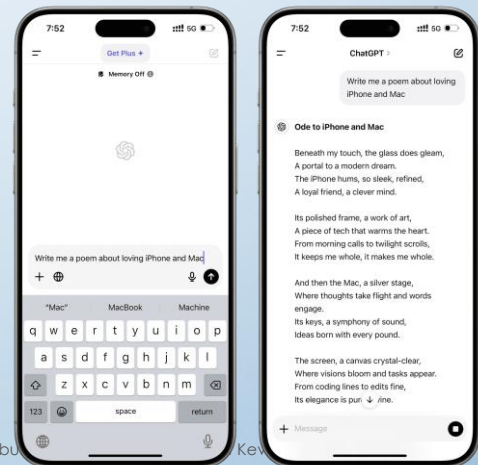
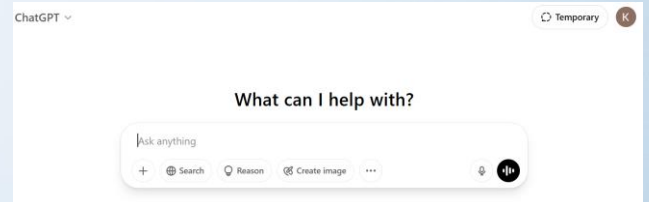
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What is ChatGPT?

- ChatGPT is an advanced LLM developed by OpenAI.
- ChatGPT is trained on large corpus of text (~300B tokens).
- Its main strength
 - the ability to generate human-like responses in various contexts.
 - The ability to understand and generate text in a wide range of domains.
- Its application
 - Inquiry/prompt (rather than search)
 - Content Generation
 - Customer support
 - Interactive Storytelling
 - Coding
 - Image Analysis
 - Art Development

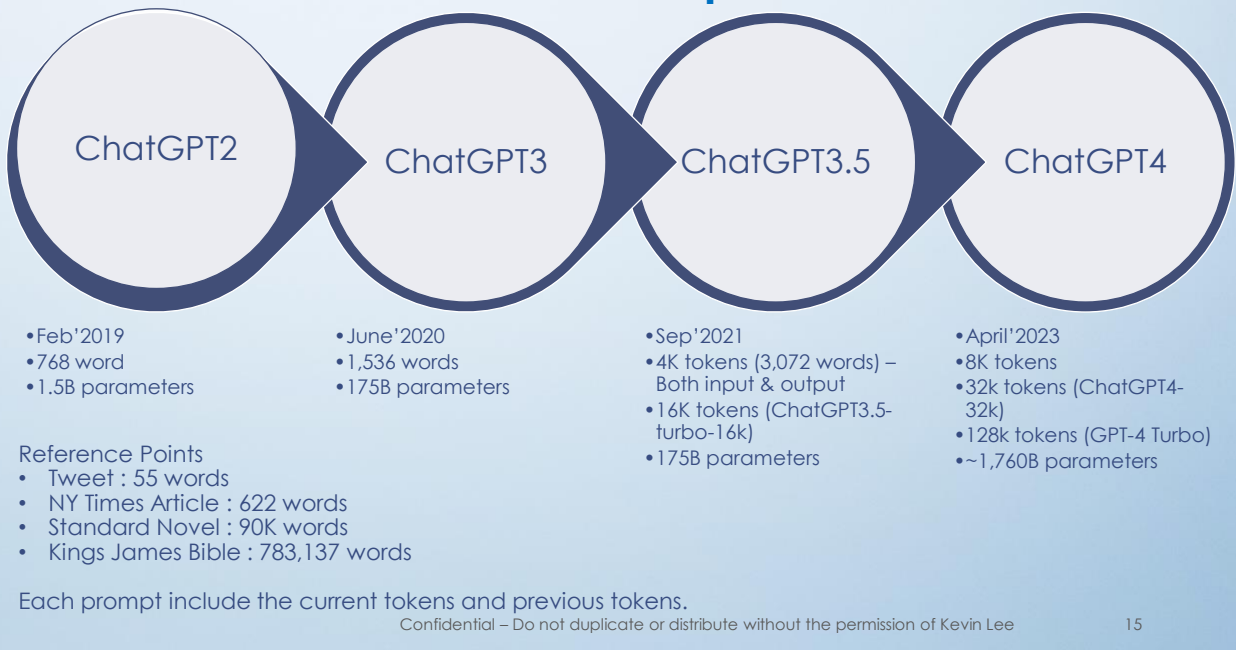


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ChatGPT Development



15

ChatGPT model

•**GPT-4.1 mini (free version)** : delivers significant improvements in instruction-following, coding, and overall intelligence .

•**GPT-4.1(paid version)** : A multimodal model capable of processing text, images, and audio, offering faster performance and improved handling of various tasks .

Specific Tasks:

•**GPT-4o**: A multimodal model capable of processing text, images, and audio, offering faster performance and improved handling of various tasks .

•**GPT-4.5**: Designed for more emotionally intelligent and nuanced conversations, making it suitable for workplace communication and sensitive dialogues .

•**o3 and o4-mini**: Advanced reasoning models optimized for complex coding, mathematics, and scientific tasks, with o4-mini being the fastest model for quick, reasoning-heavy tasks .

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Bar Exam Score Performance :

ChatGPT3.5 vs ChatGPT4

10% vs 90%

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CPA Exam Performance:

ChatGPT3.5 vs ChatGPT4

35.1% vs 85.1%

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ChatGPT Plan

- ChatGPT
 - Free Plan
 - Access to GPT-4.1mini model
 - Use custom GPTs
 - Standard voice mode
- ChatGPT Plus
 - Paid service : \$20/month
 - Access to o3, o4-mini, o1, advanced voice mode
 - Faster Response Speed
 - Exclusive access to beta features like GPT4.5
- ChatGPT Pro
 - Paid service : \$200/month
 - Unlimited Access to all reasoning models and GTP 4o
 - Extended access to Sora Video

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19

ChatGPT Plan for Enterprise

- ChatGPT API
 - Pay as you go :
 - GPT3.5-Turbo : \$0.5/ 1m tokens (input) and \$1.5 / 1m tokens (output)
 - GPT-4o : \$2.5/ 1m tokens (input) and \$10 / 1m tokens (output)
 - An interface that allows developers to integrate the ChatGPT model into their own applications, software, or platforms.
- ChatGPT Team
 - Paid service : \$25/month
 - Access to GPT-4 with 32k context window, DALL-E, GPT-4 with Vision
 - No training your business data
 - Secure workspace for your team
 - Create and share custom GPTS with your workspace
- ChatGPT Enterprise
 - Paid service :
 - The most Enterprise-grade Security & Privacy
 - Unlimited access to GPT-4, Code Interpreter
 - Free credits to use GPT API

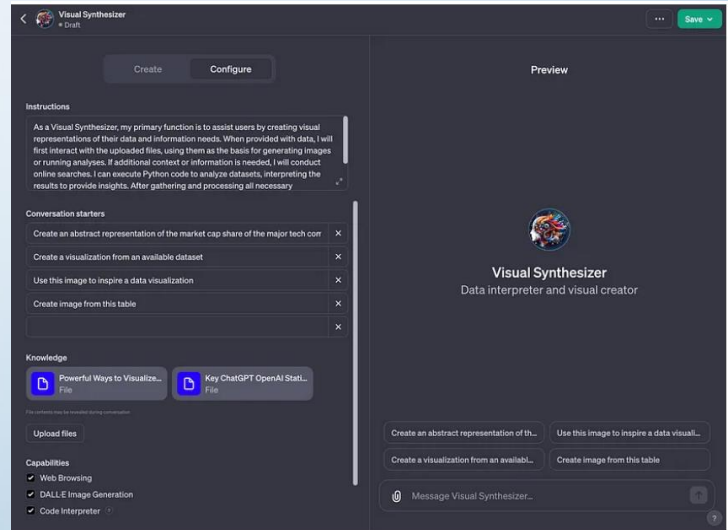
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20

20

GPTs (GPT store)

- ChatGPT user platform
- Build your own ChatGPT without coding
- GPT Store / GPT Apps
- ChatGPT Plus Subscription
- Configuration
 - Instructions : Prompt
 - Knowledge : contents
- Use Case
 - Regulatory Response
 - Drug Compound Information



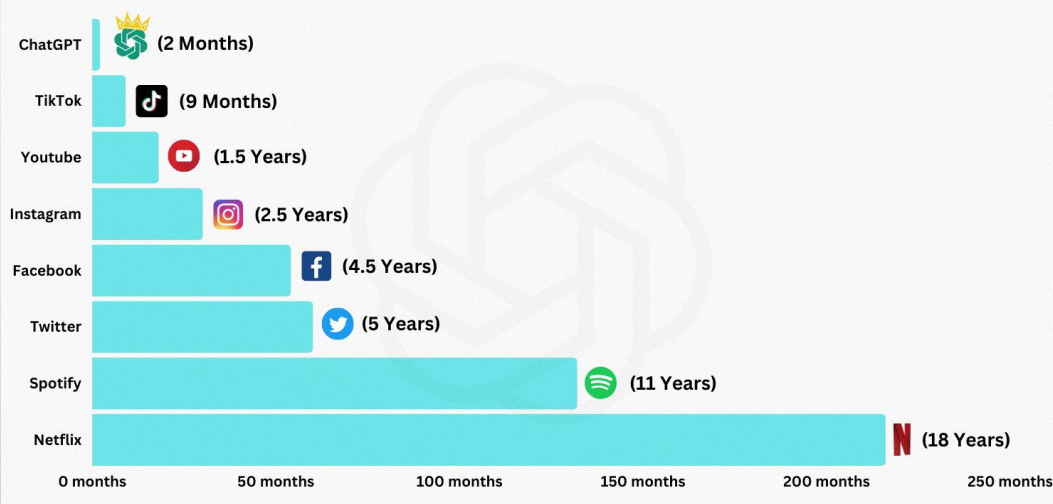
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21

ChatGPT Popularity

Road To 100 Million Users For Various Platforms



ChatGPT gained 100M users within 2 months since its release.

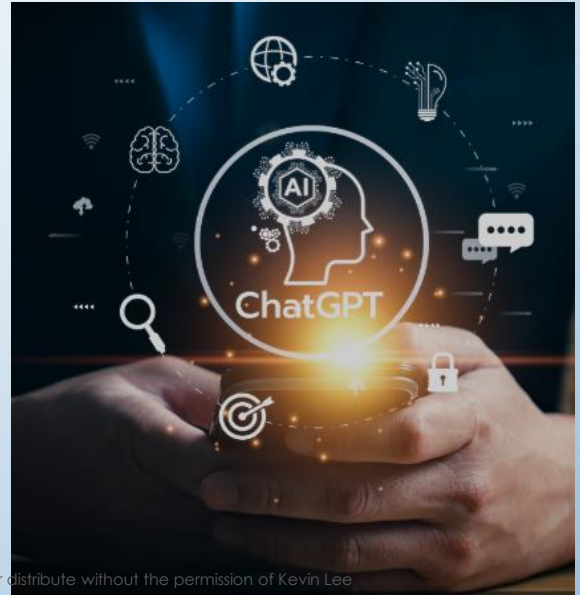
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Reasons that ChatGPT is **not** being used

- Concerns/Risks using ChatGPT
- A lack of understanding of ChatGPT Use Cases
- A lack of imagination of impact
- A lack of understanding of Prompt



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Potential Benefits using ChatGPT

- Higher Productivity / Efficiency
- Better Patient Engagement
- Patient's Recruitment
- Enhancement in Image
- Market Research & Insights



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ChatGPT Users vs Non-Users Comparison in Boston Consulting

- Results of ChatGPT Users
 - finished **12.2%** more tasks on average
 - completed tasks **25.1%** more quickly
 - produced **40%** higher quality results



25

25

Patients & HCP Support and Engagement

- Better Support to Patients & HCP :
Instant Responses to frequently asked questions
 - New Medicine
 - Dosage
 - Side effects
 - Usage instructions
 - Mechanism
 - Ongoing Research / Clinical Trials
- Better Patient's engagement thru digital platforms (Chat Bot)

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Better Patient Recruitments

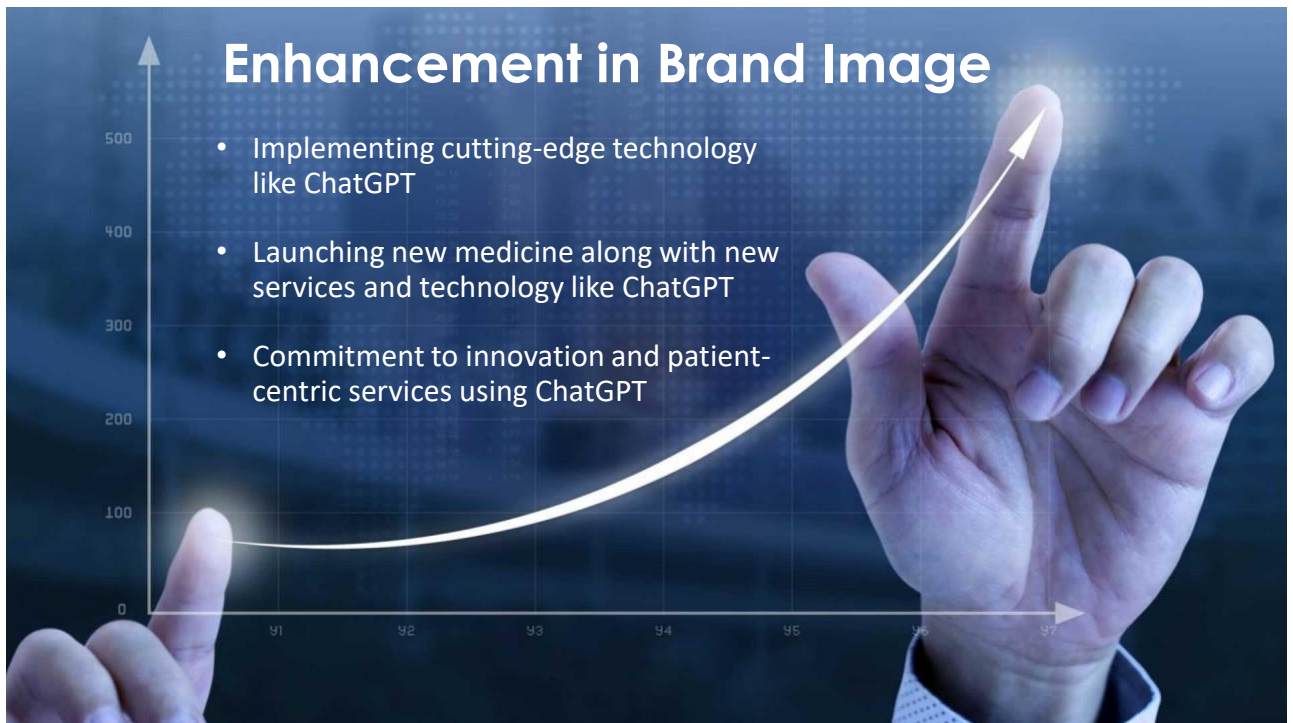
- Better Patients Recruitment finding through LLM(ChatGPT) from Large EHR using Inclusion/Exclusion criteria
- User-friendly AI interface
 - More efficient and cost-effective recruitment process using ChatGPT
 - Higher awareness and conception about ongoing and future clinical trials
 - Better education
 - Higher participation



27

Enhancement in Brand Image

- Implementing cutting-edge technology like ChatGPT
- Launching new medicine along with new services and technology like ChatGPT
- Commitment to innovation and patient-centric services using ChatGPT



28

Market Research and Insights



- Customer Feedback Collection and Sentiment Analysis
- Analysis in Patients Interaction and Preferences
- Valuable Insights into Market Trends and Optimizing Marketing Strategies
- Competitor Analysis
- Forecasting Demand

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Drug Discovery

- Quicker Scientific Literature Review and Information Retrieval
- New ideas for Drug Targets, Compounds and Experimental Approaches
- Identification of the binding affinity of compounds to targets
- Drug-Target Pairing Predictions
- Personalized medicine through biomarkers and genetic factors



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30

30

ChatGPT Application Guideline

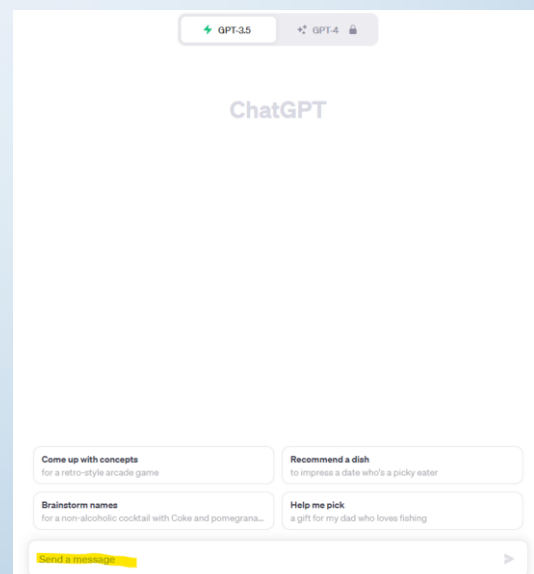
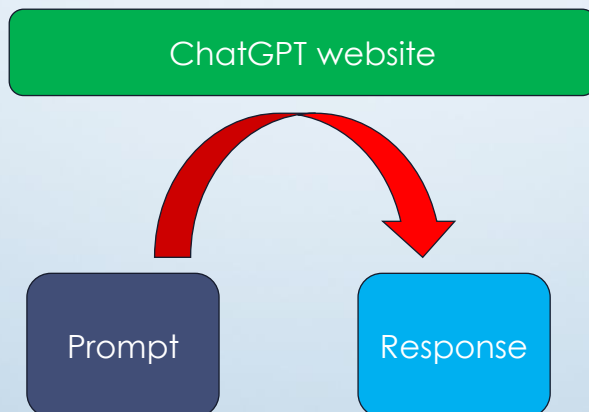
- Simple Prompt
- Application Development using ChatGPT API
- Application Development using fine-tuned ChatGPT API

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31

31

Simple Prompt in OpenAI ChatGPT website

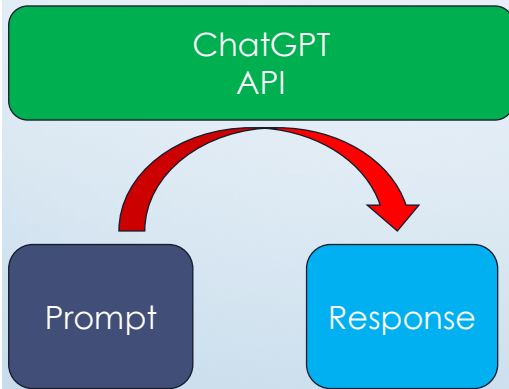


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32

Application using ChatGPT API



```
%let api_key = xxxxxxx ;
```

```
filename prompt  
"F:\biometrics\project\GenAI\prompt.txt";
```

```
filename resp  
"F:\biometrics\project\GenAI\resp.json";
```

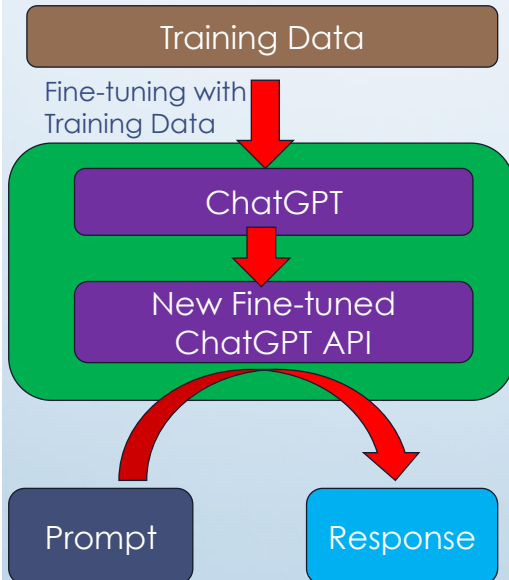
```
proc http  
  method = "POST"  
  url="https://api.openai.com/v1/chat/completions"  
  ct="application/json"  
  in=prompt  
  out=resp;  
  Headers "Authorization" = "Bearer &api_key.";  
Run;
```

```
libname response JSON fileref=resp;  
data r_sas;  
  set response.choices_message;  
run;
```

33

33

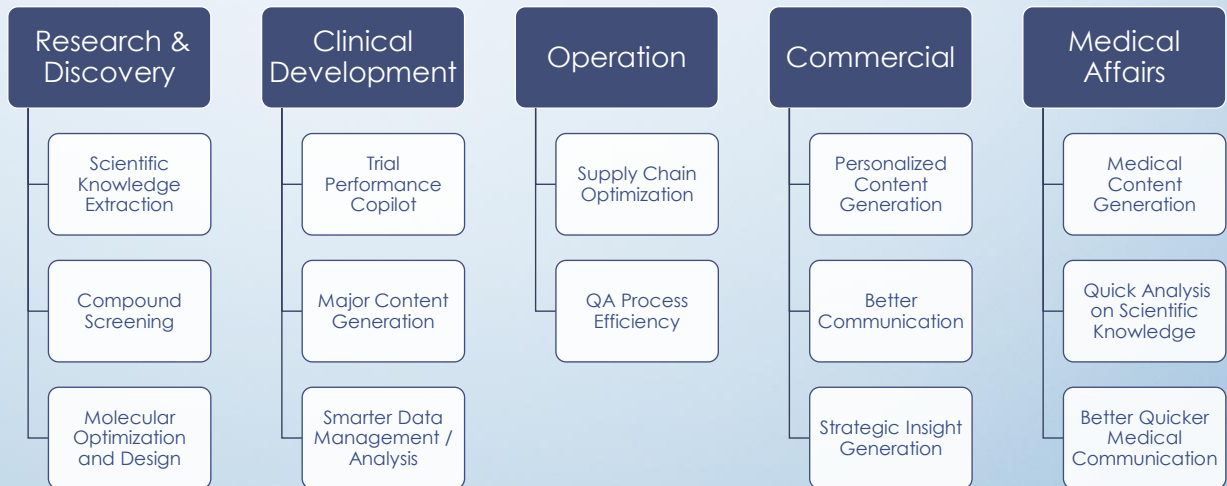
Application using fine-tuned ChatGPT API



34

34

Gen AI Impact in Each Function



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35

How Gen AI can help Pharma Function



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ChatGPT Use Cases

- Simple Inquiry
- Content Development
- Coding
- Comparison
- Patient Profiles
- Safety Cases
- Image Analysis
- Art Generation

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37

Simple Inquiry 1

Prompt :

What is
schizophrenia?
Tell me in 200
words.

Output :

Schizophrenia is a complex and chronic mental disorder that profoundly affects a person's thinking, emotions, and behavior. It typically emerges in late adolescence or early adulthood, and its exact cause remains unclear, though it is thought to involve a combination of genetic, environmental, and neurobiological factors.

Individuals with schizophrenia often experience a range of symptoms, including hallucinations (false sensory perceptions), delusions (false beliefs), disorganized thinking, and reduced emotional expression. They may also struggle with cognitive impairments, such as difficulty with memory, attention, and decision-making.

Schizophrenia can have a profound impact on a person's daily life and functioning, making it challenging to maintain relationships, hold a job, or pursue personal goals. It can lead to significant distress and impairment in social, occupational, and educational domains.

Treatment for schizophrenia typically involves a combination of antipsychotic medications, psychotherapy, and psychosocial support. While these interventions can help manage symptoms and improve the quality of life for individuals with schizophrenia, the condition is often chronic and requires ongoing care and support.

Early intervention and a comprehensive, individualized treatment plan are crucial for managing schizophrenia and helping affected individuals lead fulfilling lives while minimizing the impact of their symptoms.

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38

Simple Inquiry 2

Prompt :
Provide the information about CDISC. Tell me in 200 words.

Output :
CDISC, or Clinical Data Interchange Standards Consortium, is a global non-profit organization focused on developing and advancing data standards for the clinical research and healthcare industries. Founded in 1997, CDISC aims to facilitate the efficient exchange and integration of clinical research data to enhance collaboration, streamline processes, and ultimately improve patient outcomes.

CDISC standards provide a common framework for organizing and formatting data, ensuring consistency and interoperability across diverse research studies and healthcare settings. The organization's flagship standards include the Study Data Tabulation Model (SDTM), which standardizes the structure and content of clinical trial data, and the Analysis Data Model (ADaM), which defines dataset and metadata standards for analysis.

By adopting CDISC standards, researchers, pharmaceutical companies, and regulatory agencies can expedite the regulatory submission process, enhance data quality, and foster data sharing and reuse. CDISC collaborates with stakeholders globally, including academia, industry, and regulatory bodies, to develop and maintain these standards, promoting transparency and efficiency in the generation and analysis of clinical data. The widespread adoption of CDISC standards contributes to advancing clinical research practices, enabling more robust and interoperable data exchange in the pursuit of improved healthcare outcomes.

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39

Personal Question

Prompt :

Based on the interaction with ChatGPT, what is my personality type?

Use MBTI, and provide the reasons

Output :

Personality Type: **INTJ (The Architect)**

Introversion (I): vs Extraversion (E)

You prefer working independently and are energized by deep, focused tasks. Your conversations show a strong inclination toward thoughtful reflection rather than social interaction.

Intuition (N): vs Sensing (S)

You focus on long-term goals, systems, and innovative ideas. You consistently explore how to improve processes using advanced tools like AI and automation, indicating a forward-thinking mindset.

Thinking (T): vs Feeling (F)

You rely on logic and structure in your decision-making. Your work in programming, system validation, and analytics demonstrates a preference for objective analysis over personal values.

Judging (J): vs Perceiving (P)

You show a high level of organization and planning. Whether you're creating templates, managing clinical trial workflows, or mentoring others, your approach is methodical and goal-driven.

Summary:

You are likely an INTJ—strategic, independent, and analytical. You enjoy solving complex problems, optimizing systems, and leading through expertise rather than authority.⁴⁰

40

Brain Storming

Prompt :

I want to generate creative ideas for **promoting a Gen AI integration** in my department.

Please suggest **10 different ideas** that are easy to understand, eye-catching, and easy to follow.

Consider the audience is mostly SAS programmers who are very knowledgeable in clinical trial programming.

Think outside the box and include social media, influencers, and offline ideas too

41

41

Brain Storming

Output :

Here are **10 creative and easy-to-understand promotional ideas** to excite and engage **SAS programmers in a clinical trials department** about **Generative AI integration**—with a mix of digital, social, and offline strategies:

1. "GPT vs. SAS" Showdown

Format: Host a live internal webinar or recorded video comparing how a Gen AI tool solves a clinical programming task (e.g., writing a simple SDTM mapping) vs. how a SAS programmer would approach it.

Hook: "Can GPT keep up with a veteran programmer?"

Bonus: Post clips on LinkedIn or internal social platforms with polling: "Who did it better?"

2. "1-Minute AI Tips" Video Series

Format: Launch a weekly series of 1-minute internal (or LinkedIn) videos showing bite-sized Gen AI use cases (e.g., automating reviewer guides, generating shell specs,

checking code logic).

Easy: Screen-recorded + voiceover.

Amplify: Invite internal team members to guest-host episodes.

3. "Prompt Cookbook" for SAS Programmers

Format: Create a visually appealing PDF or web-based *Gen AI Prompt Cookbook* with categorized prompts:

Writing SDTM specs

Reviewing define.xml logic

Creating test data

Title: "From Prompt to Production: Gen AI for SAS Programmers"

Share: Email it, print and leave copies in break rooms, or upload to your team SharePoint/Slack.

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42

42

Content Development - Email

Prompt :

I am salesperson.

I need to contact the doctors about new medicine, which is 3 times more effective to lower blood pressure.

write marketing emails within 100 words.

Output :

Subject: Revolutionize Blood Pressure Management with Our New Medicine!

Dear Dr. [Doctor's Last Name],

I hope this message finds you well. I am excited to introduce our breakthrough medication, [Medicine Name], designed to transform blood pressure management. With its cutting-edge formula, it is proven to be three times more effective in lowering blood pressure compared to existing solutions.

[Medicine Name] offers a unique opportunity for your patients to achieve better health outcomes. Its advanced technology ensures quicker results and fewer side effects, leading to enhanced patient compliance.

We would be honored to provide you with more information and samples for your evaluation. Together, we can make a significant impact on improving patients' lives.

Let's revolutionize hypertension treatment together. Please reply to this email, and I'll be happy to arrange a meeting at your convenience.

Best Regards,

[Your Name]

[Your Title]

[Your Contact Information]

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43

43

Content Development - Email

Prompt :

I am stat programmer.

I am waiting for the final raw data from data managers. I have communicated with them couple times already, but I have not received the response yet.

Could you write the email for me? Write it in politely and professionally.

Output :

Dear [Data Manager's Name],

I hope this email finds you well. I wanted to follow up on the status of the final raw data for [Study Name/Project]. As we are preparing for the next steps in our analysis, I would appreciate any updates on the expected timeline for data availability.

Please let me know if there are any issues or if further clarification is needed from my end. I appreciate your support and look forward to your response.

Best regards,

[Your Name]

[Your Position]

[Your Company]

.

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44

44

Content Development - Simple protocol synopsis

- Prompt :

Write clinical trial protocol synopsis based on below "title of study".

"title of study" : A phase 3, randomized, double-blind, placebo controlled study to evaluate the efficacy of drug A in schizophrenia patients.

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- Output :

****Clinical Trial Protocol Synopsis: Phase 3 Study on the Efficacy of Drug A in Schizophrenia Patients****

****1. Study Title:****

A phase 3, randomized, double-blind, placebo-controlled study to evaluate the efficacy of drug A in schizophrenia patients.

****2. Objective:****

The primary objective of this clinical trial is to assess the efficacy of drug A in treating schizophrenia in comparison to a placebo. Secondary objectives include evaluating the safety, tolerability, and overall clinical benefit of drug A.

****3. Study Design:****

This study will adopt a randomized, double-blind, placebo-controlled design. Participants will be randomly assigned to either the treatment group receiving drug A or the control group receiving a placebo. The double-blind nature of the study ensures that both investigators and participants are unaware of the assigned treatment.

****4. Participants:****

The study will enroll adult participants diagnosed with schizophrenia according to standardized diagnostic criteria. Inclusion and exclusion criteria will be defined to ensure the homogeneity of the study population.

****5. Interventions:****

- Treatment Group: Participants will receive drug A in the prescribed dosage.
- Control Group: Participants will receive a placebo identical in appearance to drug A.

****6. Primary Outcome Measure:****

The primary efficacy endpoint is the reduction in the severity of schizophrenia symptoms, as measured by a standardized assessment tool, at the end of the treatment period.

****7. Secondary Outcome Measures:****

Secondary endpoints include:

- Changes in cognitive function.
- Improvement in overall functioning and quality of life.
- Incidence and severity of adverse events.
- Assessment of treatment adherence.

****8. Study Duration:****

The trial will be conducted over a specified period, with defined time points for participant recruitment, treatment administration, and follow-up assessments. The duration of the study will be determined based on the anticipated rate of symptom improvement and the need for robust data collection.

****9. Statistical Analysis:****

Statistical methods, including intention-to-treat analysis, will be employed to evaluate the primary and secondary endpoints. Sample size calculations will ensure the study's power to detect meaningful differences between the treatment and control groups.

****10. Ethical Considerations:****

The study will be conducted in accordance with ethical principles outlined in the Declaration of Helsinki and local regulatory requirements. Informed consent will be obtained from all participants, and measures will be taken to ensure participant confidentiality and safety.

****11. Data Monitoring and Safety:****

An independent Data Monitoring Committee (DMC) will oversee the trial's progress, ensuring participant safety and data integrity. Adverse events will be monitored and reported according to regulatory guidelines.

****12. Conclusion:****

This phase 3 clinical trial aims to contribute valuable insights into the efficacy and safety of drug A in the treatment of schizophrenia. The results of this study may have implications for improving the management of schizophrenia and enhancing the overall well-being of affected individuals.

46

Codes conversion

Prompt :

Convert below SAS codes to R codes:

```
data dm;
  merge demo rand;
  by usubjid ;
run;
```

Output :

You can convert the SAS code to R using the `merge` function in R to perform the equivalent operation. Assuming you have two data frames `demo` and `rand` that you want to merge by the `usubjid` variable, here's the equivalent R code:

```
```
Assuming you have loaded your data into two data frames: demo
and rand
Merge the data frames by usubjid
dm <- merge(demo, rand, by="usubjid")
```
```

In this code:

- `demo` and `rand` are your input data frames.
- We use the `merge` function to merge the data frames.
- The `by` parameter specifies the key variable by which to merge the data frames, which in this case is `usubjid`.

After running this code, the merged data frame `dm` will contain the result of the merge operation, similar to what you would achieve with the SAS code you provided.

47

47

Codes Creation 1

Prompt :

Write a **R script** that groups data by "trta" and "avisitn" columns in "advx.xpt" and calculates the mean of the "aval" column for each group at parmcd = 'WEIGHT'

Output :

```
```R
Load necessary libraries
library(haven)
library(dplyr)

Read the XPT file
advx <- read_xpt("path/to/advx.xpt")

Filter the dataset for parmcd == 'WEIGHT'
advx_weight <- advx %>% filter(parmcd == 'WEIGHT')

Group by 'trta' and 'avisitn' and calculate the mean of 'aval' for each group
mean_aval <- advx_weight %>%
 group_by(trta, avisitn) %>%
 summarise(mean_aval = mean(aval, na.rm = TRUE))

Print the result
print(mean_aval)
```
```

Make sure to replace `"path/to/advx.xpt"` with the actual file path of your `advx.xpt` dataset. This script will output a data frame with the mean `aval` for each combination of `trta` and `avisitn` where `parmcd` is 'WEIGHT'.

48

48

Codes Creation 2

Prompt :

Write a **SAS script** that groups data by "trta" and "avisitn" columns in "advs.xpt" and calculates the mean of the "aval" column for each group at parmcd = 'WEIGHT'

Output :

```
```sas
/* Step 1: Import the advs.xpt file */
libname mylib xport "path_to_your_advs.xpt";

/* Step 2: Create a SAS dataset from the imported data */
data advs;
 set mylib.advs;
run;

/* Step 3: Filter the data for parmcd = 'WEIGHT' */
data advs_weight;
 set advs;
 where parmcd = 'WEIGHT';
run;

/* Step 4: Group the data by trta and avisitn, and calculate the mean of aval */
proc means data=advs_weight noprint;
 class trta avisitn;
 var aval;
 output out=mean_aval (drop=_TYPE_ _FREQ_) mean=mean_aval;
run;

/* Step 5: Display the result */
proc print data=mean_aval;
run;
```

49

49

## Codes Creation 3

### Prompt :

Write a **Python script** that groups data by "trta" and "avisitn" columns in "advs.xpt" and calculates the mean of the "aval" column for each group at parmcd = 'WEIGHT'

### Output :

```
```bash
pip install pandas xport

```python
import pandas as pd
import xport

Load the .xpt file
with open("advs.xpt", "rb") as file:
 data = xport.to_dataframe(file)

Filter the data where parmcd is 'WEIGHT'
weight_data = data[data['parmcd'] == 'WEIGHT']

Group by 'trta' and 'avisitn' and calculate the mean of 'aval'
grouped_data = weight_data.groupby(['trta', 'avisitn'])['aval'].mean().reset_index()

Rename the column for clarity
grouped_data = grouped_data.rename(columns={'aval': 'mean_aval'})

Display the result
print(grouped_data)
```
```

50

50

Metadata driven Codes Development

Prompt :

Write a SAS script that reads input data and generates an output dataset based on the following #metadata; Ensure that the script correctly derives each variable from the input data according to the metadata specifications.

- Set up two libname
 - libname raw " " for input data folder path
 - Libname SDTM " " for output data folder path
- Refer to #reference to create each variables;
- Find metadata of the same input data. For your reference, in demo.study_id, demo is input data and study_id is variable from demo input data, so all the identical input data will be processed in the same data procedure.
- Create the new "Variables_Name" from the same input data.
- Merge all the new variables to create the final output dataset.
- Assign "Variable_Label" to "Variable_Name"
- Keep only "Variable_Name" in #metadata.

Assumption:

- ☐ We are using sas datasets as input data, so we will use libname raw and directly read from the folder.
- ☐ We will write back to

```
#metadata:
[ { "Variable_Name":"STUDYID",
  "Variable_Label":"Study Identifier",
  "Type":"Char",
  "Derivation":"demo.study_id",
  "example":"101",
  } ]
```

```
#reference:
Libname raw " ";
Libname sdtm " ";

Data _dm;
Set raw.demo;
STUDYID = study_id;
DOMAIN = "DM"
Run;

Data _rand;
Set raw.rando;
ARMCD = trt_cd;
Run;

Data dm_derived;
Merge _dm (in=a) _rand (in=b);
By subjid;
Run;

Data sdtm.dm;
Set dm_derived;
Keep STUDYID ...;
Label ...;
Run;
```

```
gput;
rappor;
koeb 2110DAID ...;
261 qu' deunee;
```

51

51

Metadata driven Codes Development

Output :

```
/* Replace with actual input data folder path */
libname raw "path_to_input_data";
/* Replace with actual output data folder path */
libname sdtm "path_to_output_data";

/* Extract and derive variables from raw.demo */
data _dm;
set raw.demo;
STUDYID = study_id;
DOMAIN = "DM";
USUBJID = cats(study_id, '-', site_id, '-', subjid);
SUBJID = subjid;
SITEID = site_id;
BRTHDTC = birth_dt;
SEX = sex;
RACE = race;
run;

/* Extract and derive variables from raw.expo */
data _expo;
set raw.expo;
retain RFSTDTC RFXSTDTC;
RFSTDTC = exdtc; /* First exposure date */
RFXSTDTC = exdtc; /* First exposure date */
RFENDTC = exdtc; /* Last exposure date */
RFXENDTC = exdtc; /* Last exposure date */
run;

/* Extract and derive variables from raw.desp */
data _desp;
set raw.desp;
RFCDTC = cons_dt;
DTHDTC = death_dt;
if not missing(death_dt) then DTHFL = "Y";
Run;
```

```
/* Extract and derive variables from raw.rand */
data _rand;
set raw.rand;
ARMCD = trt_cd;
ARM = trt;
ACTARMCD = trt_cd;
ACTARM = trt;
ARMNRSN = trt_reas;
ACTARMUD = trt_reas2;
run;

/* Merge all derived datasets */
data dm_merged;
merge
  _dm (in=a)
  _expo (in=b keep=subjid RFSTDTC RFENDTC
RFXSTDTC RFXENDTC)
  _desp (in=c keep=subjid RFCDTC DTHDTC
DTHFL)
  _rand (in=d keep=subjid ARMCD ARM
ACTARMCD ACTARM ARMNRSN ACTARMUD);
by subjid;
if a;
run;
```

```
/* Finalize the dataset */
data sdtm.dm;
set dm_merged;
COUNTRY = "USA";
AGE = year(input(RFCDTC, yymmdd10.)) -
year(input(BRTHDTC, yymmdd10.));
AGEU = "YEARS";
keep STUDYID DOMAIN USUBJID SUBJID RFSTDTC RFENDTC
RFXSTDTC RFXENDTC RFCDTC REPNDTC DTHDTC DTHFL SITEID
BRTHDTC AGE AGEU SEX RACE ARMCD ARM ACTARMCD
ACTARM ARMNRSN ACTARMUD COUNTRY;
label
  STUDYID = "Study Identifier"
  DOMAIN = "Domain Abbreviation"
  USUBJID = "Unique Subject Identifier"
  SUBJID = "Subject Identifier for the Study"
  RFSTDTC = "Subject Reference Start Date/Time"
  RFENDTC = "Subject Reference End Date/Time"
  RFXSTDTC = "First Date/Time of Exposure"
  RFXENDTC = "Last Date/Time of Exposure"
  RFCDTC = "Informed Consent Date/Time"
  REPNDTC = "Date/Time of End of Participation"
  DTHDTC = "Date/Time of Death"
  DTHFL = "Subject Death Flag"
  SITEID = "Study Site Identifier"
  BRTHDTC = "Date/Time of Birth"
  AGE = "Age"
  AGEU = "Age Units"
  SEX = "Sex"
  RACE = "Race"
  ARMCD = "Planned Arm Code"
  ARM = "Description of Planned Arm"
  ACTARMCD = "Actual Arm Code"
  ACTARM = "Description of Actual Arm"
  ARMNRSN = "Reason Arm and/or Actual Arm is Null"
  ACTARMUD = "Description of Unplanned Actual Arm"
```

52

Codes / Comments Creation

Prompt :

See below comments and convert them in SAS codes.

Show only SAS codes comments:

Merge demo and rand dataset by usubjid and name it as dm

Output :

```
data dm;
  merge demo rand;
  by usubjid;
run;
```

Prompt :

See below SAS codes and provide SAS comments. Refer to examples. Show only SAS comments.

SAS codes:

```
data dm; merge demo rand; by usubjid; run;
```

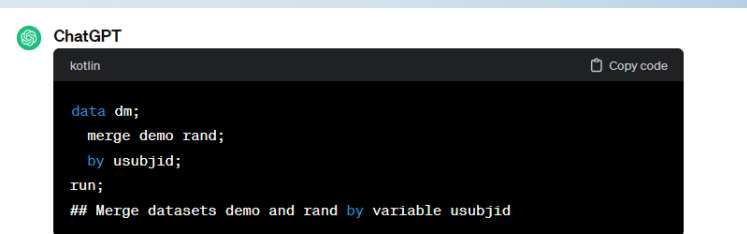
Examples:

SAS codes:

```
proc sort data=dm; by usubjid; run;
```

SAS comments:

```
## Sort dm by usubjid
```



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53

Comparison between SAS and R

Prompt :

I want you to compare SAS and R programming. Provide me with differences, present in a table format

Output :



Certainly! Here is a comparison of SAS and R programming in a table format highlighting the key differences between the two:

| Aspect | SAS | R |
|-----------------------------|--|---|
| Type of Language | Proprietary language | Open-source language |
| Cost | Commercial software with licensing fees | Free and open-source software |
| Data Manipulation | Excellent data manipulation and transformation capabilities | Strong data manipulation capabilities |
| Data Visualization | Basic data visualization features; often requires external tools like SAS Visual Analytics | Rich data visualization libraries (e.g., ggplot2) |
| Statistical Analysis | Comprehensive built-in statistical procedures | Extensive statistical packages and libraries |
| Machine Learning | Limited machine learning capabilities | Extensive machine learning libraries (e.g., scikit-learn, TensorFlow) |
| Community Support | Smaller user community and less open-source collaboration | Large and active user community with extensive online resources |
| Customization | Limited customization options compared to open-source alternatives | Highly customizable with the ability to create packages and functions |

54

Patient Profile Development

- Prompt :
Use below data to provide patient profiles:
{ 'patient' : '01-001', 'sex' : 'male', 'age' : 25, 'race' : 'White' }
- Output :
Based on the provided data, here is a patient profile for '01-001':
 - Patient ID: 01-001
 - Sex: Male
 - Age: 25
 - Race: White

This patient is a 25-year-old male of White race. Please let me know if you need any additional information or if you would like to create profiles for more patients.

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55

55

Data Analysis (1)

- Prompt :
Select patients whose sex is male. Use below "Patient Data". Provide me the answers only.

"Patient Data":
{ 'patient' : '01-001', 'sex' : 'male', 'age' : 25, 'race' : 'White'},
{ 'patient' : '01-002', 'sex' : 'female', 'age' : 28, 'race' : 'Asian'},
{ 'patient' : '02-001', 'sex' : 'female', 'age' : 50, 'race' : 'White'},
{ 'patient' : '02-002', 'sex' : 'male', 'age' : 40, 'race' : 'Black' }
- Output :
The selected patients whose sex is male are:
 1. { 'patient': '01-001', 'sex': 'male', 'age': 25, 'race': 'White' }
 2. { 'patient': '02-002', 'sex': 'male', 'age': 40, 'race': 'Black' }

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56

56

Data Analysis (2)

- Prompt :

Select patients whose age is older than or equal to 40. Use below "Patient Data". Provide me the answers only.

"Patient Data":

```
{ 'patient' : '01-001', 'sex' : 'male', 'age' : 25, 'race' : 'White'},
{ 'patient' : '01-002', 'sex' : 'female', 'age' : 28, 'race' : 'Asian'},
{ 'patient' : '02-001', 'sex' : 'female', 'age' : 50, 'race' : 'White'},
{ 'patient' : '02-002', 'sex' : 'male', 'age' : 40, 'race' : 'Black'}
```

- Output :

Answers:

```
- { 'patient' : '02-001', 'sex' : 'female', 'age' : 50, 'race' : 'White'}
- { 'patient' : '02-002', 'sex' : 'male', 'age' : 40, 'race' : 'Black'}
```

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57

57

Data Analysis (3)

- Prompt :

Sort "Patient Data" by age. Provide me the answers only.

"Patient Data":

```
{ 'patient' : '01-001', 'sex' : 'male', 'age' : 25, 'race' : 'White'},
{ 'patient' : '01-002', 'sex' : 'female', 'age' : 28, 'race' : 'Asian'},
{ 'patient' : '02-001', 'sex' : 'female', 'age' : 50, 'race' : 'White'},
{ 'patient' : '02-002', 'sex' : 'male', 'age' : 40, 'race' : 'Black'}
```

- Output :

Sorted "Patient Data" by age:

```
1. { 'patient' : '01-001', 'sex' : 'male', 'age' : 25, 'race' : 'White'}
2. { 'patient' : '01-002', 'sex' : 'female', 'age' : 28, 'race' : 'Asian'}
3. { 'patient' : '02-002', 'sex' : 'male', 'age' : 40, 'race' : 'Black'}
4. { 'patient' : '02-001', 'sex' : 'female', 'age' : 50, 'race' : 'White'}
```

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58

58

Data Analysis (4)

- Prompt :

Count below "Patient Data" by race

"Patient Data":

```
{ 'patient' : '01-001', 'sex' : 'male', 'age' : 25, 'race' : 'White'},
{ 'patient' : '01-002', 'sex' : 'female', 'age' : 28, 'race' : 'Asian'},
{ 'patient' : '02-001', 'sex' : 'female', 'age' : 50, 'race' : 'White'},
{ 'patient' : '02-002', 'sex' : 'male', 'age' : 40, 'race' : 'Black'}
```

- Output :

Certainly! Let's count the number of patients by race in the given "Patient Data":

- White: 2
- Asian: 1
- Black: 1

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59

59

Data Analysis (5)

- Prompt :

Sort "Patient Data" by age. Provide me the answers only.

"Patient Data":

```
patient,sex,age,race
01-001,male,25,White
01-002,female,28,Asian
02-001,female,50,White
02-002,male,40,Black
```

- Output :

Sorted "Patient Data" by age:

1. 01-001, male, 25, White
2. 01-002, female, 28, Asian
3. 02-002, male, 40, Black
4. 02-001, female, 50, White

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60

60

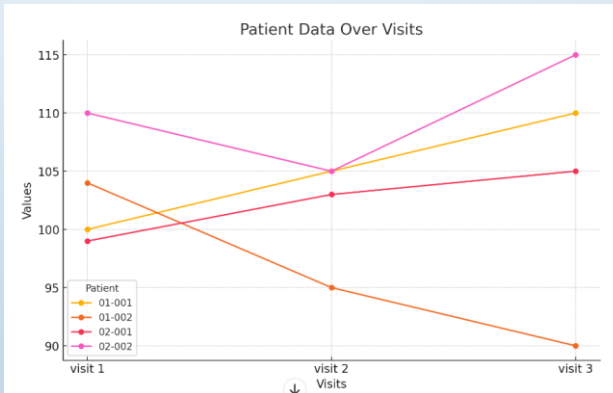
Data Visualization

- Prompt :
Create line graph of "Patient Data"

"Patient Data":

```
{ 'patient': '01-001', 'visit 1': 100, 'visit 2': 105, 'visit 3': 110},
{ 'patient': '01-002', 'visit 1': 104, 'visit 2': 95, 'visit 3': 90},
{ 'patient': '02-001', 'visit 1': 99, 'visit 2': 103, 'visit 3': 105},
{ 'patient': '02-002', 'visit 1': 110, 'visit 2': 105, 'visit 3': 115}
```

- Output :



61

61

Safety Cases Identification

- Prompt :
Classify the safety cases of the text.
Follow these steps in the classification.
1. Identify adverse event in text.
2. Identify patient information in text
3. Identify medication information in text

If text answers to "Yes" on all the steps, classify as 'Safety Issues' Otherwise 'No Safety Issues'

Let's think step by step for classification of the text.

Text: Patient 001 has adverse events after taking the medication.

- Output :

Let's follow the steps you provided to classify the safety issues in the text:

Step 1: Identify adverse event in text

- The text mentions "Patient 001 has adverse events after taking the medication." This indicates that there is an adverse event mentioned in the text.

Step 2: Identify patient information in text

- The text mentions "Patient 001," which is patient information.

Step 3: Identify medication information in text

- The text mentions "taking the medication," which is medication information.

Now, let's evaluate the results of each step:

- Step 1: Adverse event identified

- Step 2: Patient information identified

- Step 3: Medication information identified

Since all three steps have been answered "Yes," we can classify this text as 'Safety Issues' because it contains information about a patient experiencing adverse events after taking medication.

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62

62

Mental Health Assessment 1 – Prompt

- Prompt :

I am a psychiatrist, and I am interviewing a patient for his PANSS score based on answers from the patient.

Use 'Anxiety Answers from patient' and match the 'PANSS Score Anxiety Score' of Absent, Minimal, Mild, Moderate, Moderate severe, Severe and Extreme based on its definition.

Anxiety Answers from patient:

1. Question : Have you been feeling worried or nervous in the past week? Answer : **No**
2. Question : Would you say that you are usually calm and relaxed? Answer : **Yes**
3. Question : What has been making you feel nervous (worried, not clam, not relaxed) Answer : **I have not felt nervous.**
4. Question : Just how nervous (worried, etc.) have you been feeling? Answer : **I have not felt nervous**
5. Question : Have you been shaking at times, or has your heart been racing? Answer : **No**
6. Question : Do you get into a state of panic? Answer : **No**
7. Question : Has your sleep, eating, or participation in activities been affected? Answer : **No**

'PANSS Score Anxiety Score' : Subjective experience of nervousness, worry, apprehension, or restlessness, ranging from excessive concern about the present or future to feelings of panic.

1. Absent - Definition does not apply
2. Minimal - Questionable pathology; may be at the upper extreme of normal limits
3. Mild - Expresses some worry, overconcern, or subjective restlessness, but no somatic and behavioral consequences are reported or evidence.
4. Moderate - Patient reports distinct symptoms of nervousness, which are reflected in mild physical manifestations such as fine hand tremor and excessive perspiration.
5. Moderate severe - Patient reports serious problems of anxiety which have significant physical and behavioral consequences, such as marked tension, poor concentration, palpitations, or impaired sleep.
6. Severe - Subjective state of almost constant fear associated with phobias, marked restlessness, or numerous somatic manifestations.
7. Extreme - Patient's life is seriously disrupted by anxiety, which is present almost constantly and at times reaches panic proportion or is manifested in actual panic attacks.

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63

63

Mental Health Assessment 1 – Answer

- Answer :

Based on the patient's answers to the anxiety-related questions, you can match the PANSS Score Anxiety Score as follows:

1. **Question: Have you been feeling worried or nervous in the past week?**
- Answer: No
- ****PANSS Score Anxiety Score: Absent****
2. **Question: Would you say that you are usually calm and relaxed?**
- Answer: Yes
- ****PANSS Score Anxiety Score: Absent****
3. **Question: What has been making you feel nervous (worried, not calm, not relaxed)?**
- Answer: I have not felt nervous.
- ****PANSS Score Anxiety Score: Absent****
4. **Question: Just how nervous (worried, etc.) have you been feeling?**

- Answer: I have not felt nervous.
- ****PANSS Score Anxiety Score: Absent****

5. **Question: Have you been shaking at times, or has your heart been racing?**

- Answer: No
- ****PANSS Score Anxiety Score: Absent****

6. **Question: Do you get into a state of panic?**

- Answer: No
- ****PANSS Score Anxiety Score: Absent****

7. **Question: Has your sleep, eating, or participation in activities been affected?**

- Answer: No
- ****PANSS Score Anxiety Score: Absent****

Based on the patient's responses, it seems that the subjective experience of anxiety is absent, and there are no signs or symptoms that warrant a PANSS Score Anxiety Score other than **"Absent."**

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64

64

Mental Health Assessment 2 – Prompt

- Prompt :

I am a psychiatrist, and I am interviewing a patient for his PANSS score based on answers from the patient.

Use 'Anxiety Answers from patient' and match the 'PANSS Score Anxiety Score' of Absent, Minimal, Mild, Moderate, Moderate severe, Severe and Extreme based on its definition. Final PANSS score will be the most severe assessment.

Anxiety Answers from patient:

1. Question : Have you been feeling worried or nervous in the past week? Answer : **Yes**
2. Question : Would you say that you are usually calm and relaxed? Answer : **Not Applicable**
3. Question : What has been making you feel nervous (worried, not clam, not relaxed) Answer : **worried**
4. Question : Just how nervous (worried, etc.) have you been feeling? Answer : **a little**
5. Question : Have you been shaking at times, or has your heart been racing? Answer : **No**
6. Question : Do you get into a state of panic? Answer : **No**
7. Question : Has your sleep, eating, or participation in activities been affected? Answer : **No**

'PANSS Score Anxiety Score' : Subjective experience of nervousness, worry, apprehension, or restlessness, ranging from excessive concern about the present or future to feelings of panic.

1. Absent - Definition does not apply
2. Minimal - Questionable pathology; may be at the upper extreme of normal limits
3. Mild - Expresses some worry, overconcern, or subjective restlessness, but no somatic and behavioral consequences are reported or evidence.
4. Moderate - Patient reports distinct symptoms of nervousness, which are reflected in mild physical manifestations such as fine hand tremor and excessive perspiration.
5. Moderate severe - Patient reports serious problems of anxiety which have significant physical and behavioral consequences, such as marked tension, poor concentration, palpitations, or impaired sleep.
6. Severe - Subjective state of almost constant fear associated with phobias, marked restlessness, or numerous somatic manifestations.
7. Extreme - Patient's life is seriously disrupted by anxiety, which is present almost constantly and at times reaches panic proportion or is manifested in actual panic attacks.

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65

65

Mental Health Assessment 2 – Answer

- Answer :

Based on the patient's answers and the PANSS Score Anxiety definitions, we can match the responses to the corresponding scores:

1. Question: Have you been feeling worried or nervous in the past week?

Answer: Yes

PANSS Score Anxiety: Mild

2. Question: Would you say that you are usually calm and relaxed?

Answer: Not Applicable

PANSS Score Anxiety: Absent

3. Question: What has been making you feel nervous (worried, not calm, not relaxed)

Answer: worried

PANSS Score Anxiety: Moderate

4. Question: Just how nervous (worried, etc.) have you been feeling?

Answer: a little

PANSS Score Anxiety: Minimal

5. Question: Have you been shaking at times, or has your heart been racing?

Answer: No

PANSS Score Anxiety: Absent

6. Question: Do you get into a state of panic?

Answer: No

PANSS Score Anxiety: Absent

7. Question: Has your sleep, eating, or participation in activities been affected?

Answer: No

PANSS Score Anxiety: Absent

The final PANSS score would be the most severe assessment, which in this case is Moderate (Question 3). Therefore, the overall PANSS Score Anxiety for this patient based on the provided answers is **Moderate**.

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66

66

Mental Health Assessment 3 – Prompt

- Prompt :

I am a psychiatrist, and I am interviewing a patient for his PANSS score based on answers from the patient.

Use 'Anxiety Answers from patient' and match the 'PANSS Score Anxiety Score' of Absent, Minimal, Mild, Moderate, Moderate severe, Severe and Extreme based on its definition. Final PANSS score will be the most severe assessment.

Anxiety Answers from patient:

1. Question : Have you been feeling worried or nervous in the past week? Answer : **Yes**
2. Question : Would you say that you are usually calm and relaxed? Answer : **Not Applicable**
3. Question : What has been making you feel nervous (worried, not clam, not relaxed) Answer : **worried**
4. Question : Just how nervous (worried, etc.) have you been feeling? Answer : **a lot**
5. Question : Have you been shaking at times, or has your heart been racing? Answer : **Yes**
6. Question : Do you get into a state of panic? Answer : **Yes**
7. Question : Has your sleep, eating, or participation in activities been affected? Answer : **Yes**

'PANSS Score Anxiety Score' : Subjective experience of nervousness, worry, apprehension, or restlessness, ranging from excessive concern about the present or future to feelings of panic.

1. Absent - Definition does not apply
2. Minimal - Questionable pathology; may be at the upper extreme of normal limits
3. Mild - Expresses some worry, overconcern, or subjective restlessness, but no somatic and behavioral consequences are reported or evidence.
4. Moderate - Patient reports distinct symptoms of nervousness, which are reflected in mild physical manifestations such as fine hand tremor and excessive perspiration.
5. Moderate severe - Patient reports serious problems of anxiety which have significant physical and behavioral consequences, such as marked tension, poor concentration, palpitations, or impaired sleep.
6. Severe - Subjective state of almost constant fear associated with phobias, marked restlessness, or numerous somatic manifestations.
7. Extreme - Patient's life is seriously disrupted by anxiety, which is present almost constantly and at times reaches panic proportion or is manifested in actual panic attacks.

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67

67

Mental Health Assessment 3 – Answer

- Answer :

Based on the patient's answers, let's match them to the corresponding PANSS Score Anxiety Scores:

1. Question: Have you been feeling worried or nervous in the past week?

Answer: Yes

PANSS Score: Mild

2. Question: Would you say that you are usually calm and relaxed?

Answer: Not Applicable

PANSS Score: Absent

3. Question: What has been making you feel nervous (worried, not calm, not relaxed)?

Answer: Worried

PANSS Score: Mild

4. Question: Just how nervous (worried, etc.) have you been feeling?

Answer: I was worried a lot

PANSS Score: Moderate

5. Question: Have you been shaking at times, or has your heart been racing?

Answer: Yes

PANSS Score: Moderate Severe

6. Question: Do you get into a state of panic?

Answer: Yes

PANSS Score: Severe

7. Question: Has your sleep, eating, or participation in activities been affected?

Answer: Yes

PANSS Score: Severe

Based on the most severe assessment, the final PANSS Score for anxiety would be **"Severe"** as it is the highest level mentioned in the responses.

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68

68

Image and Graphic recognition using ChatGPT4

- ChatGPT 4 can analyze and comment on images and graphics while ChatGPT3.5 mainly focuses on text
- ChatGPT4
 - Describe the content of phone,
 - Identify trends in graph
 - Generate captions for images.

| | | | |
|--------|-----------|---------|--------|
| Name : | Teacher : | Score : | Date : |
| 18.18 | 61.53 | 34.28 | 88.62 |
| +22.74 | +88.24 | +96.41 | +44.37 |
| 70.65 | 37.59 | 89.54 | 87.17 |
| +73.85 | +88.88 | +22.17 | +33.84 |
| 71.86 | 74.94 | 86.27 | 30.17 |
| +68.33 | +40.34 | +65.14 | +71.48 |
| 65.61 | 12.57 | 63.46 | 26.12 |
| +33.26 | +30.32 | +82.11 | +14.11 |

Give me the answers.

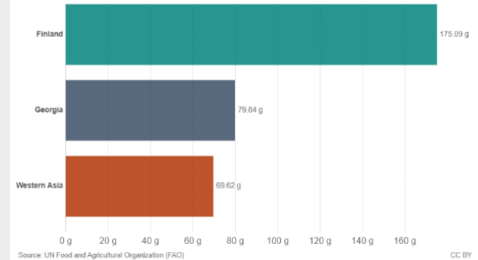
Certainly! Let's solve these addition problems:

1. $18.18 + 22.74 = 40.92$
2. $61.53 + 80.24 = 141.77$
3. $34.28 + 96.41 = 130.69$
4. $88.62 + 44.37 = 132.99$
5. $70.65 + 73.85 = 144.50$
6. $37.59 + 88.88 = 126.47$

User: What is the sum of average daily meat consumption for Georgia and Western Asia? Provide a step-by-step reasoning before providing your answer.

Daily meat consumption per person, 1997

Average daily meat consumption per person, measured in grams per person per day. Countries with daily meat consumption greater than the expected EU average of 165g per person are shown in red, yellow are those countries below 165g but exceeding the more ambitious limit of 40g per person, and in blue are those below 40g per person.



GPT-4: To find the sum of average daily meat consumption for Georgia and Western Asia, we will follow these steps:

1. Identify the average daily meat consumption for Georgia.
2. Identify the average daily meat consumption for Western Asia.
3. Add the values from steps 1 and 2.

Step 1: From the provided chart, we see that the average daily meat consumption for Georgia is 79.84 grams per person per day.

Step 2: From the chart, we see that the average daily meat consumption for Western Asia is 69.62 grams per person per day.

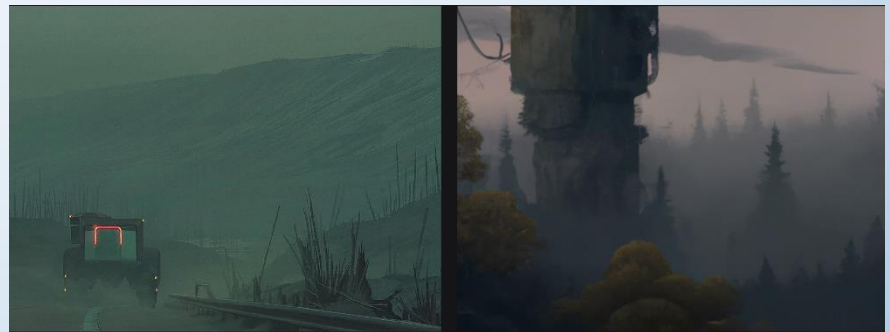
Step 3: Add the values from steps 1 and 2.

$79.84 \text{ g (Georgia)} + 69.62 \text{ g (Western Asia)} = 149.46 \text{ g}$

69

Art Generation

- ChatGPT can generate creative, realistic visuals and art from a simple prompt by mixing concepts, characteristics and styles.



- A left drawing is the original artwork by Simon Stalenhag
- A right drawing from the prompt:
Monumental old ruins tower of a dark misty forest, overcast, sci-fi digital painting by Simon Stalenhag

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70

70

Video Generation using Sora

- OpenAI can generate short video(60 second) based on simple prompt.

- Prompt :

A bicycle race on ocean with different animals as athletes riding the bicycles with drone camera view



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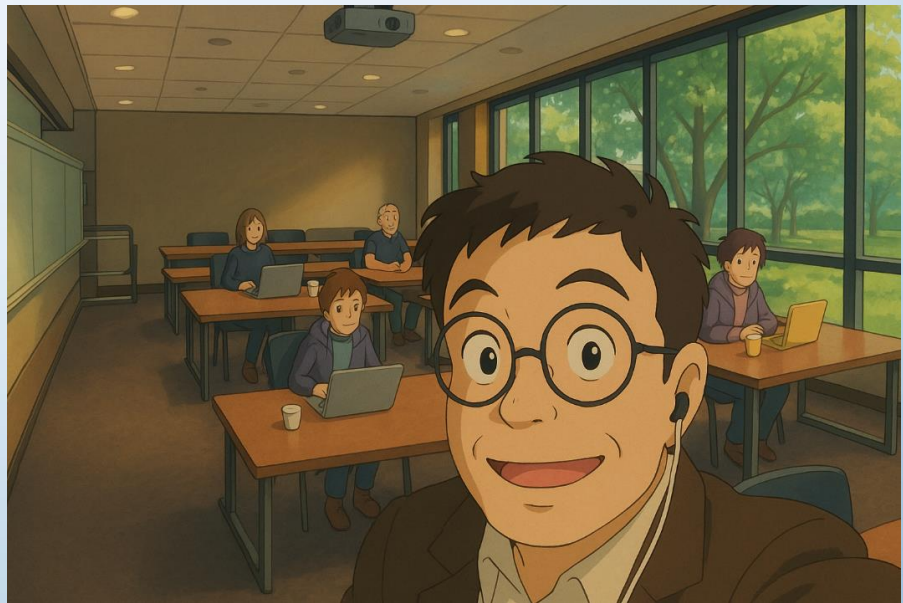
71

71

Image creation using ChatGPT

Prompt:

Convert the picture in Ghibli animation format



72

How to use ChatGPT

- What is Prompt?
- Introduction of Prompt Engineering

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73

73

Prompt it is the new ~~Google~~ it

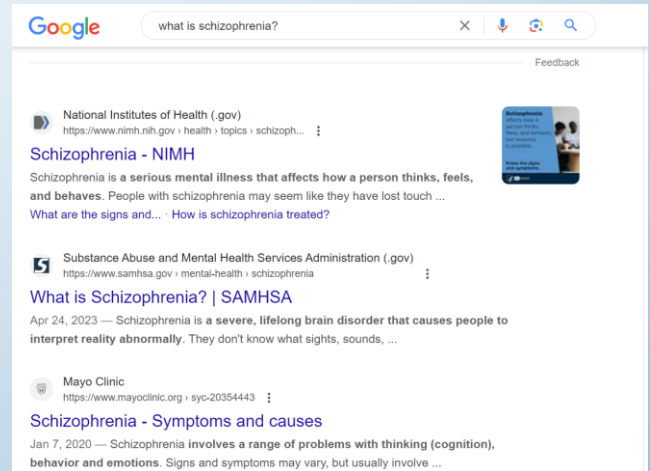
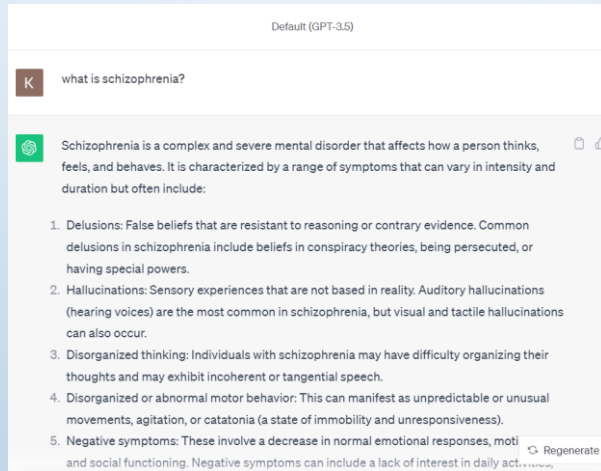


74

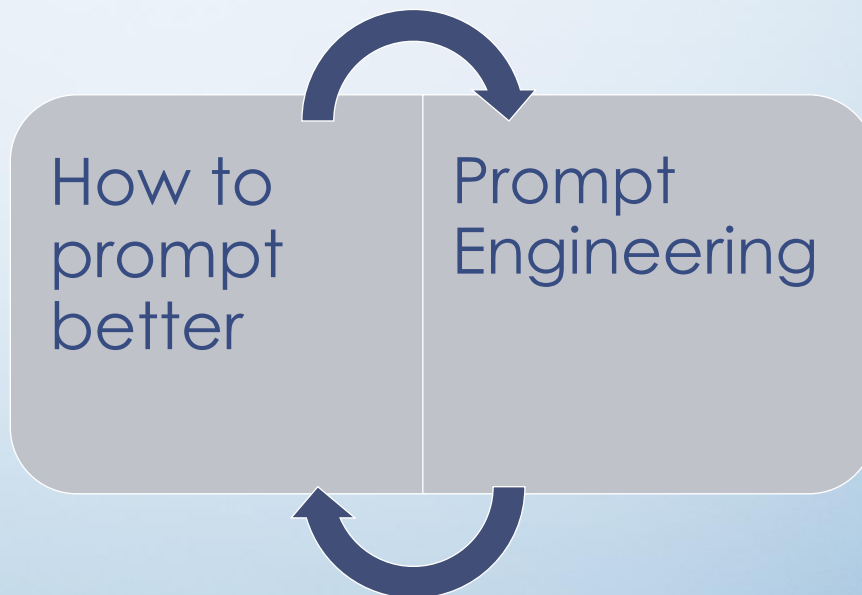
74

What is Prompt?

- Prompt – A tool to communicate with Gen AI (ChatGPT)



75



76

76

What is Prompt Engineering?

- Prompt Engineering – the **art** of creating effective communication/ prompt that guide Gen AI (ChatGPT) to generate **desired response**.

Bad Prompt – Bad Response
(e.g., Hallucination)



Good Prompt – Good Response



77

**ChatGPT is like a very smart, capable,
recently graduated intern/new hire.**



78

78

How to work with the recently graduated, smart intern at work?

How to make an intern competent or incompetent?



79

79

How to prompt better in ChatGPT (How to communicate better)

- Get Specific
- Add Constraints
- Add more contexts
- Set the role
- Coach with feedback
- Probe further
- Step by step introduction

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80

80

Get specific

- Generic inquiry will provide generic (and less impressive) answers

| Generic Prompt | Specific Prompt |
|-------------------------------|---|
| Write an essay about flowers. | Write an essay about flowers at 9 th grade level |
| Write me a rap song | Write me a rap song using references to SpongeBob Square Pants. |
| Write me a promotion request. | Write me a promotion request as Manager of Clinical Data Manager. |

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81

81

Add Parameter or Constraints

- Restriction and constraints breed a creativity
- Example of parameters
 - Under 100 words
 - Include word of "creative" / do not use a word of "creative"
 - Make it under \$100
 - Make it work for 15 people

| Generic Prompt | Prompt with parameter |
|---|--|
| Write an essay about flowers. | Write an essay about flowers at 9 th grade level within 200 words |
| Tell me the best places to visit in Prague. | Tell me the best places to visit in Prague within 4 days. |
| Tell me the best places to visit in Prague. | Tell me the best places to visit in Prague with \$500 budget. |

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82

82

Add more context

- If one adds more background, ChatGPT can provide more relevant, customized answers.
- Provide the reason that I am using ChatGPT

| Generic Prompt | Prompt with more context |
|-------------------------|--|
| Write a greeting email. | I have not seen my friend for a long time. I want to know how he is doing. Write a greeting email. |
| | |
| | |

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83

83

Set the role

- If one gives a role to ChatGPT and ask, ChatGPT play a role given and provide the answers.
- Example of roles : boss, parent, salesperson,

| Generic Prompt | Prompt with role |
|---|---|
| Write a greeting email. | I am a salesperson who markets new medicine. Write a greeting email. |
| Review this proposal and provide your feedback. | Pretend you are the head of IT. Review this proposal and provide your feedback. |
| How do you respond below proposal? | You are Steve Jobs. How do you respond below proposal? |

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84

84

Coach ChatGPT with feedback

- After ChatGPT provide an answer, one can provide a feedback and ask again.
- By iterating, one can get the better answer

| Generic Prompt | Prompt with feedback |
|-------------------------|--------------------------|
| Write a greeting email. | |
| | Make it in 50 words. |
| | Make it more formal. |
| | Make it more persuasive. |

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85

85

Provide a reference

- This technique is called 'Few-shot Learning'
- By providing reference/ examples, one could receive customized response.
- It will help ChatGPT to mimic writing style and format.

| Generic Prompt | Prompt with feedback |
|-------------------------|--|
| Write a greeting email. | Read the following passage and analyze its style. I want you to copy this style to write a greeting email. |
| | |
| | |

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86

86

Probe further

- After receiving the answers, one could probe further in the current chat.
- One can find more detail information from the certain part of previous response.

| Generic Prompt | Prompts with probing |
|----------------|---|
| | |
| | Tell me more about paragraph 2. |
| | Tell me more about option 1 and 2. |
| | Please explain the attraction in paragraph 2. |

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87

87

Provide step by step instruction

- One can give step by step instruction on how ChatGPT think the answer "step by step"
- This will help ChatGPT to perform better.

| Generic Prompt | Prompt with step by step introduction |
|----------------|---------------------------------------|
| | Perform the following actions: |
| | 1. Summarize the below text |
| | 2. Translate in French |
| | 3. Convert into JSON format |
| | |
| | |

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88

Prompt Engineering Technique

- Zero Shot
- Few Shot
- Reflection
- Chain of Thought
- Tree of Thought

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89

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Prompt Technique : Zero-Shot

- Purpose : One time Query
- Method : Straight-up Query
- Examples
 - Prompt : **What is 2 + 8?**
 - Output : **2 + 8 equals 10.**

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90

90

Prompt Technique : Few-Shot (1)

- Purpose : Teaching AI models to follow specific patterns
- Method : Feeding model with examples before asking the desired questions
- Examples
 - Prompt :
 - 2@4 = 8
 - 3@3 = 9
 - 4@2 = 8
 - 5@3 = 15
 - 4@2 =
 - Output :

It seems like you have a pattern where you're using "@" as an operator. Based on the examples you provided:

2@4 = 8 (2 multiplied by 4)

3@3 = 9 (3 multiplied by 3)

4@2 = 8 (4 multiplied by 2)

5@3 = 15 (5 multiplied by 3)

So, following this pattern, 4@2 would be equal to 4 multiplied by 2, which is 8.

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Prompt Technique : Few-Shot Examples (2)

- | | |
|---|--|
| <ul style="list-style-type: none"> • Prompt : <p>Classify these conversations:</p> <p>Text: Patients has Adverse event.
Label: Safety Issue</p> <p>Text: Patients does not have Adverse event.
Label: No Safety Issue</p> <p>Text: Patients has a headache.
Label: Safety Issue</p> <p>Text: Patients feel sick after taking the medication
Label: Safety Issue</p> <p>Text: Patients does not have any Adverse Event
Label: No Safety Issue</p> <p>Text: Patients feel good after taking the medication
Label:</p> | <ul style="list-style-type: none"> • Output : <p>Based on the provided texts and labels, here is the classification of these conversations:</p> <ol style="list-style-type: none"> 1. Text: Patients has Adverse event.
Label: Safety Issue 2. Text: Patients does not have Adverse event.
Label: No Safety Issue 3. Text: Patients has a headache.
Label: Safety Issue 4. Text: Patients feel sick after taking the medication.
Label: Safety Issue 5. Text: Patients does not have any Adverse Event.
Label: No Safety Issue 6. Text: Patients feel good after taking the medication.
Label: No Safety Issue <p>It appears that the classification is based on whether the patient experienced an adverse event (Safety Issue) or not (No Safety Issue) after taking the medication or as described in the text.</p> |
|---|--|

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92

Prompt Technique : Chain of Thought

- Purpose :
 - Teaching AI model the thought process it should use.
 - It is a subset of few-shot.
- Method : Breaking the prompt into simpler steps – step 1, step 2, step 3, step 4 and so on
- Examples
 - Prompt :

Provide the answers

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can have 3 tennis balls. How many tennis balls does he have now?

A: Rogers started with 5 balls.
2 cans of 3 tennis balls each is 6 tennis balls.
 $5 + 6 = 11$. The answer is 11.

Q: The cafeteria has 23 oranges. If they use 20 and buy 6 more. How many oranges do they have?
 - Output :

A: The cafeteria started with 23 oranges.
They used 20, so they have $23 - 20 = 3$ oranges left.
Then, they bought 6 more.
So, they now have $3 + 6 = 9$ oranges. The answer is 9

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93

Prompt Technique : Chain of Thought 2

- **Prompt :**
Explain why SAS programmers need to learn prompt engineering.
- Follow below steps:
 1. Provide the introduction
 2. Provide the two examples
 3. Provide the conclusion with introduction and two examples
- **Output :**
 - **Introduction:****
SAS (Statistical Analysis System) programmers play a crucial role in extracting meaningful insights from data through programming and analytics. One essential skill for SAS programmers is prompt engineering, which involves creating interactive prompts to enhance the flexibility and usability of their programs. This skill is particularly important as it allows users to customize their analyses, making the programming process more dynamic and user-friendly.
 - **Two Examples:****
 - *Example 1: Enhancing User Interaction***
Consider a scenario where a SAS programmer develops a program to analyze sales data. Without prompt engineering, the program might be fixed to analyze a specific time period or product category. However, by incorporating prompts, the programmer can allow users to input their preferred date range
 - or choose specific product categories. This enhances user interaction, making the program adaptable to different business scenarios and reducing the need for multiple versions of the same code.
 - *Example 2: Streamlining Data Selection***
In another example, a SAS programmer is tasked with creating a report that summarizes employee performance metrics. Instead of hardcoding specific employee IDs or performance criteria, prompt engineering enables the programmer to incorporate prompts for users to input the desired employee IDs, performance thresholds, or other relevant parameters. This ensures that the program can be reused for various teams or departments without the need for extensive code modification, thereby streamlining the data selection process.
 - **Conclusion:****
In conclusion, prompt engineering is a valuable skill for SAS programmers due to its ability to enhance user interaction and streamline data selection. By incorporating interactive prompts into their programs, SAS programmers empower users to customize analyses based on specific requirements, making the programs more versatile and user-friendly. The examples provided illustrate how prompt engineering enables adaptability, reducing the need for repeated code development and maintenance. Ultimately, mastering prompt engineering is essential for SAS programmers aiming to create robust and flexible solutions for data analysis and reporting.

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Prompt Technique : Chain of Thought 3

- Prompt :

Classify the safety issues of the text.
Follow these steps in the classification.

 1. Identify adverse event in text.
 2. Identify patient information in text
 3. Identify medication information in text

If text answers to "Yes" on all the steps, classify as 'Safety Issues' Otherwise 'No Safety Issues'

Let's think step by step for classification of the text.

Text: Patient 001 has adverse events after taking the medication.
- Output :

Let's follow the steps you provided to classify the safety issues in the text:

Step 1: Identify adverse event in text
- The text mentions "Patient 001 has adverse events after taking the medication." This indicates that there is an adverse event mentioned in the text.

Step 2: Identify patient information in text
- The text mentions "Patient 001," which is patient information.

Step 3: Identify medication information in text
- The text mentions "taking the medication," which is medication information.

Now, let's evaluate the results of each step:

 - Step 1: Adverse event identified
 - Step 2: Patient information identified
 - Step 3: Medication information identified

Since all three steps have been answered "Yes," we can classify this text as 'Safety Issues' because it contains information about a patient experiencing adverse events after taking medication.

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Prompt Technique : Reflection

- Purpose : To understand how ChatGPT4 generates the output.
- Method : Ask the question to ChatGPT, and ask 'Helper Agent' to explain the output / process of ChatGPT
- Examples
 - Prompt :

ChatGPT and Helper Agent, I'd like to ask both of you. ChatGPT, provide answers to my questions, and Helper Agent, reflect on ChatGPT's responses to help optimize the conversation. After the Helper Agent's reflection, ChatGPT, provide a final response to the initial question that incorporates the insights from the Helper Agent's analysis. Let's begin. \[Your question here\]"
 - Output :
 - ChatGPT:
 - Helper Agent:

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96

Prompt Template

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97

Prompt Template

| Categories | Explanation | Examples |
|-------------------------|---|--|
| # Identity | Identity or the roles that AI will play | A short sentence.
I am SAS programmer. |
| # Task | Main task that AI will do for me. | A short sentence.
Create SAS codes. |
| # Sub tasks | Subtasks, Step by Step Tasks that AI will follow to do main task. Refer to #Context and #Guidance and examples for tasks. | Multiple sentences.
First read ## Guidance and examples for the reference.
Do Step 1, 2 and 3. |
| # Context | Background of this task.
Who you are
State of projects
Goal of projects | Multiple sentences.
This is SAS codes that create SDTM DM domain from EDC raw data. |
| # Guidance and examples | Guidance or Examples that AI use or refer to for the task | Current SOP examples for the new SOP development task
SDTM and raw metadata for SDTM development task |

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98

Prompt Template Examples

- Prompt :
- # Identity
- # Task
- # Sub Tasks
- # Context
- # Guidance and Example

Prompt Template Example - Prompt

| | | |
|--|--|--|
| <p># Identity
I am SOP developer.</p> | <p>(https://database.ich.org/sites/default/files/E6_R2_Addendum.pdf)</p> | <ul style="list-style-type: none">•SDTM Developer creates SDTM datasets using SAS programming techniques.•Ensure adherence to SDTM variable naming conventions and data types.•Include variable labels, formats, and definitions within the SAS datasets.•Implement data cleaning and validation procedures to ensure data quality. |
| <p># Task
I am writing SOP for TFL development</p> | <p>4. Responsibilities
•SDTM Developer:</p> <ul style="list-style-type: none">• Develops SDTM mapping specification.• Develops SDTM datasets based on the approved mapping document. <p>•SDTM Validator:</p> <ul style="list-style-type: none">• Validate SDTM datasets based on SDTM specification.• Reviews and approves the final SDTM datasets. | |
| <p># Sub Tasks
Refer to SOP for SDTM Development in #Guidance. Write SOP for TFL development</p> | | <p>5.4. Review and Approval
•SDTM validator reviews the developed SDTM datasets for accuracy and completeness.</p> |
| <p># Context
TFL is the abbreviation of Table, Figure and Listing. SAS programmers create TFL from ADaM and SDTM datasets based on SAP. TFL should be validated by independent SAS programmers.</p> | <p>5. Procedure
5.1. Pre-development Activities
•SDTM Developer reviews the clinical trial protocol and data collection forms.
•Identify data elements for SDTM mapping.
•Select the appropriate CDISC SDTM Implementation Guide based on the study type.</p> <p>5.2. SDTM Mapping
•Map each data element from the clinical trial forms to the corresponding SDTM domain and variable.
•Utilize controlled terminologies (e.g., MedDRA) for coded data elements.
•Document variable labels, formats, definitions, and value ranges in SDTM mapping specification.</p> | <p>6. Training
•Train personnel involved in data analysis and reporting on the structure and content of the SDTM datasets.</p> <p>7. Records
•Maintain records of all SDTM development documents, including mapping documents, programming code, and data cleaning logs.</p> |
| <p># Guidance
SOP for SDTM Development</p> <p>1. Purpose
This Standard Operating Procedure (SOP) outlines the process for developing Study Data Tabulation Model (SDTM) datasets for clinical trials, ensuring data consistency, regulatory compliance, and efficient analysis.</p> <p>2. Scope
This SOP applies to all personnel involved in SDTM development, including SAS programmers, data managers, and clinical trial leads.</p> | <p>5.3. SDTM Dataset Development</p> | <p>8. Deviations
•Any deviations from this SOP must be documented with justification and approval from the Programming Lead.</p> <p>9. Revision History
•Maintain a revision history of this SOP, including the date of each revision and a brief description of the changes made.</p> |
| <p>3. References
•ICH E6 Good Clinical Practice</p> | | |

Prompt Template Examples - Output

SOP for TFL Development

1. Purpose

This Standard Operating Procedure (SOP) outlines the process for developing Table, Figure, and Listing (TFL) outputs from Analysis Data Model (ADaM) and Study Data Tabulation Model (SDTM) datasets for clinical trials. The objective is to ensure the creation of accurate, consistent, and regulatory-compliant TFLs that support efficient data analysis.

2. Scope

This SOP applies to all personnel involved in TFL development, including SAS programmers, data managers, and clinical trial leads.

3. References

- ICH E6 Good Clinical Practice (https://database.ich.org/sites/default/files/E6_R2_Addendum.pdf)
- SDTM Development SOP

4. Responsibilities

- ### TFL Developer:
 - Develops TFL mapping specification based on the Analysis Plan (SAP).
 - Creates TFL outputs using SAS programming techniques.
 - Implements validation checks to ensure TFL

accuracy and consistency.

Independent SAS Programmer (TFL Validator):

- Validates TFL outputs based on TFL mapping specification.
- Reviews and approves the final TFL outputs.

5. Procedure

5.1. Pre-development Activities

- TFL Developer reviews the SAP and identifies variables for TFL mapping.
- Identify any specific requirements or considerations for TFL development outlined in the SAP.

5.2. TFL Mapping

- Map each variable identified in the SAP to the corresponding TFL output.
- Utilize controlled terminologies when applicable.
- Document output labels, formats, definitions, and any statistical methodology used in the TFL mapping specification.

5.3. TFL Development

- TFL Developer creates TFL outputs using SAS programming techniques.
- Ensure adherence to TFL output naming conventions and formatting.
- Include output labels, formats, and definitions within the output files.

- Implement data quality checks and validation procedures.

5.4. Review and Approval

- Independent SAS Programmer (TFL Validator) reviews the developed TFL outputs for accuracy and completeness.
- Any discrepancies or issues identified during the validation process are documented and addressed by the TFL Developer.

6. Training

- Train personnel involved in TFL development and validation on the structure and content of TFL outputs.

7. Records

- Maintain records of all TFL development documents, including mapping documents, programming code, and validation logs.

8. Deviations

- Any deviations from this SOP must be documented, providing justification and obtaining approval from the Programming Lead.

9. Revision History

- Maintain a revision history of this SOP, including the date of each revision and a brief description of the changes made.

101

Prompt Template Examples

- Prompt 1: Pareto Principle (80/20 Rule)
"Using the Pareto Principle, help me identify the 20% of activities in my [personal life/professional life/business goals] that yield 80% of my results. Provide a step-by-step guide for analyzing my current efforts, pinpointing high-impact tasks, and creating an action plan to prioritize these tasks. Additionally, suggest ways to delegate, automate, or minimize the remaining 80% of lower-impact activities to free up time for what truly matters."
- Prompt 2: SMART Goals
"Help me create three SMART goals for [specific time frame] that align with my [personal/professional/fitness/etc.] objectives. For each goal, provide a clear breakdown of the goal itself, milestones to track progress, specific metrics to measure success, potential challenges I may face, and strategies to overcome these challenges and stay on track."
- Prompt 3: The 5 Whys Method
"Using the 5 Whys technique, help me identify the root cause of a recurring issue I face in [specific area, e.g., productivity, personal relationships, project management]. Provide a detailed step-by-step approach to uncover the core problem by repeatedly asking 'Why?' until the root cause is clear. Then, offer actionable solutions to address the root issue and prevent similar challenges in the future."
- Prompt 4: Atomic Habits
"Using the principles from Atomic Habits by James Clear, help me build a sustainable habit for [specific goal]. Provide a comprehensive framework that includes habit stacking ideas to make this habit easier to integrate into my daily routine, clear cues and triggers to initiate the habit, strategies for tracking progress and staying motivated, and a reinforcement plan to make the habit stick over the next [time frame]."
- Prompt 5: Ikigai Framework
"Using the Ikigai framework, help me find clarity about how to align [my career/personal life/business] with my purpose by identifying what I love doing, what I'm skilled at, what the world needs, and what I can be paid for. Based on this alignment, provide actionable steps for creating a fulfilling [career/life plan] that incorporates all four elements of Ikigai."

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102

102

Prompt Formula

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103

103

6 Prompt Components

1. Task
2. Context
3. Example
4. Persona
5. Format
6. Tone

More important



Optional

Task:

Generate an open source strategy for the biometrics department.

Context:

The user is a senior biometrics professional working in a regulated clinical research setting, aiming to reduce reliance on commercial tools and promote innovation through open source adoption.

Example:

For instance, the strategy could include evaluating tools like R for statistical programming, Git for version control, and defining contribution guidelines for internal package development.

Persona:

You are an experienced open source strategist familiar with clinical data workflows, regulatory environments, and open science principles.

Format:

Provide a structured outline in bullet points, organized into key sections: Goals, Current State, Tool Evaluation, Governance, Implementation Plan, and Recommendations.

Tone:

Professional, clear, and practical—suitable for presentation to department leadership.

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104

104

Task

- Always start “Task sentence” with an Action Verb
 - Generate
 - Give
 - Write
 - Analyze...
- One Simple Task
 - Generate Open Source strategy for biometrics department.
- Multi-Task Request
 - Assess current tools and needs for biometrics department
 - Research and Evaluate Open Source Solutions
 - Draft Open Source Strategy Document

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105

105

Context

- To generate a great context, ask three questions to self.
 1. What is the user's background?
 2. What does success look like?
 3. What environment are they in?
- Example context
 1. background : I am the head of Biometrics Department.
 2. Success Criteria : I want to build a clear, compliant, practical open source strategy that improves efficiency and encourages the innovation within 1 or 2 years.
 3. Environment : The work should be done in a regulated clinical trial setting with a mix of proprietary tools like SAS and open source tools like R and Python.
- Then, add “Generate Open Source strategy for biometrics department” to the end.

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106

106

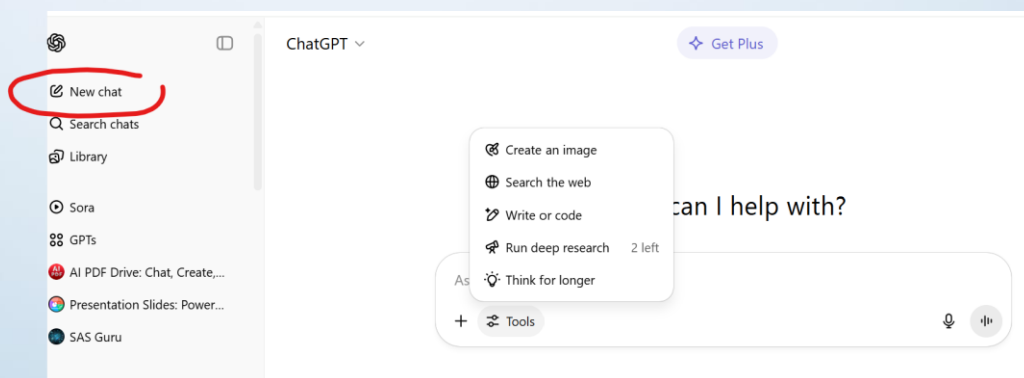
ChatGPT Functionality

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107

107

Start new chat



- Go to <https://chatgpt.com/?model=auto>
- Click "New Chat" to start the new chat.

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108

108

Search

I am visiting Paris for 5 days with my family. Tell me about the itinerary.

Visiting Paris with your family for five days offers a wonderful opportunity to explore the city's rich history, vibrant culture, and family-friendly attractions. Here's a thoughtfully curated itinerary to help you make the most of your trip:

FR Day 1: Iconic Landmarks & Seine River Cruise

- Morning:** Begin your adventure at the **Eiffel Tower**. To avoid long queues, it's advisable to book tickets in advance. Children under 4 years old can enter for free. Afterward, enjoy a leisurely walk through the **Champ de Mars**, where kids can play and you can relax with views of the tower.
- Afternoon:** Visit the **Aquarium de Paris**, located near the Trocadéro. It's home to over 10,000 fish and invertebrates, including a shark tunnel that's sure to captivate the children.
- Evening:** Embark on a **Seine River cruise**. Companies like Vedettes de Paris offer family-friendly cruises with engaging commentary suitable for children aged 6 to 12. As night falls, find a spot to watch the Eiffel Tower's sparkling light show, which occurs every hour on the hour after sunset.

Day 2: Art, Gardens & Cultural Exploration

- Morning:** Head to the **Louvre Museum**. To make the visit engaging for kids, consider a scavenger hunt or a guided family tour focusing on highlights like the Mona Lisa and Egyptian antiquities.
- Afternoon:** Stroll through the **Tuileries Garden**, where children can enjoy playgrounds and a carousel. Continue to the **Champs-Élysées** for some shopping or a treat at one of the many cafés.
- Evening:** Dine in the **Saint-Germain-des-Près** district, known for its charming streets and family-friendly restaurants.

Day 3: Historic Sites & Montmartre

- Morning:** Visit **Notre-Dame Cathedral**. Although it's undergoing restoration, the exterior and surrounding area remain impressive.

Search the web

Find the best restaurant in Paris in Day1 itinerary.

For your Day 1 itinerary in Paris, after visiting the Eiffel Tower and enjoying a Seine River cruise, here are some excellent dining options to consider:



1. Le Jules Verne
Open now until 9:30 PM · French · \$\$\$\$ · 4.3 (511 reviews)
A Michelin-starred restaurant located on the second floor of the Eiffel Tower, offering panoramic views and refined French cuisine crafted by Chef Frédéric Anton.

2. Madame Brasserie
Restaurant · 3.8 (20 reviews)
Situated on the first floor of the Eiffel Tower, this brasserie offers sustainable French cuisine with views of the Trocadéro. led by Chef Thierry Marx.

3. Chez Pippo
Open now until 11:00 PM · Italian · \$\$\$ · 4.0 (21 reviews)
A family-friendly Italian trattoria near the Eiffel Tower, serving homemade pasta, pizzas, and bruschetta in a cozy setting.

4. Le Casse Noix
Closed until Mon 12:00 PM · French · \$\$\$ · 4.6 (86 reviews)
A charming bistro offering traditional French dishes with a seasonal menu, known for accommodating families and providing excellent service.

Search the web

- Click "Search" and prompt to have response and search link together.

109

Reason

Reasoning Prompt :

You are reviewing an SDTM AE (Adverse Events) dataset. You notice that for a subject, AEENDTC (Adverse Event End Date/Time) is missing, but AEOU (Outcome of Adverse Event) is recorded as "RECOVERED/RESOLVED".

Based on CDISC SDTM standards and your knowledge of clinical trial data, should this be flagged as a potential data issue? Why or why not? How would you verify or correct this? Explain your reasoning process.

What can I help with?

You are reviewing an SDTM AE (Adverse Events) dataset. You notice that for a subject, AEENDTC (Adverse Event End Date/Time) is missing, but AEOU (Outcome of Adverse Event) is recorded as "RECOVERED/RESOLVED". Based on CDISC SDTM standards and your knowledge of clinical trial data, should this be flagged as a potential data issue? Why or why not? How would you verify or

+ Think x

Thought for 5 seconds >

According to CDISC SDTM conventions, an adverse-event record whose outcome is "RECOVERED/RESOLVED" implies that the event has an end date (and possibly a time), so a missing AEENDTC in conjunction with AEOU="RECOVERED/RESOLVED" is indeed a data inconsistency that should be flagged. Below is the step-by-step reasoning and a suggested approach for verification and correction:

1. Why this appears to be a data issue

1. SDTM Variable Definitions

Ask anything

permission of Kevin Lee

110

110

Reason

Output :

In SDTM AE domains, consistency between date/time variables and corresponding outcomes is key to accurately characterizing the event. Here's a detailed breakdown of the issue and the recommended steps:

Identifying a Potential Data Issue

•Expectation from Standards:

According to CDISC SDTM standards, if an adverse event is recorded as "RECOVERED/RESOLVED" (via AEOUT), it implies that the adverse event has ended. Therefore, there should typically be a corresponding AEENDTC (Adverse Event End Date/Time) value that marks when the event resolved.

•Flagging the Inconsistency:

When AEENDTC is missing despite AEOUT indicating that the event resolved, it can be flagged as a potential data quality issue because:

- It may signal incomplete or inaccurately captured event timelines.
- It could affect the analyses that depend on the duration of events, such as time-to-resolution or incidence rates.
- It may lead to challenges in regulatory review, where precise event timing is necessary for safety assessments.

Verification and Correction Process

1.Review Source Documentation:

- Case Report Forms (CRFs) or Medical Records:** Verify the original source documents to see if an end date was recorded at the time of data entry.
- Adverse Event Narratives:** Check if there is additional descriptive information or notes indicating when the event resolved, even if the structured field is missing.

2.Consult Data Management or Clinical Teams:

- Engage with the data management or clinical safety team to confirm if the absence of AEENDTC is a data entry oversight, a coding anomaly, or if there's an approved rationale (e.g., if the end date wasn't captured due to a protocol-specific condition).

3.Data Review and Cross-Checks:

- Cross-Reference with Other Variables:** Look at related variables such as AESTDTC (start date/time of the adverse event) and any flags or comments that might hint at the

timeline of the event.

- Consistency Rules:** Apply internal consistency checks used in your study's data review procedures to see if similar issues occur across other records.

4.Correcting Data:

- Updating Missing Dates:** If source documentation confirms an event resolution date, update AEENDTC accordingly.
- Documentation of the Change:** Ensure that any changes are well-documented in your data review logs, and that a query is raised (and resolved) through your study's quality control processes.
- Data Cleaning SOPs:** Follow your sponsor's or regulatory guidance for data correction in SDTM datasets, ensuring compliance with validation and audit trail requirements.

Reasoning Process Recap

•Standards Compliance:

The SDTM standards stipulate that an adverse event's timeline is essential for its characterization. Since AEOUT indicates the event resolved, logically, AEENDTC should not be missing.

•Impact on Analysis:

Missing resolution dates can affect safety analyses, signal detection, and regulatory reviews, warranting further investigation.

•Data Verification:

The process involves cross-referencing source documents, consulting with study teams, and applying consistency checks—all designed to ensure that the dataset accurately reflects the clinical scenario.

Conclusion

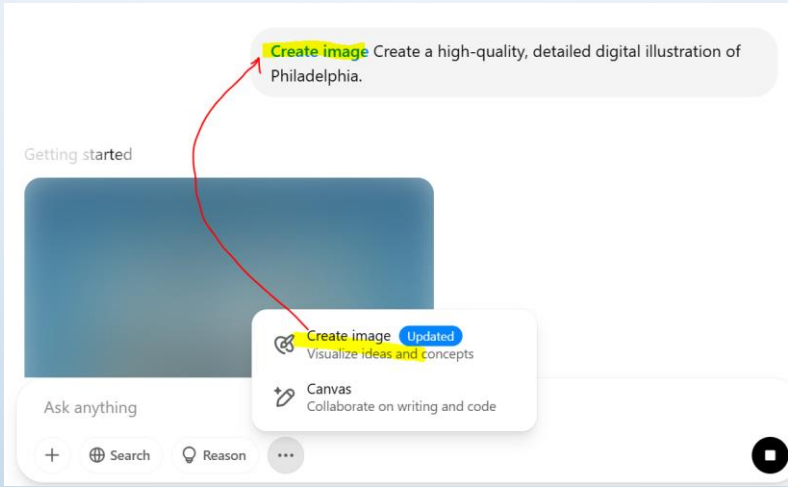
Yes, the situation should be flagged as a potential data issue because the absence of AEENDTC contradicts the indication that the event resolved. Verifying the source data and discussing the discrepancy with data management are necessary steps to either confirm an error or understand the rationale behind the missing date. Correcting it—if necessary—by updating the dataset based on authenticated source documentation will improve the dataset's integrity and ensure compliance with CDISC SDTM standards.

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Create Image



This will take some time to create a image



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Create Image



convert this picture into a Studio Ghibli-style animation

This is a famous Ghibli-style animation prompt.

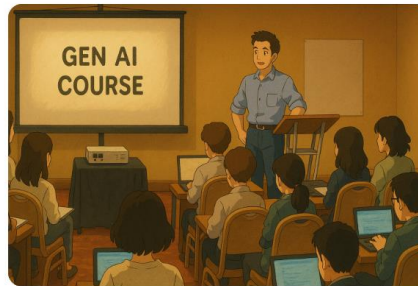
Image created



🔄 📄 📌

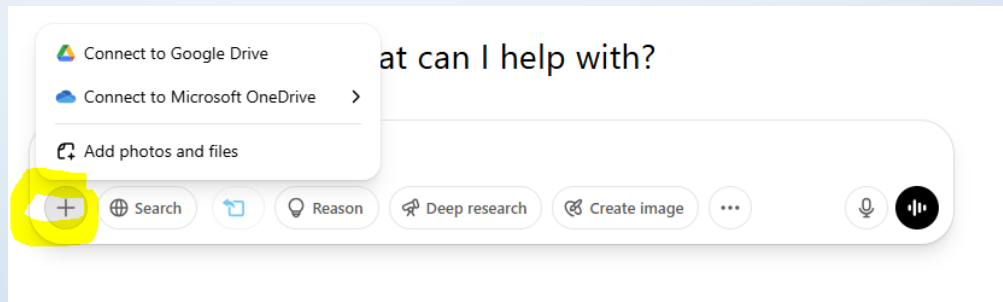
In the writing board, please add Gen AI course and create another Ghibli-style animation.

Image created



113

Upload the document



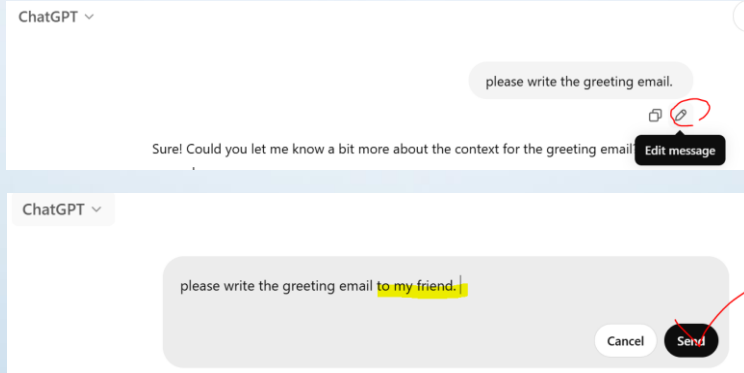
- Click + sign, or paste over prompt panel.

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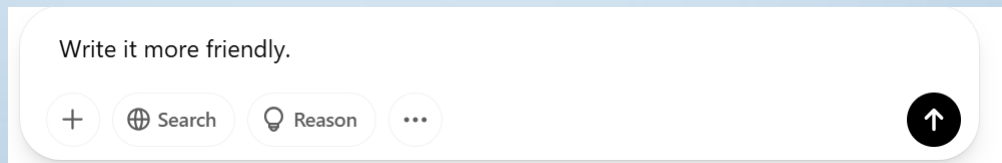
114

114

Modify and resubmit the inquiry



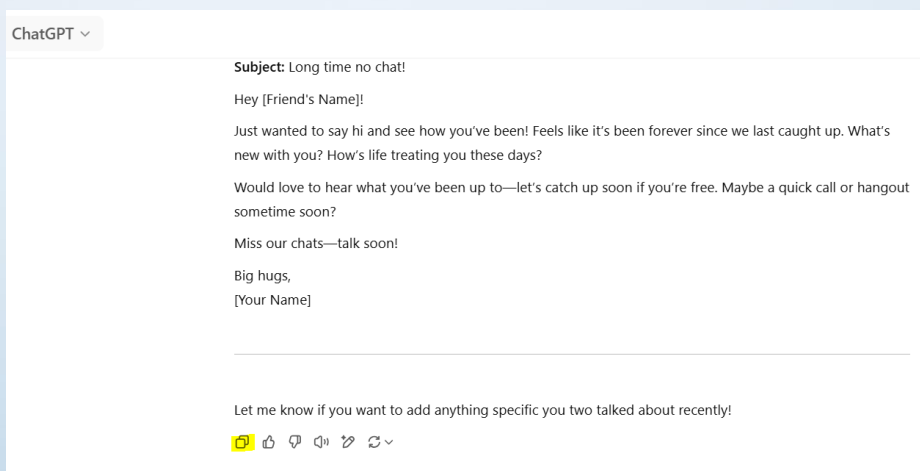
- We can also re-prompt to make it better.



115

115

Copy the answer



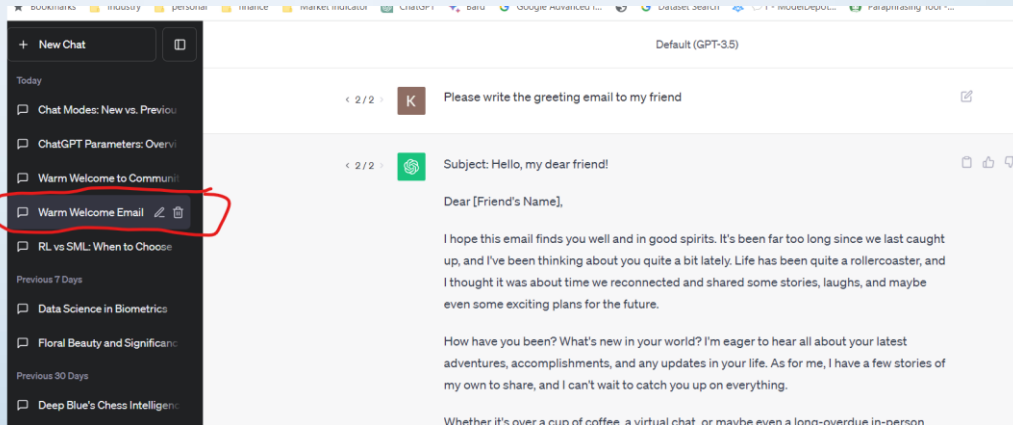
- Click "Copy" icon.
- Paste the saved texts in other application.

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116

Continue the previous Chat



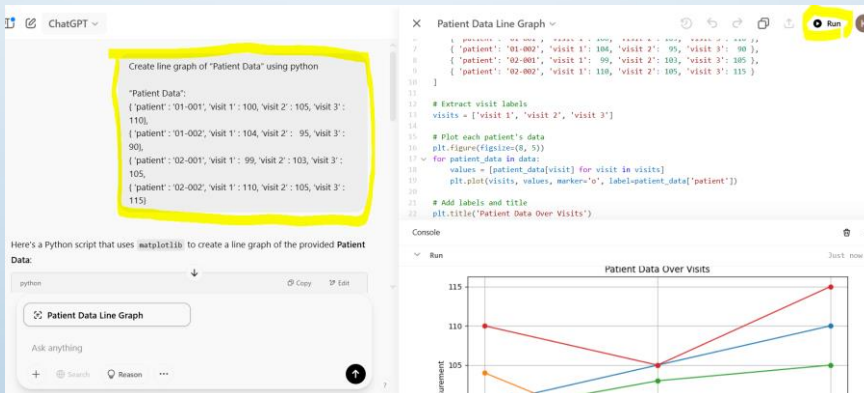
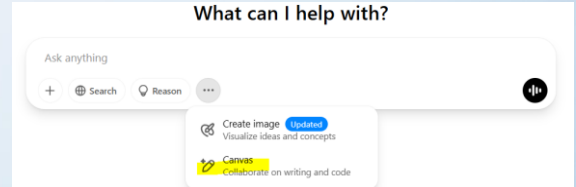
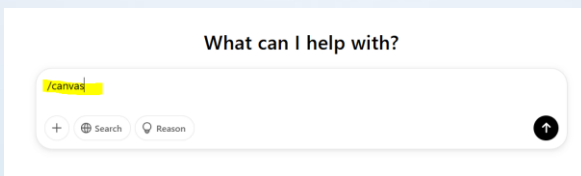
- Click previous chat.
- In summary, "new chat" starts a conversation from scratch, while "previous chat" continues an ongoing conversation within the same session, leveraging the context and information established in earlier messages.

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117

Using OpenAI Canvas (writing and coding)



- Start Canvas.
- prompt for codes
- Run the code, and you will see the results in Console.
- Note: There is a limit to use Canvas in free version (few times a day)

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118

Using OpenAI Canvas (prototyping in html)

Canvas

Create line graph of "Patient Data" in html.

```

"Patient Data":
[
  { 'patient': '01-001', 'visit 1': 100, 'visit 2': 105, 'visit 3': 110 },
  { 'patient': '01-002', 'visit 1': 104, 'visit 2': 95, 'visit 3': 90 },
  { 'patient': '02-001', 'visit 1': 99, 'visit 2': 103, 'visit 3': 105 },
  { 'patient': '02-002', 'visit 1': 110, 'visit 2': 105, 'visit 3': 115 }
]

```

Answer in chat instead >

Your HTML file with a line graph of the patient data is ready. It's using Chart.js to plot the measurements over three visits. Let me know if you'd like to add interactivity, export features, or style changes!

Ask anything

+ Search Reason ...

Patient Data Linegraph

```

1 <!DOCTYPE html>
2 <html>
3 <head>
4   <meta charset="UTF-8">
5   <title>Patient Data Line Graph</title>
6   <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
7 </head>
8 <body>
9   <h2>Patient Data Over Visits</h2>
10  <canvas id="patientChart" width="800" height="400"></canvas>
11  <script>
12    const ctx = document.getElementById('patientChart').getContext('2d');
13
14    const data = {
15      labels: ['Visit 1', 'Visit 2', 'Visit 3'],
16      datasets: [
17        {
18          label: '01-001',
19          data: [100, 105, 110],
20          borderColor: 'blue',
21          fill: false
22        },
23        {
24          label: '01-002',
25          data: [104, 95, 90],
26          borderColor: 'red',
27          fill: false
28        },
29        {
30          label: '02-001',
31          data: [99, 103, 105],
32          borderColor: 'green',
33          fill: false
34        },
35        {
36          label: '02-002',
37          data: [110, 105, 115],
38          borderColor: 'purple',
39          fill: false
40        }
41      ]
42    };
43
44    new Chart(ctx, {
45      type: 'line',
46      data,
47      options: {
48        responsive: true,
49        plugins: {
50          legend: {
51            position: 'top',
52          },
53        },
54      },
55    });
56  </script>
57 </body>
58 </html>

```

Patient Data Linegraph

Patient Data Over Visits

| Patient | Visit 1 | Visit 2 | Visit 3 |
|---------|---------|---------|---------|
| 01-001 | 100 | 105 | 110 |
| 01-002 | 104 | 95 | 90 |
| 02-001 | 99 | 103 | 105 |
| 02-002 | 110 | 105 | 115 |

Note : The limited trials in free version (a couple of times per a day

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119

Deep Research

What are you researching?

Build AI Strategy for Biostatistical department in Pharma Industry

+ Research

To tailor the AI strategy effectively, could you please clarify a few points:

Copy

1. What is the size and structure of the biostatistics department (e.g., global vs. local, number of team members)?
2. What are the primary responsibilities of the department (e.g., clinical trial data analysis, regulatory submissions, RWE)?
3. What types of data are typically handled (e.g., clinical trial data, omics data, EHRs)?
4. Are there existing AI initiatives or tools in use?
5. What are the key goals or challenges you want the AI strategy to address (e.g., automation, predictive modeling, regulatory compliance)?

This will help in designing a strategy that aligns with your operational needs and long-term vision.

📄 📄

Please use below answer.

1. 500 stat programmers, biostatisticians, data managers
2. clinical trial data analysis, submission, efficacy and safety analysis, RWE
3. clinical trial data
4. Not yet
5. automation, faster time of drug development, regulatory compliance

uplicate or distribute

Thanks for the detailed context. I'll develop an AI strategy tailored for a biostatistics department in the pharma industry with 500 professionals, focusing on clinical trial data analysis, regulatory submissions, and RWE, aiming to improve automation, speed up drug development, and maintain compliance.

I'll structure this strategy around key pillars such as data infrastructure, model deployment, regulatory considerations, and team upskilling. I'll get back to you shortly with a comprehensive strategy document.

Research completed in 6m · 34 sources · 79 searches

AI Strategy for Biostatistics in Pharma

AI Strategy for Biostatistics Department in Pharma

Executive Summary: This strategy outlines how the biostatistics department can embed AI across the clinical trial lifecycle to automate manual tasks, speed development, and maintain compliance. Key elements include deploying AI use cases (e.g. automated SDTM/ADaM mapping, anomaly detection, trial simulation), building a secure, GxP-ready AI infrastructure, upskilling staff and recruiting data scientists, and instituting a governance framework aligned with FDA/EMA guidelines. Short-term pilots will demonstrate quick wins in data quality and reporting; medium-term goals focus on expanding to design optimization and RWE analytics; and long-term goals aim for full AI integration in data analysis and regulatory submissions. The proposed roadmap and table below summarize prioritized use cases, impact, and feasibility.

AI Use Cases Across Clinical Trial Lifecycle

- **SDTM/ADaM Automation:** Use machine learning to auto-map clinical data into CDISC standards. AI + human-in-the-loop approaches can **clean, map and validate** source data to SDTM, dramatically reducing manual effort and error. [AbbVie's pilots](#) show such AI frees scientists for interpretation and cuts SDTM conversion timelines/costs. [Applied Biosystems](#) and [Applied Biosystems](#) are extending to ADaM (analysis datasets) can further streamline analysis-ready data generation.
- **Data Quality & Anomaly Detection:** AI can continuously scan incoming trial data to flag outliers, inconsistencies, or protocol deviations. Algorithms trained on historical data detect subtle anomalies (e.g. implausible lab values or site irregularities) that human review might miss. [AstraZeneca](#), [AstraZeneca](#), This real-time monitoring improves data integrity, reduces rework, and shortens queries, leading to **faster, more reliable results**. [AstraZeneca](#), [AstraZeneca](#).
- **Protocol Design & Simulation:** Machine learning and simulation models can optimize trial parameters before launch. For example, discrete-event simulation of enrollment and dropout scenarios helps refine inclusion criteria and sample size. [Oxford Biomedica](#) and [Oxford Biomedica](#) reinforcement learning can adapt designs mid-trial to balance efficacy and risk. [Oxford Biomedica](#) An FDA-[Oxford Biomedica](#) endorsed Alzheimer's trial simulator showed AI-driven

120

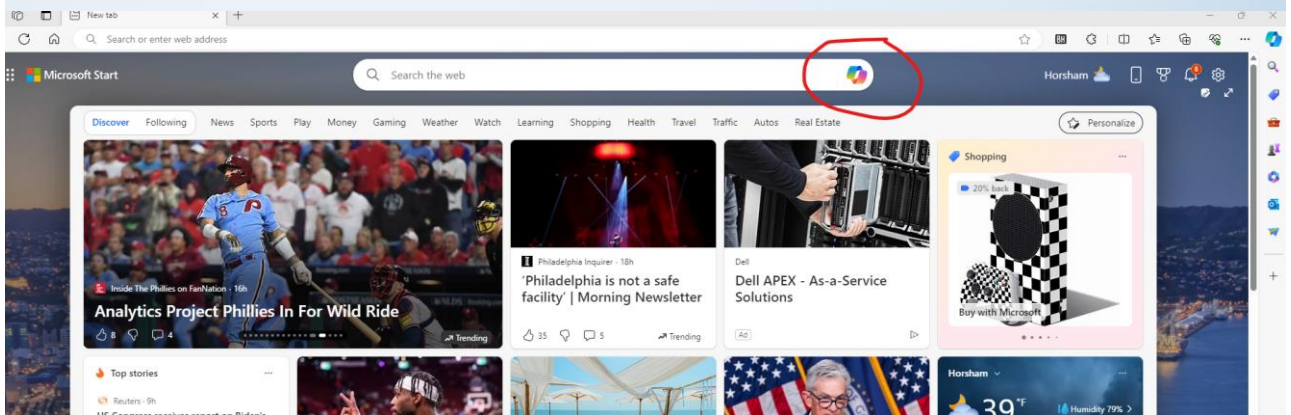
MS Bing Copilot Functionality

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121

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Start Bing Copilot (free version – Copilot with GPT-4)



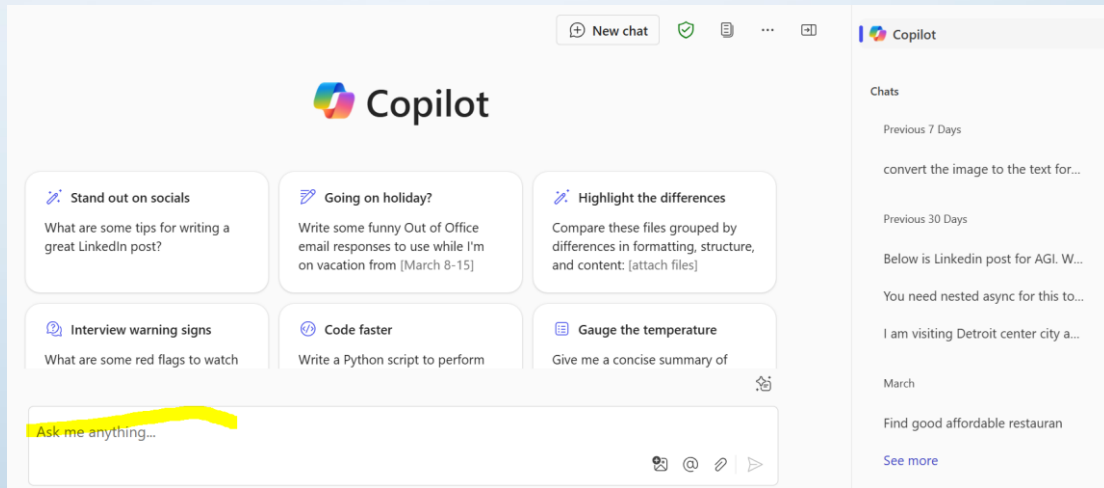
- Start "Edge" and Click "Open Copilot"
- <https://copilot.microsoft.com>

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122

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Start new chat



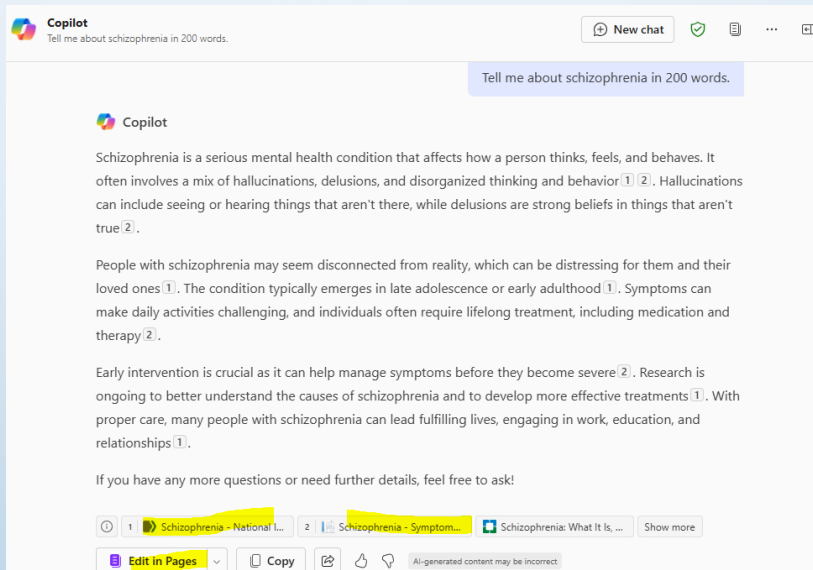
- Start your prompt in Copilot

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123

123

Results from prompt



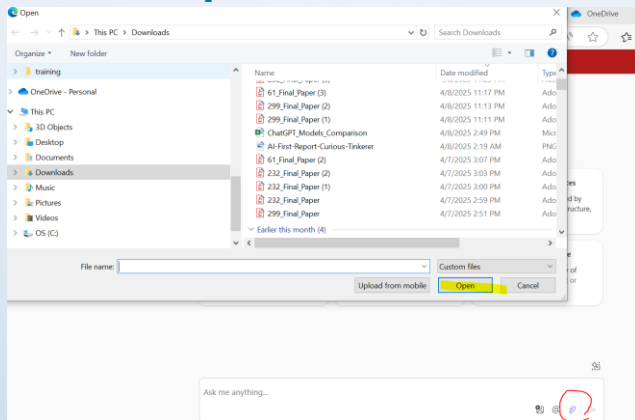
- Answers from the prompt
- Links for the prompt

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124

124

Upload the files for the analysis



Kyo S Lee

Home

My files

Shared

Favorites

Recycle bin

My files > Microsoft Copilot Chat Files

| | Name | Modified | Modified By | File size | Sharing |
|--|--------------------------------|--------------------|-------------|-----------|---------|
| | ADaMIG v1.2-Final.pdf | About a minute ago | Kyo S Lee | 1.51 MB | Private |
| | ChatGPT_Models_Comparison.xlsx | 23 minutes ago | Kyo S Lee | 6.94 KB | Private |

125

Results from prompt

Copilot
Tell me about schizophrenia in 200 words.

Tell me about schizophrenia in 200 words.

Copilot

Schizophrenia is a serious mental health condition that affects how a person thinks, feels, and behaves. It often involves a mix of hallucinations, delusions, and disorganized thinking and behavior ^[1] ^[2]. Hallucinations can include seeing or hearing things that aren't there, while delusions are strong beliefs in things that aren't true ^[2].

People with schizophrenia may seem disconnected from reality, which can be distressing for them and their loved ones ^[1]. The condition typically emerges in late adolescence or early adulthood ^[1]. Symptoms can make daily activities challenging, and individuals often require lifelong treatment, including medication and therapy ^[2].

Early intervention is crucial as it can help manage symptoms before they become severe ^[2]. Research is ongoing to better understand the causes of schizophrenia and to develop more effective treatments ^[1]. With proper care, many people with schizophrenia can lead fulfilling lives, engaging in work, education, and relationships ^[1].

If you have any more questions or need further details, feel free to ask!

1 Schizophrenia - National L... 2 Schizophrenia - Symptom... Schizophrenia: What It Is, ... Show more

Edit in Pages Copy Share Like Comment AI-generated content may be incorrect

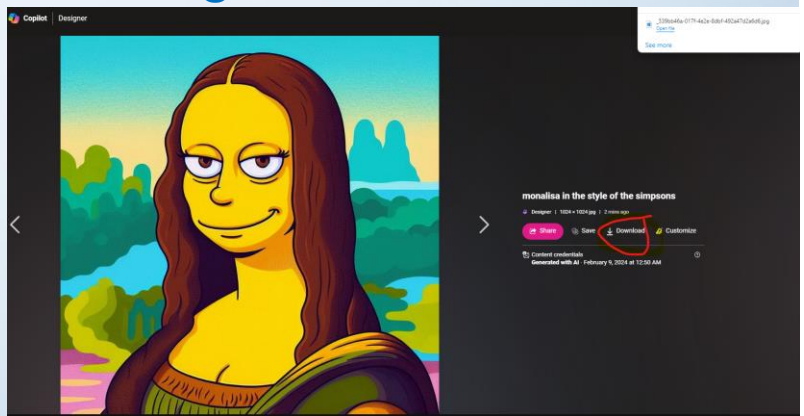
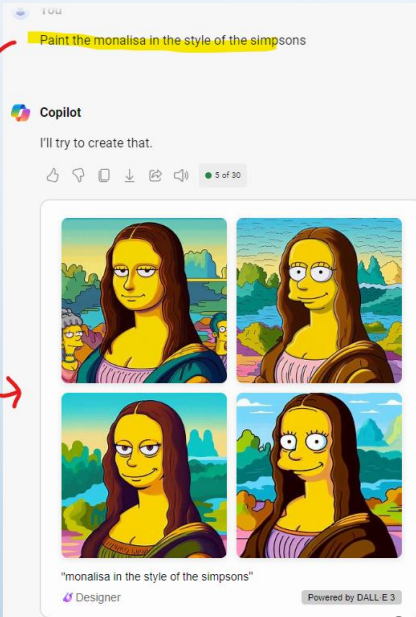
- Answers from the prompt
- Links for the prompt

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Create the image



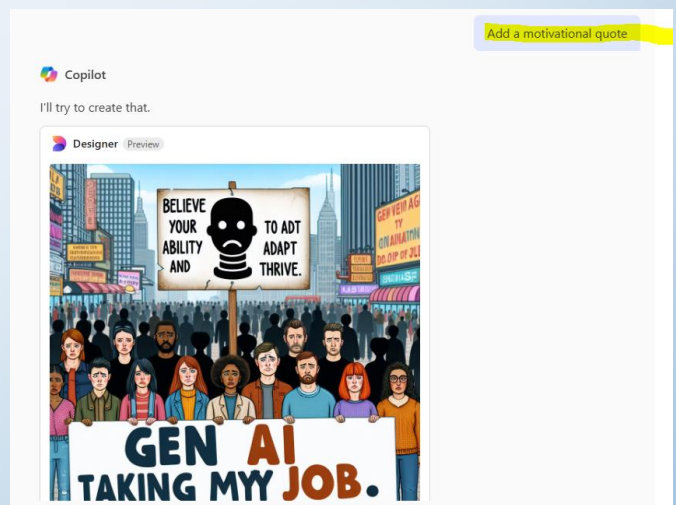
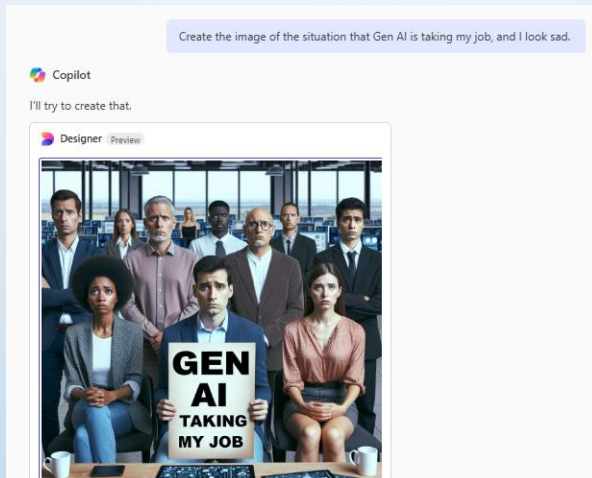
- Create the image with DALL-E 3

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127

127

Create the image (2)



- Create the image with DALL-E 3

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Summarize the article in link

You

Summarize the article in <https://www.fda.gov/news-events/fda-voices/fda-releases-two-discussion-papers-spur-conversation-about-artificial-intelligence-and-machine> with two paragraphs.

Copilot

The U.S. Food and Drug Administration (FDA) has recently released two discussion papers to foster dialogue about the integration of **artificial intelligence (AI)** and **machine learning (ML)** in drug development and manufacturing. These papers aim to engage stakeholders, including pharmaceutical companies, ethicists, academia, patients, and global regulatory authorities, in a thoughtful exploration of how AI/ML can revolutionize the field.

The first paper, titled "Using Artificial Intelligence and Machine Learning in the Development of Drug and Biological Products," highlights the potential of AI/ML to accelerate the delivery of safe and effective treatments. For instance, AI/ML could analyze medical literature to identify relevant findings, predict individual responses to therapies, and even facilitate patient engagement through conversational agents. However, challenges such as ethical considerations, data security, and algorithmic transparency must be addressed to harness AI/ML's full potential. The second paper, "Artificial Intelligence in Drug Manufacturing," delves into how AI/ML can enhance drug production processes. By fostering informed discussions, the FDA aims to promote equitable and responsible adoption of these transformative technologies ¹ ².



Learn more [1 fda.gov](#) [2 bing.com](#) [3 fda.gov](#) [4 biospace.com](#) [5 prnewswire.com](#) [6 gettyimages.com](#)

1 of 30

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129

Convert the image to the text

| DATE | TIME | ID# | NOTES SHOULD BE DATED, TIMED, AND SIGNED BY PHYSICIAN |
|----------|------|-----|---|
| 12/21/18 | 435P | | Critical Care Medicine |
| | | | Pt is doing well this am |
| | | | VSS |
| | | | Gen: MAO |
| | | | Chest: CTG |
| | | | W: rxn 5'2 91c |
| | | | Abd: soft, NT, ND +BS |
| | | | Eyes: +edema |
| | | | Labs reviewed |
| | | | Imp: |
| | | | 1. Acute Rep Failure |
| | | | 2. Septic Shock |
| | | | 3. AKI |
| | | | 4. Met Encephalopathy |
| | | | - Cont ABX, pressors, vent |
| | | | - Follow renal fxn |
| | | | - Check labs |

PHY-0008 rev1/2016
Form # 20041 / # 136

PLEASE TURN OVER

You

convert the image to text

| DATE | TIME | ID# | NOTES SHOULD BE DATED, TIMED, AND SIGNED BY PHYSICIAN |
|----------|------|-----|---|
| 12/21/18 | 435P | | Critical Care Medicine |
| | | | Pt is doing well this am |
| | | | VSS |
| | | | Gen: MAO |
| | | | Chest: CTG |
| | | | W: rxn 5'2 91c |
| | | | Abd: soft, NT, ND +BS |
| | | | Eyes: +edema |
| | | | Labs reviewed |
| | | | Imp: |
| | | | 1. Acute Rep Failure |
| | | | 2. Septic Shock |
| | | | 3. AKI |
| | | | 4. Met Encephalopathy |
| | | | - Cont ABX, pressors, vent |
| | | | - Follow renal fxn |
| | | | - Check labs |

Copilot

Certainly! Here's the text extracted from the image:

12/21/18
435P
Critical Care Medicine
Pt is doing well this am
VSS
Gen: MAO
Chest: LTG
W: cm 5'2 91c
Abd: soft, NT, ND +BS
Eyes: +edema
Labs reviewed
Imp:
1. Acute Rep Failure
2. Septic Shock
3. AKI
4. Met Encephalopathy
Cont ABX per resp/vent
Follow renal fxn
Check labs

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130

Create the spreadsheet and export it to excel

convert this in table format.

| ORDER | DOMAIN | OBSERVATION CLASS | VARIABLE NAME | VARIABLE LABEL | DATA TYPE | LENGTH | ROLE |
|-------|--------|-------------------|---------------|-----------------------------------|-----------|--------|------------------|
| 1 | DM | Special-Purpose | STUDYID | Study Identifier | Text | 21 | Identifier |
| 2 | DM | Special-Purpose | DOMAIN | Domain Abbreviation | Text | 8 | Identifier |
| 3 | DM | Special-Purpose | USUBJID | Unique Subject Identifier | Text | 30 | Identifier |
| 4 | DM | Special-Purpose | SUBJID | Subject Identifier for the Study | Text | 8 | Topic |
| 5 | DM | Special-Purpose | RFSTDTC | Subject Reference Start Date/Time | Datetime | 19 | Record Qualifier |
| 6 | DM | Special-Purpose | RFENDTC | Subject Reference End Date/Time | Datetime | 19 | Record Qualifier |
| 7 | DM | Special-Purpose | RFSTDTC | Subject Reference Start Date/Time | Datetime | 19 | Record Qualifier |
| 8 | DM | Special-Purpose | RFENDTC | Subject Reference End Date/Time | Datetime | 19 | Record Qualifier |
| 9 | DM | Special-Purpose | RFSTDTC | Subject Reference Start Date/Time | Datetime | 19 | Record Qualifier |
| 10 | DM | Special-Purpose | RFENDTC | Subject Reference End Date/Time | Datetime | 19 | Record Qualifier |
| 11 | DM | Special-Purpose | SITEID | Study Site Identifier | Text | 8 | Topic |

Is there anything else you need help with?

Sure! Here's the information from the image converted into a table format:

| ORDER | DOMAIN | OBSERVATION CLASS | VARIABLE NAME | VARIABLE LABEL | DATA TYPE | LENGTH | ROLE |
|-------|--------|-------------------|---------------|-----------------------------------|-----------|--------|------------------|
| 1 | DM | Special-Purpose | STUDYID | Study Identifier | Text | 21 | Identifier |
| 2 | DM | Special-Purpose | DOMAIN | Domain Abbreviation | Text | 8 | Identifier |
| 3 | DM | Special-Purpose | USUBJID | Unique Subject Identifier | Text | 30 | Identifier |
| 4 | DM | Special-Purpose | SUBJID | Subject Identifier for the Study | Text | 8 | Topic |
| 5 | DM | Special-Purpose | RFSTDTC | Subject Reference Start Date/Time | Datetime | 19 | Record Qualifier |
| 6 | DM | Special-Purpose | RFENDTC | Subject Reference End Date/Time | Datetime | 19 | Record Qualifier |
| 7 | DM | Special-Purpose | RFSTDTC | Subject Reference Start Date/Time | Datetime | 19 | Record Qualifier |
| 8 | DM | Special-Purpose | RFENDTC | Subject Reference End Date/Time | Datetime | 19 | Record Qualifier |
| 9 | DM | Special-Purpose | RFSTDTC | Subject Reference Start Date/Time | Datetime | 19 | Record Qualifier |
| 10 | DM | Special-Purpose | RFENDTC | Subject Reference End Date/Time | Datetime | 19 | Record Qualifier |
| 11 | DM | Special-Purpose | SITEID | Study Site Identifier | Text | 8 | Topic |

Export to Excel

Copy page link to here

Move block

Delete

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131

Use the editor

The table is metadata for CDISC DM. I want to add the sample data. Can you create the sample data?

Copilot

Got it! Here's a sample dataset based on the CDISC DM metadata:

| STUDYID | DOMAIN | USUBJID | SUBJID |
|----------|--------|----------|---------|
| STUDY001 | DM | USUBJ001 | SUBJ001 |
| STUDY002 | DM | USUBJ002 | SUBJ002 |
| STUDY003 | DM | USUBJ003 | SUBJ003 |

Feel free to adjust the sample data as needed. Is there anything else you need help with?

+ Add to page

Copy

AI-generated content may be incorrect.

Can you explain the significance of each field?

Add more rows to this sample data.

ask me anything...

convert this in table format

Just start typing...

Got it! Here's a sample dataset based on the CDISC DM metadata:

| STUDYID | DOMAIN | USUBJID | SUBJID |
|------------|--------|----------|---------|
| 1 STUDY001 | DM | USUBJ001 | SUBJ001 |
| 2 STUDY002 | DM | USUBJ002 | SUBJ002 |
| 3 STUDY003 | DM | USUBJ003 | SUBJ003 |

+ New

Feel free to adjust the sample data as needed. Is there anything else you need help with?

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132

Prompt for search results

news-events/press-announcements/fda-releases-artificial-intelligence-machine-learning-action-plan

here's how you know...

Press Announcements / FDA Releases Artificial Intelligence/Machine Learning Action Plan

FDA NEWS RELEASE

FDA Releases Artificial Intelligence/Machine Learning Action Plan

Share Post LinkedIn Email Print

For Immediate Release: January 12, 2021

Today, the U.S. Food and Drug Administration released the agency's first [Artificial Intelligence/Machine Learning \(AI/ML\)-Based Software as a Medical Device \(SaMD\) Action Plan](#). This action plan describes a multi-pronged approach to advance the Agency's oversight of AI/ML-based medical software.

"This action plan outlines the FDA's next steps towards furthering oversight for AI/ML-based SaMD," said Bakul Patel, director of the Digital Health Center of Excellence in the Center for Devices and Radiological Health (CDRH). "The plan outlines a holistic approach based on total product lifecycle oversight to further the enormous potential that these technologies have to improve patient care while delivering safe and effective software functionality that improves the quality of care that patients receive. To stay current and address patient safety and improve access to these promising technologies, we anticipate that this action plan will continue to evolve over time."

Content current as of: 01/12/2021

Regulated Product(s)
Medical Devices

Follow FDA
Follow @US_FDA
Follow @FDA
Follow @FDA Media

Copilot

Here is a summary of the web page:

- FDA's AI/ML Action Plan:** The web page announces the release of the FDA's first Artificial Intelligence/Machine Learning (AI/ML)-Based Software as a Medical Device (SaMD) Action Plan, which describes the agency's approach to advance the oversight of AI/ML-based medical software.
- Five Actions:** The action plan outlines five actions that the FDA intends to take, such as further developing the proposed regulatory framework, supporting the development of good machine learning practices, fostering a patient-centered approach, developing methods to evaluate and improve machine learning algorithms, and enhancing regulatory science methods for AI/ML.
- Stakeholder Feedback:** The action plan is a response to stakeholder feedback received from the April 2019 discussion paper, Proposed Regulatory Framework for Modifications to Artificial Intelligence/Machine Learning-Based Software as a Medical Device.
- Digital Health Center of Excellence:** The web page also introduces the CDRH Digital Health Center of Excellence, which is committed to advancing science and

Ask me anything...

- Your search in Bing Search
- Prompt (e.g. summary) for your search results
- In Copilot

133

Upload document in Edge and prompt in Copilot

File | D:\zother\industry\conference\CDISC\ADaM\ADaMIG_v1.3_Release_Package\ADaM_v2.1.pdf

Ask Copilot

CDISC Analysis Data Model Version 2.1

CDISC

Analysis Data Model (ADaM)

Prepared by the
CDISC Analysis Data Model Team

Notes to Readers

This is Version 2.1 of the Analysis Data Model (ADaM) Document. It includes modifications so that it corresponds to Version 1.0 of the Analysis Data Model Implementation Guide (ADaMIG).

Revision History

| Date | Version | Summary of Changes |
|---------------|-----------|--|
| Dec. 17, 2009 | 2.1 Final | Released version reflecting all changes and corrections identified during comment period. |
| May 30, 2008 | 2.1 Draft | Draft for comment. |
| Aug. 11, 2006 | 2.0 Final | Final document |
| 31 May 2006 | 2.0 Draft | Incorporate comments from public review |
| 15 Feb 2006 | 2.0 Draft | Reformatted from General Considerations v1.0, incorporating Subject level model, emphasizing requirements and naming and content rules and guidelines. |

Copilot

Unsolicited Conversation

You

Based on this page, what are main ADaM data set models?

Copilot

This conversation may have info that's not public.
Conversation will not be saved after this point.
Based on this page, the main ADaM data set models are:

- Subject-Level Analysis Dataset (ADSL):** This is a dataset that has one record per subject and contains variables such as subject-level population flags, treatment variables, demographic information, and important dates (1). ADSL is required in a CDISC-based submission of data from a clinical trial (2) (3).
- Basic Data Structure (BDS):** This is a dataset that has one or more records per subject, per analysis parameter, per analysis timepoint (4). This structure contains variables that describe the analysis parameter, the value being analyzed, and other information that support the analysis (5). The BDS supports various types of statistical analyses, such as ANOVA, ANCOVA, logistic regression, time-to-event analysis, etc.

References:

1: Page 11 - Line 24 2: Page 3 - Line 14
3: Page 11 - Line 26 4: Page 11 - Line 13

Ask me anything...

- Upload/ Open document (e.g pdf) in Edge
- Prompt and get the answers from the document

134

Find source codes in the website

The screenshot shows a web browser at <https://clinvia.com>. The website has a blue header with the text "WELCOME TO CLINVIA" and "Our services lead you to success". Below this, it says "Clinvia is a data science and technology company that plans to reduce costs and speed up the clinical trials by bringing faster insights and real time analytics." and a "Contact Us" button. A Copilot chat window is open on the right, showing the HTML source code for the page. The code includes the DOCTYPE, language, charset, viewport, title, and a link to styles.css.

```

1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>Clinvia</title>
7   <link rel="stylesheet" href="styles.css">
8 </head>

```

Below the code, there are prompts: "Can you add CSS for styling?", "How to make it responsive?", and "What JavaScript features can I include?". There is also a "View prompts" button and an input field "Ask me anything...".

- We could find html source codes for the web-page.

135

135

Find source codes in the website

The screenshot shows a web browser displaying the "R Package Validation" page. The page has a white background with a blue header. A Copilot chat window is open on the right, showing the HTML and CSS source code for the page. The code includes the DOCTYPE, language, charset, viewport, title, and a link to styles.css. Below the code, there are prompts: "Can you add CSS for styling?", "How to make it responsive?", and "What JavaScript features can I include?". There is also a "View prompts" button and an input field "Ask me anything...".

```

1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>R Package Validation</title>
7   <style>
8     body {
9       font-family: Arial, sans-serif;
10      margin: 20px;
11    }
12    h1, h2 {
13      color: #333;
14    }
15    table {

```

Below the code, there is a "Show more lines" button. At the bottom of the chat window, there is a "Feel free to customize the content and style as needed. If you have any specific requirements or need further assistance, let me know!" message and a row of buttons: "Edit in Pages", "Copy", "Share", "Like", and "Dislike". A small disclaimer "AI-generated content may be incorrect." is also present.

- We could find html source codes for the web-page.

136

136

Google Gemini Functionality

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137

137

Google Gemini (Bard)

The screenshot displays the Google Gemini (Bard) web application. On the left, a sidebar lists various models: Gemini 2.0 Flash (Fast all-around help), Gemini 2.5 Flash (preview) (Our next reasoning model built for speed), Gemini 2.5 Pro (preview) (Reasoning, math & code), Personalization (preview) (Based on your Search history), and Gemini Advanced. The main chat area shows a greeting 'Hello Kevin' and a 'Generate Audio Overview' button. On the right, a 'Files' panel is open, showing a list of files. A red checkmark is next to a PDF file named 'ADaMIG_v1.3'. A red circle highlights the file management icon in the top right corner of the interface.

- <https://gemini.google.com/app>
- You could sign in with current google account.

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138

138

Strengths in Gemini

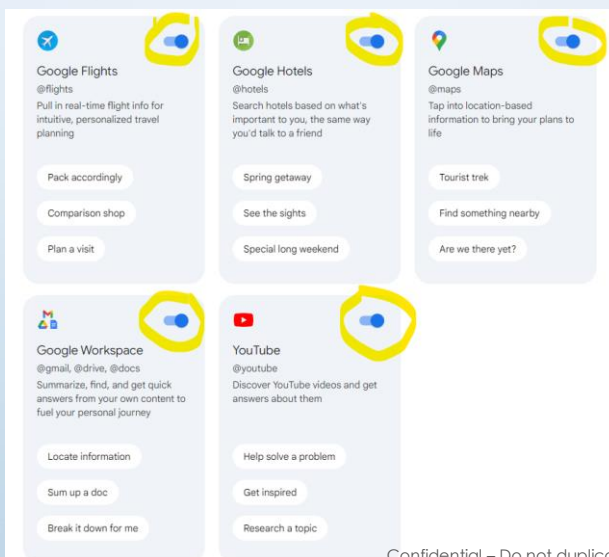
1. Can connect to other Google tools.
2. Faster than other Gen AI
3. Usually free

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139

139

Add extensions in gemini.google.com/extensions



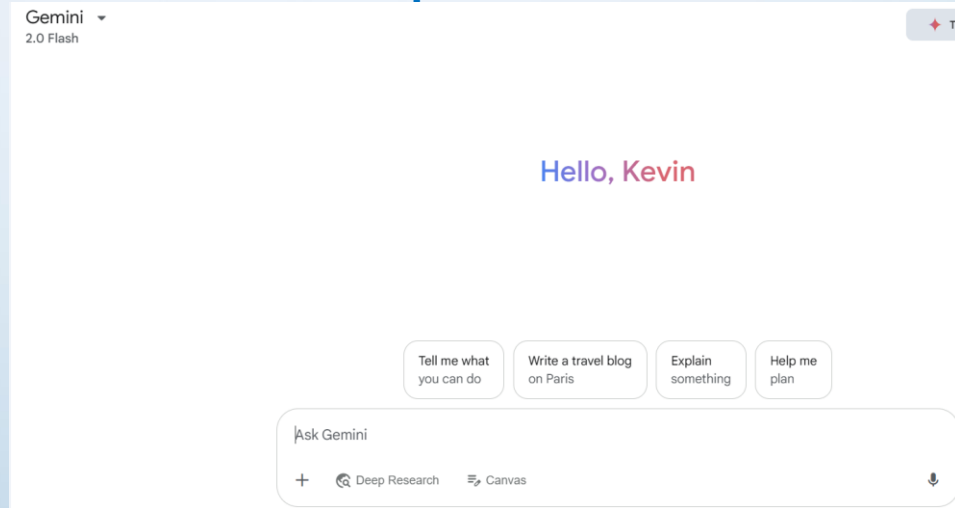
- Pick five extensions to add apps functionality.

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140

140

Prompt – 2.0 Flash



- You can search sample prompts or simply ask Gemini.
- Click the sample, and it will give you prompt example

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141

141

Prompts – 2.5 Flash (Reasoning)

Gemini
2.5 Flash (preview)

Try Gemini Advanced

I am statistical programmer. I want to know if I need to learn open-source programming or stay in SAS programming. Provide your reasoning in 100 words.

Show thinking

While SAS remains strong in regulated fields like pharma, the industry is rapidly adopting open-source languages like R and Python. These offer greater flexibility, cutting-edge tools, and are increasingly required in job postings across various sectors. Learning open-source expands your skillset, making you more adaptable and competitive in an evolving statistical programming landscape. While maintaining SAS skills is valuable, adding open-source proficiency is crucial for broader career prospects and accessing modern analytical techniques.

Show thinking

I will start by performing a Google search to understand the current landscape and trends in statistical programming, specifically comparing the use of SAS and open-source languages like R and Python in the industry and also search for information on the job market and employer preferences for statistical programmers, looking for mentions of required skills in SAS versus open-source technologies.

Finally, I will synthesize the gathered information on industry trends and job market demands to formulate a concise, less than 100-word answer advising the user on whether to learn open-source programming or stay in SAS programming, based on career prospects and industry direction.

I have gathered information on the trends in statistical programming, comparing SAS with open-source languages like R and Python, and examined the current job market demands. The search results indicate that while SAS remains prevalent, particularly in regulated industries like pharmaceuticals, there is a significant and growing demand for skills in R and Python across various sectors. Many job postings now require proficiency in multiple languages, and open-source tools are increasingly used for advanced analytics, machine learning, and data integration. The future suggests a hybrid approach where knowing open-source languages alongside SAS will be highly beneficial, offering greater flexibility and career opportunities.

Therefore, I have sufficient information to answer the user's question within the requested word count. I will synthesize these findings to explain why learning open-source programming is advisable while acknowledging the continued relevance of SAS.

We can get the answer.

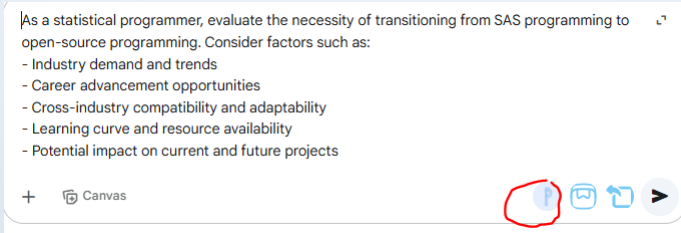
We can also get the reasoning behind.

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142

142

Prompt – enhancing prompt



- We can edit plan or start research
- “Deep Search” take some time – 2 to 3 minutes.
- Prompt:

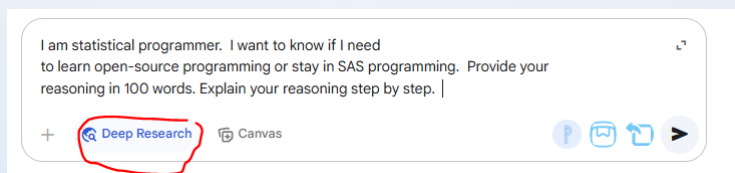
I am statistical programmer. I want to know if I need to learn open-source programming or stay in SAS programming. Provide your reasoning in 100 words. Explain your reasoning step by step.

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143

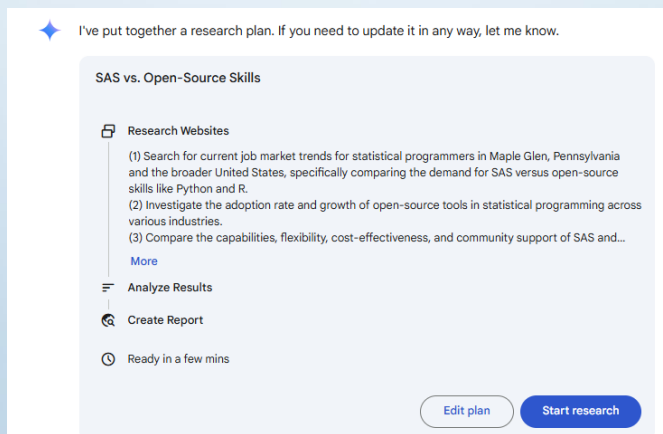
143

Prompt – Deep Research in “2.0 Flash”



- We can edit plan or start research
- “Deep Search” take some time – 2 to 3 minutes.
- Prompt:

I am statistical programmer. I want to know if I need to learn open-source programming or stay in SAS programming. Provide your reasoning in 100 words. Explain your reasoning step by step.



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144

144

Prompt – Deep Research (thinking)

Gemini Deep Research

Start research

I'm on it. I'll let you know when your research is done. In the meantime, you can leave this chat.

SAS vs. Open-Source Skills
Researching 104 websites...

Ask Gemini
+ Deep Research

SAS vs. Open-Source Skills | Show thinking

Initial Research Focus
I'm starting my research by looking into the current job market to see how strong the demand is for both SAS and open-source programming skills in the field of statistical programming. This will help me understand the immediate relevance of learning new tools.

Researching websites

| | | | |
|-------------------------------|--------------------------------|---------------------------------|------------------------------|
| handsonsystem.com Clini... | ziprecruiter.com \$34-\$76/... | mindmajix.com Python vs ... | ziprecruiter.com The 9 Hi... |
| sas.com SAS 2025 predict... | m.economicpoint.com Pyt... | datasciencecentral.com P... | mobility.com Hire SAS ... |
| index.dev Programming L... | communities.sas.com SA... | optiontrain.com Demand ... | r4stats.com The Popularit... |
| reddit.com [C] Does the c... | edvancer.in R, Python or S... | reddit.com Is it worth it to... | bacancytechnology.com P... |
| communities.sas.com Wh... | scholar.smu.edu Data Sci... | bomberbot.com R vs Pyth... | reddit.com [D] How popul... |
| reddit.com Should I learn ... | syntelli.com Python & R v... | datacamp.com Top 12 Pro... | reddit.com SAS vs Python... |
| coursera.org 7 In-Demand... | coursera.org Python or R f... | youtube.com Comparing ... | reddit.com What are some... |
| stxnext.com R vs. Python: ... | | | |

Job Market Insights
I've found that SAS is still a key player, especially in big organizations like those in finance and pharmaceuticals, often due to existing systems and regulations. There are definitely jobs out there for SAS programmers, and they can pay well. However, it seems fewer new professionals are learning SAS the other hand, Python is becoming increasingly popular and is in high demand for data science, machine learning, and even web development. It's also considered easier to learn, which is a plus.

Open Source Growth

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145

Prompt – Deep Research (final output)

Gemini Deep Research

Start research

I've completed your research. Feel free to ask me follow-up questions or request changes.

SAS vs. Open-Source Skills
Apr 14, 12:32 PM

Ask Gemini
+ Deep Research

SAS vs. Open-Source Skills | Export to Docs

Navigating the Landscape of Statistical Programming: SAS and the Rise of Open-Source Alternatives

The field of statistical programming is dynamic, with continuous shifts in technology and industry demands. For professionals skilled in SAS, a long-established and powerful tool, the question of whether to embrace open-source programming languages like Python and R is increasingly pertinent. This report examines the current job market trends, the capabilities of these languages, their cost implications, flexibility, integration potential, learning curves, and the impact on career growth to provide a reasoned perspective on this critical decision.

I. Analyzing the Current Job Market for Statistical Programmers

An examination of the present job market reveals a complex landscape where SAS maintains a significant foothold while open-source languages are experiencing rapid growth. Despite the increasing adoption of open-source alternatives, SAS skills remain in considerable demand. This is evidenced by its strong ranking in the "Jobs" category of the IEEE Spectrum 2024 list. Furthermore, numerous job postings on platforms such as Indeed.com consistently seek professionals with SAS expertise. This sustained demand is particularly notable in specialized areas like clinical SAS programming, driven by ongoing developments in pharmaceutical research and clinical trials. SAS's long-standing reputation as a global leader in data science, known for its extensive statistical functions and reliable performance, contributes to its continued prevalence in many organizations. Large corporations often favor SAS due to the robust customer service and technical support it offers, which are crucial in enterprise settings where operational reliability is paramount. This indicates that despite the growing popularity of open-source languages, SAS maintains a strong and active presence in the current job market, especially within established and regulated industries with substantial investments in SAS infrastructure and validated processes. Professionals with specialized SAS knowledge in fields like clinical trials or financial modeling may continue to find numerous opportunities.

Conversely, the job market is also witnessing a significant surge in demand for open-source programming skills, particularly in Python and R. Surveys and job market analyses highlight a tremendous increase in job openings requiring these skills, signifying a strong competitive

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146

Using Canvas (writing)

Write the paper about how biostatistical programmers can use open source as well as SAS.

+ Canvas

- Click "canvas" to start
- You could see the prompt in the left and answer to the right,
- You can continue to prompt in the left side while seeing the response in the right side.
- You can also highlight and prompt to modify them.

biostatistical programmers can use open source as well as SAS.

The Dual Toolkit: Leveraging Open Source and SAS in Biostatistical Programming
May 18, 2:09 PM

Try again without Canvas

This document provides a comprehensive overview of the topic. Let me know if you would like any specific sections expanded or modified!

Ask Gemini

+ Canvas

The Dual Toolkit: Leveraging Open Source and SAS in Biostatistical Programming

Abstract

Biostatistical programming is a critical function in clinical research and drug development, primarily focused on the analysis and reporting of clinical trial data for regulatory submissions. Historically, SAS has been the undisputed standard in this field due to its robust validation framework, regulatory acceptance, and comprehensive suite of statistical procedures. However, the landscape of data science and statistical computing has evolved dramatically with the rise of powerful and flexible open-source languages like R and Python. This paper explores the complementary roles of SAS and open-source tools in modern biostatistical programming, discussing their respective strengths and weaknesses, potential strategies for integration, and the implications for programmers and organizations in the life sciences industry. We argue that a "dual toolkit" approach, strategically leveraging the strengths of both environments, represents the most effective path forward for enhancing efficiency, innovation, and reproducibility in biostatistical programming while maintaining regulatory compliance.

1. Introduction

Biostatistical programmers play a pivotal role in translating raw clinical trial data into meaningful statistical results and regulatory submission deliverables. Their work involves data manipulation, statistical analysis, table, listing, and figure (TLF) generation, and the creation of submission-ready datasets (e.g., CDISC SDTM and ADaM). For decades, SAS has been the de facto standard in this domain, largely due to its widespread adoption by regulatory agencies like the FDA and its perceived reliability and validation capabilities.

However, the broader data science community has increasingly embraced open-source languages such as R and Python, which offer unparalleled flexibility, cutting-edge statistical and machine learning packages, advanced visualization capabilities, and vibrant community support. This divergence presents both challenges and opportunities for the biostatistical programming field. Relying solely on SAS may limit access to modern analytical techniques and efficient workflows, while a complete shift to open source faces hurdles related to validation, regulatory acceptance, and the existing infrastructure within pharmaceutical and biotech companies.

This paper advocates for a pragmatic approach: recognizing the enduring value of SAS, particularly for regulatory deliverables, while strategically integrating open-source tools to leverage their strengths in areas like exploratory data analysis, advanced modeling, and visualization.

Ask Gemini

SAS has earned its dominant position in biostatistical programming for several key reasons:

- **Regulatory Acceptance:** Regulatory bodies worldwide have a long history of accepting SAS-based analyses and submissions. This provides a level of confidence and reduces regulatory risk.
- **Validation and Documentation:** SAS has a well-established framework for software validation, and pharmaceutical companies have developed extensive internal validation procedures around SAS programs and macros.
- **Comprehensive Statistical Procedures:** SAS offers a wide range of well-tested and documented statistical procedures relevant to clinical trials (e.g., PROC MIXED, PROC GLM, PROC PHREG).
- **Data Management Capabilities:** SAS is powerful for handling large, complex datasets and performing intricate data transformations required for clinical data.
- **Industry Standard:** A large pool of experienced SAS programmers is available, and many existing systems and processes within the industry are built around SAS.

147

Using Canvas (coding)

Gemini 2.5 Pro (experimental)

Create line graph of "Patient Data" using python.

"Patient Data":

```
{ 'patient': '01-001', 'visit 1': 100,
  'visit 2': 105,
  'visit 3': 110},
{ 'patient': '01-002', 'visit 1': 104,
  'visit 2': 95, 'visit 3': 90},
{ 'patient': '02-001', 'visit 1': 99,
  'visit 2': 103, 'visit 3': 105},
{ 'patient': '02-002', 'visit 1': 110,
  'visit 2': 105,
  'visit 3': 115}
```

Ask Gemini

+ Canvas

Patient Data Line Graph

```
1 import matplotlib.pyplot as plt
2 import pandas as pd
3
4 # Corrected patient data (assuming missing closing braces)
5 patient_data = [
6     {'patient': '01-001', 'visit 1': 100, 'visit 2': 105, 'visit 3': 110},
7     {'patient': '01-002', 'visit 1': 104, 'visit 2': 95, 'visit 3': 90},
8     {'patient': '02-001', 'visit 1': 99, 'visit 2': 103, 'visit 3': 105},
9     {'patient': '02-002', 'visit 1': 110, 'visit 2': 105, 'visit 3': 115}
10 ]
11
12 # Convert the list of dictionaries to a Pandas DataFrame for easier handling
13 # Set 'patient' column as the index
14 df = pd.DataFrame(patient_data).set_index('patient')
15
16 # Transpose the DataFrame so visits are rows and patients are columns
17 # This makes plotting easier where each column is a line
18 df_transposed = df.transpose()
19
20 # Create the plot
21 plt.figure(figsize=(10, 6)) # Set the figure size for better readability
22
```

Export to Colab

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148

148

Using Canvas (coding)

Gemini 2.5 Pro (experimental)

Create line graph of "Patient Data" using python.

"Patient Data":

```
{'patient': '01-001', 'visit 1': 100, 'visit 2': 105, 'visit 3': 110},
{'patient': '01-002', 'visit 1': 104, 'visit 2': 95, 'visit 3': 90},
{'patient': '02-001', 'visit 1': 99, 'visit 2': 103, 'visit 3': 105},
{'patient': '02-002', 'visit 1': 110, 'visit 2': 105, 'visit 3': 115}
```

Ask Gemini

New notebook saved in Drive

Open Colab

Export to Colab

```
1 import matplotlib.pyplot as plt
2 import pandas as pd
3
4 # Corrected patient data (assuming missing closing braces)
5 patient_data = [
6     {'patient': '01-001', 'visit 1': 100, 'visit 2': 105, 'visit 3': 110},
7     {'patient': '01-002', 'visit 1': 104, 'visit 2': 95, 'visit 3': 90},
8     {'patient': '02-001', 'visit 1': 99, 'visit 2': 103, 'visit 3': 105},
9     {'patient': '02-002', 'visit 1': 110, 'visit 2': 105, 'visit 3': 115}
10 ]
11
12 # Convert the list of dictionaries to a DataFrame
13 # Set 'patient' column as the index
14 df = pd.DataFrame(patient_data).set_index('patient')
15 # Transpose the DataFrame so visits are columns
16 # This makes plotting easier where each column represents a visit
```

Console

Line graph generated successfully.

Patient Data Line Graph

Patient Measurements Over Visits

| Patient ID | Visit 1 | Visit 2 | Visit 3 |
|------------|---------|---------|---------|
| 01-001 | 100 | 105 | 110 |
| 01-002 | 104 | 95 | 90 |
| 02-001 | 99 | 103 | 105 |
| 02-002 | 110 | 105 | 115 |

- We could run the codes in Colab in the left side.
- We can prompt to use other packages as well.

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149

Using Canvas (coding)

Patient Data Line Graph (HTML/Chart.js)

Apr 14, 1:03 PM

This HTML file sets up a basic page structure, includes Chart.js and Tailwind CSS, defines the patient data in JavaScript, formats it for Chart.js, and then renders the line chart on the canvas. Each patient is represented by a colored line with points marking the measurements at each visit.

Ask Gemini

Patient Data Line Graph (HTML/Chart.js)

Code Preview

Patient Measurements Over Visits

| Patient ID | Visit 1 | Visit 2 | Visit 3 |
|------------|---------|---------|---------|
| 01-001 | 100 | 105 | 110 |
| 01-002 | 104 | 95 | 90 |
| 02-001 | 99 | 103 | 105 |
| 02-002 | 110 | 105 | 115 |

- We could also create it in html and JS.
- We could view actual coding in "code" tab.

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150

150

Response Search in Google

• Day 3:

- Take a Batobus cruise to Notre Dame.
- Visit Notre Dame Cathedral.
- Explore the M...
- Visit Sainte-C...
- Visit Shakespe...
- Stroll on Île S...
- Stroll along th...
- Visit the Jardi...
- Take the Bato...

• Day 4:

- Shopping at Monoprix or Le Bon Marché Department Stores.
- Play at the playground and picnic at Luxembourg Gardens.

• Day 5:

- Visit the Musée d'Orsay.
- Explore the Louvre.
- Visit the Tuileries Garden Playground.

Would you like me to find more information about any of these locations?

🔍 Sources

Notre Dame Cathedral.

AI Mode All Images News Maps Videos Short videos More Tools

Notre-Dame Cathedral of Paris

4.7 ★ (65K) · Cathedral in Paris, France · Closed

Overview Tickets Reviews



Reviews 4.7 ★ (64,906 reviews) Tickets Free from official site

Weather Sun 68° Mon 72° Tue 75°

Sponsored · Tickets & tours

| Notre-Dame Exterior & ... | Paris: Notre-Dame Outdo... | Notre-Dame Cathedral... | Paris: Sainte-Chapelle Entr... | Notre-Dame Cathedral & ... |
|---------------------------|----------------------------|-------------------------|--------------------------------|----------------------------|
| \$83 | \$44 | \$60 | \$66 | \$11 |
| TripAdvisor 4.7 ★ (91) | GetYourGuide 4.7 ★ (430) | TripAdvisor 4.5 ★ (147) | GetYourGuide 4.5 ★ (221) | Headout 4.5 ★ (1,480) |

Sponsored Headout: Notre-Dame Cathedral & Optional Île de la Cité Guided Tour \$11 4.0 ★ (1.4K) Guided tour

About Notre-Dame des Dames, often referred to simply as Notre-

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151

151

Share

K What is Large Language Model?

Show drafts 🔽 🔊

Imagine a super reader, one who's inhaled libraries of books and articles, websites and code. That's kind of what a Large Language Model (LLM) is! These brainiacs are trained on mountains of text data, learning all the tricks of the language trade – grammar, jokes, even sarcasm.

Think of an LLM as a wordsmith extraordinaire. They can translate languages like a pro, answer your questions like a whiz, and even write poems or stories that sound like they came straight from a bestseller. They're also pretty handy at summarizing stuff, coming up with cool creative text formats, and even writing different kinds of content (think poems, scripts, emails, you name it!).

But hey, even super readers aren't perfect. Sometimes they might mix things up or get stuff wrong, and their knowledge is limited to what they've read. Plus, it's important to remember that they can reflect the biases they learned from their training data, so keep that in mind.

Even with these little hiccups, LLMs are like language superheroes, opening doors to new ways of learning, talking, and creating. They're still learning and growing, and who knows what they'll be able to do in the future! So next time you chat with a helpful LLM behind the scenes, remember the cool world of LLMs behind the scenes.

Check complete

Share Export to Docs Draft in Gmail

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152

152

Better Search Google tools (AI Agent)

Gemini 2.0 Flash

Enabled

- Google Calendar
- Google Keep
- Google Tasks
- Gmail
- Google Docs
- Google Drive
- Google Flights
- Google Hotels
- Google Maps

Deep Research Canvas

Hello, Kevin

@Google Maps Find the direction from my home to new york penn staaion.

Google Maps

I will use the Google Maps app to find the directions from your home to New York Penn Station. Okay, I have the directions. It is approximately 92.6 miles and will take about 1 hour and 46 minutes to drive. Here's a link to the route on Google Maps: [Directions to New York Penn Station](#)

Google Maps

Directions from home to New York Penn Station

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153

153

Better Search with YouTube extension

Find Youtube videos for Tokyo travel.

YouTube

Here are some YouTube videos about Tokyo travel:

- [Tokyo Vacation Travel Guide | Expedia](#)
- [How to Spend 5 Days in TOKYO - Japan Travel Itinerary](#)
- [Usually harmless if you ignore them #japan #japantraveler #travel #travelvlog #tokyo #osaka](#)
- [Surviving Your First Hour In Japan](#)
- [Know before you go: Japan JP #familytraveltips #travelwithkids #japan #tokyo #japantravel #travel](#)

Would you like to watch any of these?

YouTube

Tokyo travel

Tokyo Vacation Travel Guide | Expedia

summarize the first video.

YouTube

Here's a summary of the Tokyo travel guide video:

- General Overview:** Tokyo is a large, dynamic city that beautifully combines tradition with modernity [00:38]. It's a well-organized city made up of different neighborhoods, each having its own unique vibe [01:37].
- Historical and Cultural Sites:**
 - Asakusa:** Visit the ancient Sensō-ji Temple and Nakamise-dōri market [01:58].
 - Ueno Park:** Relax and explore the cultural treasures of the Tokyo National Museum [02:45].
 - Meiji Shrine:** Pay respects to Emperor Meiji in this peaceful forest setting [03:47].
- Youth Culture and Fashion:** Explore the vibrant and unique youth culture in Harajuku [02:58].
- Shopping Districts:**
 - Shibuya:** Experience one of the world's busiest pedestrian crossings

154

Use extension for the prompt – AI agent

K I will visit Boston from 2024/04/23 to 2024/04/26. I need to attend conference in Boston Convention center. Find the hotel near by.

Show drafts ▾ 🔊



Google Maps & 1 more ▾

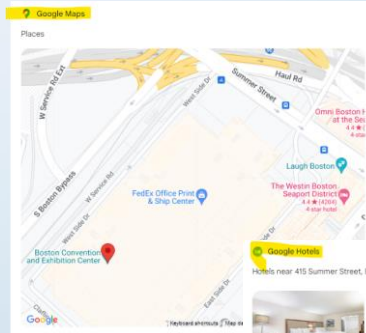
🕒 Gemini doesn't always get it right. Be sure to check the following hotel details:

Hotels near 415 Summer Street, Boston, MA - Apr 23 - Apr 26 - 2 adults

Sure, here are some hotels available near the Boston Convention Center between April 23rd and April 26th:

- [Holiday Inn Express & Suites Boston - Cambridge, an IHG Hotel](#) - Sleek rooms in a straightforward hotel with free hot breakfast buffet & WiFi. Price: \$167 per night.
- [Holiday Inn Boston Bunker Hill Area, an IHG Hotel](#) - Contemporary hotel with a bar/restaurant, plus an indoor pool, basketball court & climbing wall. Price: \$156 per night.
- [Hilton Boston Park Plaza](#) - Upmarket 1927 hotel with elegant rooms, steakhouse dining & a trendy bar, plus a golf simulator. Price: \$167 per night.
- [Courtyard by Marriott Boston-South Boston](#) - Modern hotel with a restaurant & an outdoor courtyard, plus free Wi-Fi. Price: \$179 per night.
- [Courtyard by Marriott Boston Cambridge](#) - Modern riverside hotel with 2 restaurants & a ballroom, plus free Wi-Fi. Price: \$171 per night.

I hope this helps with planning your trip to Boston!



Hotels near 415 Summer Street, Boston, MA - Apr 23 - Apr 26 - 2 adults

| | | |
|--|--|-------------|
| | Holiday Inn Express & Suites Boston - Cambridge, an IHG Hotel | \$167/night |
| | Holiday Inn Boston Bunker Hill Area, an IHG Hotel | \$156/night |
| | Hilton Boston Park Plaza | \$167/night |
| | Courtyard by Marriott Boston-South Boston | \$179/night |

155

Connect to my Google Drive thru Google Workspace extension

K what is the latest documents that I uploaded in my google doc?

Show drafts ▾ 🔊



Google Workspace ▾

Assuming by "latest" you mean the recency of creation date, then the latest documents you uploaded to your Google Drive are:

- Outline for Clinical Data Management with Python, created on July 7, 2023.
- Strategic Implementation of Open Source Software in Clinical Trials, created on May 8, 2021.
- cs329s_03_note_data_engineering, created on January 19, 2021.
- RESUME, created on May 16, 2012.
- hero.pdf, created on June 8, 2014.

I hope this helps!

Google Drive Items considered for this response. [Learn more](#)

Outline for Clinical Data Management with Python Niles Kowadkar

K summary "Outline for Clinical Data Management with Python" from Google Drive in one sentence.

Show drafts ▾ 🔊



Google Workspace ▾

"Outline for Clinical Data Management with Python" provides a comprehensive guide to using Python for clinical trial data management and analysis, covering topics from fundamental concepts to advanced analytics and regulatory considerations.

Google Drive Items considered for this response. [Learn more](#)

Outline for Clinical Data Management with Python Niles Kowadkar

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156

156

Google AI Studio – image

Google AI Studio

Get API key

Studio

Dashboard

Documentation

Settings

Profile

Chat

Stream

Starter Apps

History

Enable save

Untitled prompt

Get started with Gemini

Turn something or pick one from prompt gallery

Create a image of a majestic Bengal tiger with vibrant orange and black stripes, piercing blue eyes, and a powerful build.



Run settings

Gemini 2.0 Flash (Image Generation) Experimental

Output format

Images & text

Temperature

1

- https://aistudio.google.com/prompts/new_chat
- Much faster than ChatGPT in creating the images

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157

157

Google AI Studio – contents with image

Korean Bibimbap Recipe

Clipboard

<>

Share

↩

↻

⋮

Please create a recipe for Korean style bibimbob. create the necessary image along the way.

Korean Bibimbap: A Colorful and Delicious Mixed Rice Bowl

Bibimbap (비빔밥) literally translates to "mixed rice" and is a quintessential Korean dish. It's a vibrant bowl of warm rice topped with an assortment of seasoned vegetables (namul), gochujang (Korean chili paste), and often a fried egg and sliced meat. This recipe provides a detailed guide to creating your own delicious bibimbap at home.

Korean bibimbap recipe

Image 1: Raw short-grain white rice in a measuring cup.



Image 13: A person using chopsticks to mix all the ingredients of a bibimbap bowl together.



Run settings

Gemini 2.0 Flash (Image Generation) Experimental

Output format

Images & text

Temperature

1

158

158

Google AI Studio – Creating video

Google AI Studio

Get API key 2.4 MB • Done

Chat

Stream

Generate Media

Build

History

Generate Media > **Generate Video**

Run settings

Veo 2

Number of results 1

Aspect ratio 16:9

Video duration

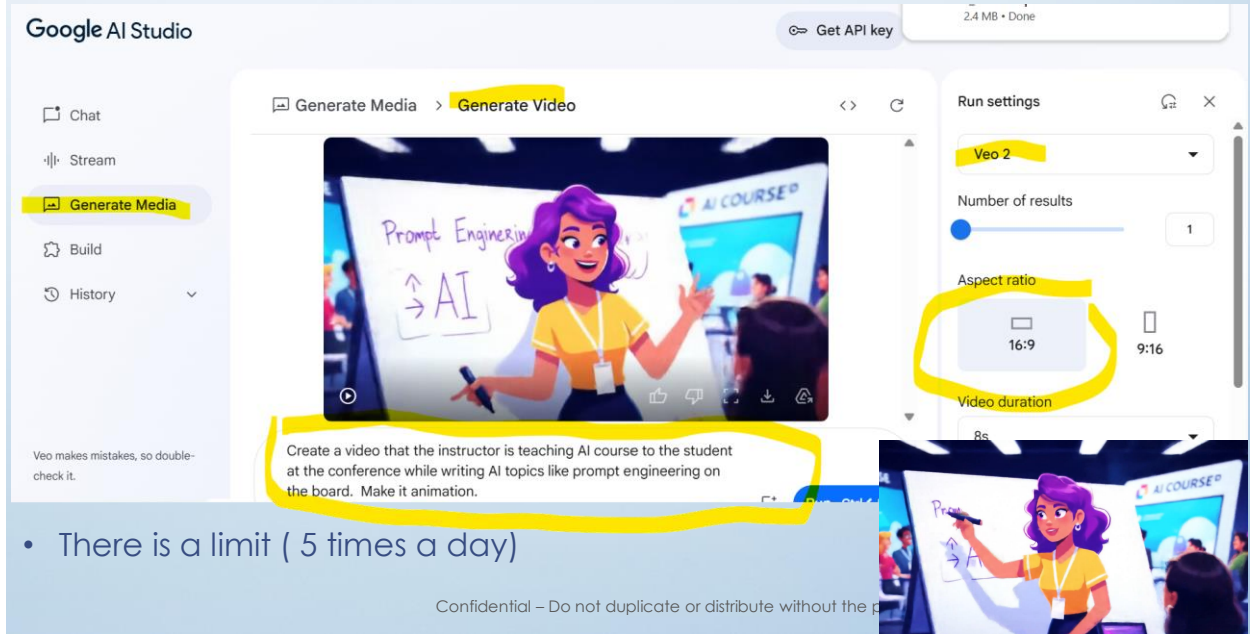
Re

Create a video that the instructor is teaching AI course to the student at the conference while writing AI topics like prompt engineering on the board. Make it animation.

Veo makes mistakes, so double-check it.

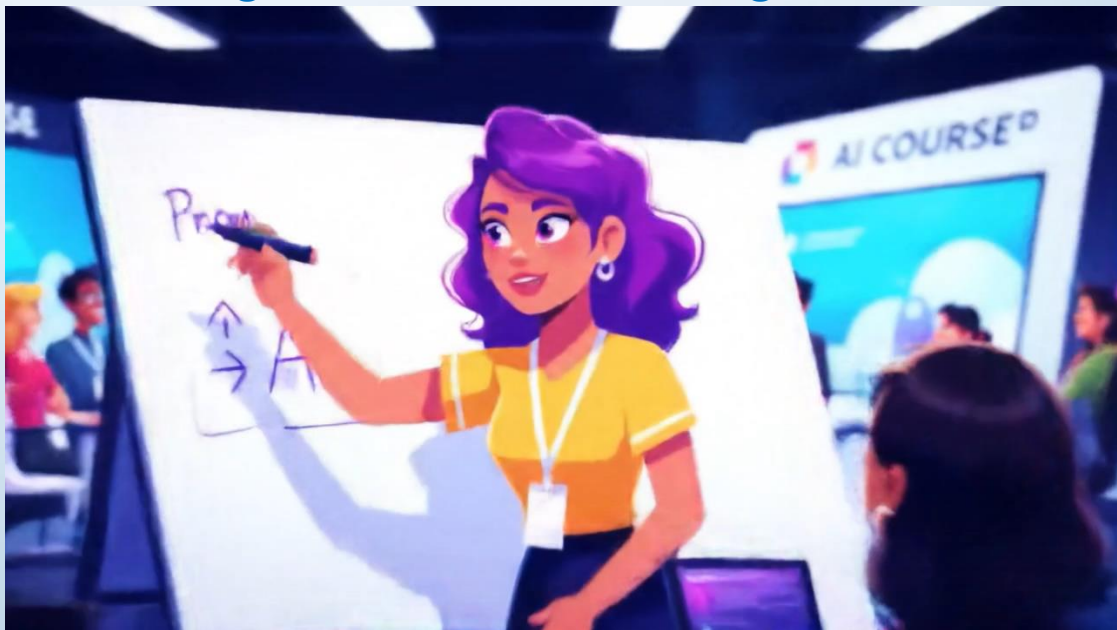
- There is a limit (5 times a day)

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Google AI Studio – Creating video



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160

160

Gems (GPTs)

The screenshot displays the Gemini 'Gems' interface. On the left is a sidebar with a 'Recent' list of prompts and a 'Gems' section where 'Gem manager' is selected. The main area is titled 'Gem manager' and shows 'Premade by Google' gems like 'Chess champ', 'Brainstormer', 'Career guide', and 'Coding partner'. Below this is 'Your Gems' section, which features a preview of a custom Gem titled 'ADaM Q&A'. The preview shows the Gem's instructions: 'You are an expert in CDISC ADaM (Analysis Data Model) with deep knowledge of the ADaM Implementation Guide (IG). Your job is to help' and a knowledge base containing a PDF file named 'ADaMIG v1.2-Final'.

components of Gems

- Title
- Instructions
- reference files

161

Claude Sonnet Functionality



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162

162

Claude Sonnet

Free plan · [Upgrade](#)

☀ What's new, Kevin?

How can I help you today?



Claude 3.7 Sonnet ▾



- <https://claude.ai/new>
- You could sign in with current google account / new userid and pw.

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163

163

Set up artifact

⌕ Settings

Profile

Billing

Account

Full name

Kevin lee

What should we call you?

Kevin

Update Name

What best describes your work?

Select your work function ▾

Show prompt suggestions

We'll show examples of starter prompts you can use based on your role

Enable artifacts

Ask Claude to generate content like code snippets, text documents, or website designs, and Claude will create an Artifact that appears in a dedicated window alongside your conversation.



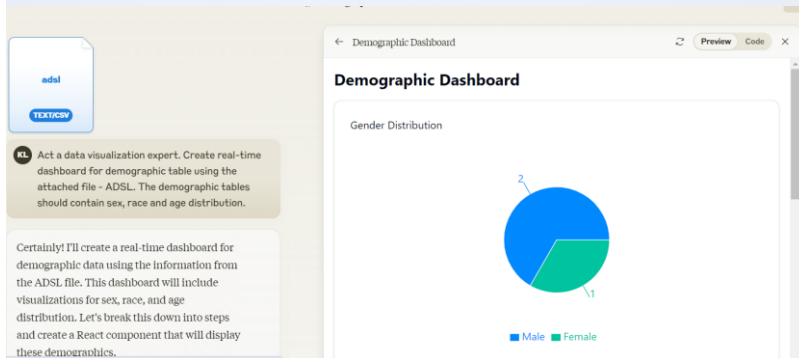
- Click the setting
- Enable artifacts
- Claude will automatically generate Artifacts for specific content types like code or graphics, displaying them in a dedicated window next to the chat.

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164

164

Data Analysis



1. Attach the file - adsl
2. Prompt : Act a data visualization expert. Create real-time dashboard for demographic table using the attached file - ADSL. The demographic tables should contain sex, race and age distribution.
3. It will create
 1. Answer.
 2. Artifacts – Preview and Codes

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165

165

Data Analysis - Answer

Certainly! I'll create a real-time dashboard for demographic data using the information from the ADSL file. This dashboard will include visualizations for sex, race, and age distribution. Let's break this down into steps and create a React component that will display these demographics.

I've created a React component that generates a real-time dashboard for demographic data based on the ADSL file. Here's a breakdown of the dashboard:

1. Gender Distribution: A pie chart showing the distribution of males and females in the study.
2. Race Distribution: A pie chart displaying the racial breakdown of participants.
3. Age Group Distribution: A bar chart illustrating the distribution of participants across different age groups.

The dashboard is designed to update every 5 seconds, simulating real-time data updates. In a real-world scenario, you would replace the simulated data fetch with an actual API call to get the latest data.

Key features of this dashboard:

1. Responsive design: The charts will resize based on the screen size.
2. Real-time updates: The data refreshes every 5 seconds.
3. Interactive charts: Users can hover over chart elements to see detailed information.
4. Clear visualization: Each demographic aspect (gender, race, age) is presented in its own card for easy comprehension.

To use this dashboard:

1. Make sure you have the necessary dependencies installed (react, recharts, and the shadcn/ui components).
2. Replace the simulated data in the `fetchData` function with your actual data fetching logic.
3. Adjust the styling as needed to fit your application's design.

This dashboard provides a clear and interactive way to visualize the demographic data from your ADSL file. Users can quickly understand the distribution of participants across different demographic categories.

166

Data Analysis – Artifacts (code)

```
import React, { useState, useEffect } from 'react';
import { PieChart, Pie, Cell, BarChart, Bar, XAxis, YAxis, CartesianGrid, Tooltip, Legend, ResponsiveContainer } from 'recharts';
import { Card, CardHeader, CardContent } from '@components/ui/card';

const COLORS = ['#0088FE', '#00C49F', '#FFBB28', '#FF8042', '#8884D8'];

const DemographicDashboard = () => {
  const [data, setData] = useState([]);

  useEffect(() => {
    // Simulating data fetch
    const fetchData = () => {
      // In a real scenario, you'd fetch this data from an API
      const rawData = `USUBJID,DISCONT,DSREASCD,DEATH,TRTP,TRTPCD,AGE,AGEGRP,RACE,SEX
01-701-1015,,Completed,,Placebo,Pbo,63,<65,"HISPANIC (MEXICAN - AMERICAN, MEXICO, CENTRAL AND SOUTH AMERICA)",F
01-701-1023,Y,Adverse Event,,Placebo,Pbo,64,<65,"HISPANIC (MEXICAN - AMERICAN, MEXICO, CENTRAL AND SOUTH AMERICA)",M
01-701-1028,,Completed,,Xanomeline High Dose,Xan_Hi,71,65-80,CAUCASIAN,M
01-701-1033,Y,Sponsor Decision,,Xanomeline Low Dose,Xan_Lo,74,65-80,CAUCASIAN,M
01-701-1034,,Completed,,Xanomeline High Dose,Xan_Hi,77,65-80,CAUCASIAN,F`;

      const parsedData = rawData.split("\n").slice(1).map(row => {
        const [,,,,,AGE,AGEGRP,RACE,SEX] = row.split(',');
        return { AGE: parseInt(AGE), AGEGRP, RACE, SEX };
      });

      setData(parsedData);
    };
  });
};
```

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167

167

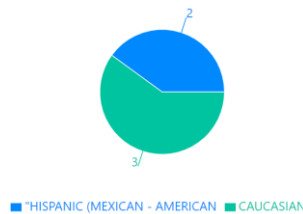
Data Analysis - Artifacts

Demographic Dashboard

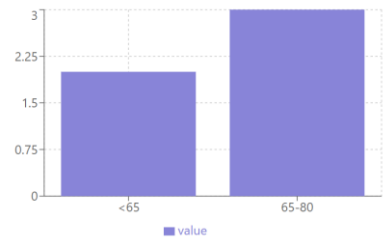
Gender Distribution



Race Distribution



Age Group Distribution



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168

168

Claude Console

Happy late night, Kevin

Write a prompt from scratch

Generate a prompt

Improve an existing prompt

Get API keys

Explore documents

Workbench

Generate a prompt

You can generate a prompt template by sharing basic details about your task.

I want to create R codes that will create SDTM data sets using data specification.

Summarize a document Write me an email Translate code

This prompt will be used with models that have thinking enabled

Cancel Generate

- <http://console.anthropic.com>
- You could sign in with current google account / new userid and pw.

- Click on "Workbench"
- Click on "Generate a Prompt"

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169

Claude Console

SDTM Dataset Generation in R

Last saved Apr 11 at 1:50 AM

claude-3-7-sonnet-20250219 { } 1 Examples

```
<data_specification>
[[DATA_SPECIFICATION Variable_Name Variable_Label]]
</data_specification>
```

Analyze the provided data specification carefully. It should contain dataset name, variables, their types, and any specific requirements.

Create the R code using the following structure:

1. Start with a comment block explaining the purpose of the code created.
2. Include any necessary library imports (e.g., tidyverse, lubridate).
3. Create a function named 'create_[dataset_name]_sdtm()' where [dataset_name] is the name of the SDTM dataset you're creating.

Pre-fill response Add message pair

1.09k

Test Case

Generate

| Variable_Name | Variable_Label | Type | Derivation |
|---------------|-----------------------------------|------|--|
| STUDYID | Study Identifier | Char | demo.study_id |
| DOMAIN | Domain Abbreviation | Char | dm |
| USUBJID | Unique Subject Identifier | Char | demo.study_id + '-' + demo.site_id + '-' + demo.subjid |
| SUBJID | Subject Identifier for the Study | Char | demo.subjid |
| RFSTDTC | Subject Reference Start Date/Time | Char | first expo.exdtc |
| RFENDTC | Subject Reference End Date/Time | Char | last expo.exdtc |

Run Ctrl + d

- We can test and update prompts very easily in the console.

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170

Claude Console - response

You are tasked with creating R code that will generate SDTM (Study Data Tabulation Model) datasets using only a provided data specification. This task is crucial for standardizing clinical trial data. Follow these instructions carefully to create accurate and efficient R code.

You will be provided with a data specification in the following format:

```
<data_specification>
{{DATA_SPECIFICATION}}
</data_specification>
```

Analyze the provided data specification carefully. It should contain information about the dataset name, variables, their types, and any specific requirements or derivations.

Create the R code using the following structure:

1. Start with a comment block explaining the purpose of the code and the dataset being created.
2. Include any necessary library imports (e.g., tidyverse, lubridate).
3. Create a function named `create_[dataset_name]_sdtm()` where `[dataset_name]` is the name of the SDTM dataset you're creating.
4. Within the function, create an empty data frame with the specified variables.
5. Add code to populate the data frame based on the specifications for each variable.
6. Include error handling and input validation where appropriate.
7. Return the completed data frame at the end of the function.

When handling different variable types, use appropriate R functions and data types:

- For character variables, use `as.character()`
- For numeric variables, use `as.numeric()`

- For date variables, use appropriate date conversion functions from the lubridate package
- For derived variables, include the derivation logic as specified

Include necessary R libraries at the beginning of your code, such as:

```
``r
library(tidyverse)
library(lubridate)
``r
```

Add error handling and input validation where appropriate, using `tryCatch()` or `if-else` statements. Include comments throughout your code to explain complex operations or derivations.

Present your final R code within `<r_code>` tags. Ensure your code is properly formatted and follows R Best practices.

```
<r_code>
# Your R code here
</r_code>
```

After the code, provide a brief explanation of how the code works and any assumptions made during its creation. Present this explanation within `<explanation>` tags.

```
<explanation>
Your explanation here
</explanation>
```

Remember to adhere strictly to the provided data specification and SDTM standards when creating the R code.

DeepSeek Functionality



DeepSeek R1



Hi, I'm DeepSeek.

How can I help you today?

Message DeepSeek

DeepThink (R1)

Search



- <https://chat.deepseek.com/>
- You could sign in with current google account / new userid and pw.

- Only for texts
- Free
- Long Text (128K tokens)
- Strong Performance in Coding & Math
- File Support – PDF, Word, Excel, PPT
- Good in Deep Research – Thinking long

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173

173

DeepSeek R1

Based on the attached file, tell me what are required variables in ADSL?

DeepThink (R1)

Search



Based on the attached file, tell me what are required variables in ADSL?

Edit



Thought for 26 seconds

Okay, let me try to figure out what the required variables in ADSL are based on the attached file. The user is asking specifically about ADSL, which is the Subject-Level Analysis Dataset. From what I remember from the document, ADSL has to have certain variables that are mandatory.

First, I'll start by recalling that in Section 3.2 of the document, there are tables that list ADSL variables. Table 3.2.1 talks about identifier variables. The first few variables listed there are STUDYID, USUBJID, SUBJID, and SITEID. The "Core" column for these says "Req" for required. So those must be required.

- Once clicking on "DeepThink(R1)", it will start "Deep Research" functionality and thinking about 26 seconds.
- We will see its thinking process, then eventually it provide the response.

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marked "Req" in the Core column and the conditional ones that are necessary based on study design.

So the final list should include the identifiers, demographics with their units, treatment variables for the first period, ARM, and at least one population flag. The date variables are required only if applicable.

Based on the ADaMIG v1.2 document, the required variables in ADSL (Subject-Level Analysis Dataset) are as follows:

1. Identifier Variables (Table 3.2.1)

- **STUDYID**: Study Identifier (from SDTM DM).
- **USUBJID**: Unique Subject Identifier (from SDTM DM).
- **SUBJID**: Subject Identifier for the Study (from SDTM DM).
- **SITEID**: Study Site Identifier (from SDTM DM).

2. Demographic Variables (Table 3.2.2)

- **AGE**: Subject's Age (from SDTM DM).
- **AGEU**: Age Units (from SDTM DM).
- **SEX**: Sex (from SDTM DM).
- **RACE**: Race (from SDTM DM).

174

Gen AI capabilities

- Reasoning
- Web Access and Research
- Image Generation
- Read documents, images, videos
-

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175

175

Reasoning

Recent Breakthrough:

- Letting an AI "think" before answering leads to better results
- The thinking process often feels surprisingly human-like

How to Use Reasoning Models

- Designed for deep thinking – like scholars, not chatty assistants
- Expect some delay – they take time to generate thoughtful answers
- Best for:
 - Academic research
 - Complex math and computer science questions
- Tip: Always provide clear, detailed context for better answers

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176

176

Model Comparison

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177

177

Data Analysis - Artifacts

	OpenAI	Gemini	Anthropic	DeepSeek
Reasoning	o1, GPT-4o	2.0 Flash Thinking	Sonnet 3.7	r1
Web Access	GPT	2.0 Flash		r1
image	sora	Imagen 3		
Code Execution	Canvas		artifact	
Live mode				

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178

178

Application Development using Gen AI API

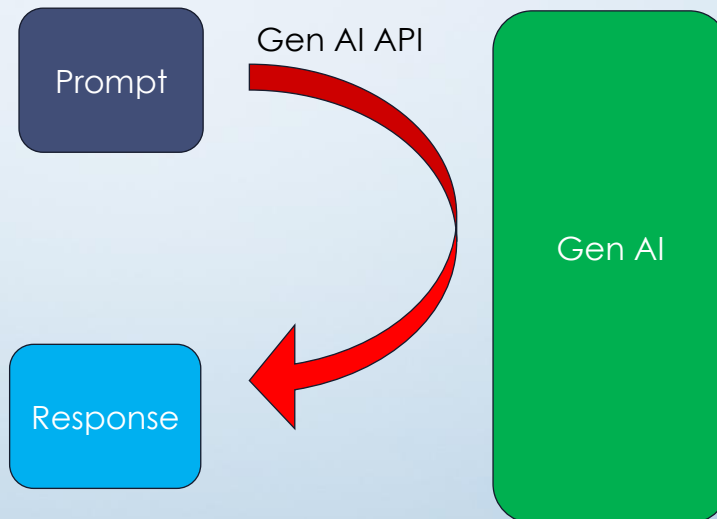
- SAS
- Python
- R

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179

179

Application Developing using Gen AI API



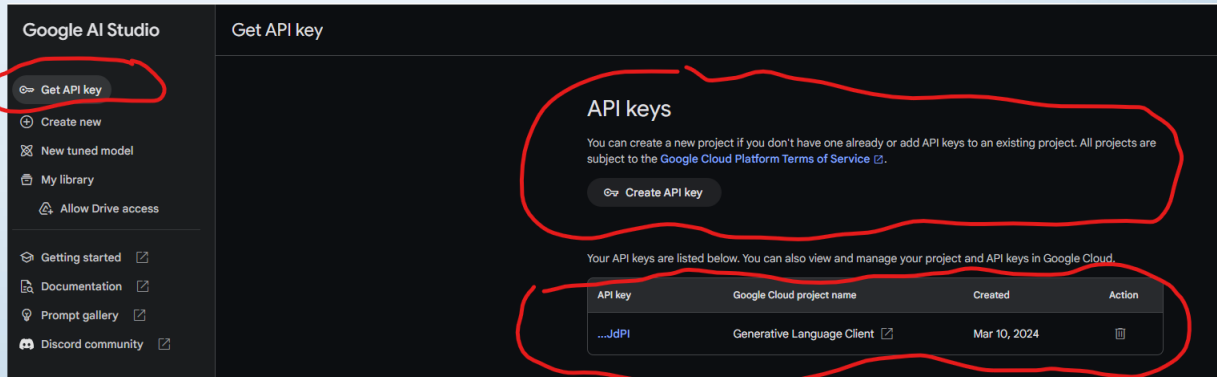
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180

180

How to create Gemini API

- API Keys in <https://aistudio.google.com/app/apikey>



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181

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SAS Application using Gemini API (free)

```
{ "contents":
  [ { "parts":
    [ { "text": <prompt> } ]
  } ]
}

{ "candidates":
  [ { "content":
    [ { "parts":
      [ { "text": <response> } ]
    } ]
  } ],
}

"usageMetadata": {
  "promptTokenCount": 12,,
}

"modelVersion": "gemini-1.5-flash"
}
```

```
%let api_key = xxxxxx;
%let url =
https://generativelanguage.googleapis.com/v1/models/gemini-1.5-
flash:generateContent;
%let url_query = &url.&key=&api_key. ;

filename prompt "F:\biometrics\project\GenAI\Gemini\prompt.txt";
filename resp "F:\biometrics\project\GenAI\ Gemini\resp.json";

Proc http
  method = "POST"
  url="&url_query"
  ct="application/json"
  in=prompt
  out=resp;
Run;

libname response JSON fileref=resp;

data r_sas;
  set response.content_parts;
run;
```

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182

182

SAS Application using Gen AI (Gemini) API

prompt.txt

```
{ "contents":
  [{ "parts":
    [{"text": "<prompt>"}]
  }
}
```

resp.json

```
{ "candidates":
  [ "content":
    { "parts":
      [{"text": "<resp>"}]
    }
  ],
}
```

```
%let api_key = xxxxxx;
%let url =
  https://generativelanguage.googleapis.com/v1/mod
  els/gemini-1.5-flash:generateContent;

%let url_query = &url.&key=&api_key.;

filename prompt
  "F:\biometrics\project\GenAI\Gemini\prompt.txt";
filename resp "F:\biometrics\project\GenAI\
  Gemini\resp.json";

Proc http
  method = "POST"
  url="&url_query"
  ct="application/json"
  in=prompt
  out=resp;
Run;

libname response JSON fileref=resp;
data _sas;
  set response.content_parts;
run;
```

183

Python Application using Gemini API

```
## import Google Gemini library
import google.generativeai as genai

## Create Prompt message
prompt = " Tell me about CDISC in 200 words" }}

## Set up Google Gemini API
genai.configure(api_key="xxxx")

## Send Prompt message to Gemini and get the response
google_model = genai.GenerativeModel('gemini-pro')

response = google_model.generate_content(prompt)
```

```
response:
GenerateContentResponse(
  done=True,
  iterator=None,
  result=glm.GenerateContentResponse({'candidates': [{'content': {'parts': [{'text': 'CDISC (Clinical Data Interchange Standards Consortium) is a non-profit organization dedicated to developing and maintaining data standards for clinical research. Its mission is to enhance data quality, facilitate data sharing, and streamline the clinical research process. CDISC standards provide a common language for researchers, data managers, and regulatory agencies to exchange and interpret clinical data. They include standards for data models, terminology, and data exchange formats, ensuring consistency and interoperability of clinical research data.'], 'role': 'model'}, 'finish_reason': 1, 'index': 0, 'safety_ratings': [{'category': 9, 'probability': 1, 'blocked': False}, {'category': 8, 'probability': 1, 'blocked': False}, {'category': 7, 'probability': 1, 'blocked': False}, {'category': 10, 'probability': 1, 'blocked': False}], 'token_count': 0, 'grounding_attributions': []}]})
```

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184

184

How to create ChatGPT API (paid)

- API Keys in <https://platform.openai.com/api-keys>

API keys

As an owner of this organization, you can view and manage all API keys in this organization.

Do not share your API key with others or expose it in the browser or other client-side code. To protect your account's security, OpenAI may automatically disable any API key that has leaked publicly.

View usage per API key on the [Usage page](#).

NAME	SECRET KEY	LAST USED	PROJECT ACCESS	CREATED BY	PERMISSIONS
HelloKevinLee	sk-...vyGg	Mar 26, 2025	Default project	Kevin Lee	All

185

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185

How to create ChatGPT API (paid)

- API Keys in <https://platform.openai.com/api-keys>

API keys

Your secret API keys are listed below. Please note that we do not display your secret API keys again after you generate them.

Do not share your API key with others, or expose it in the browser or other client-side code. In order to protect the security of your account, OpenAI may also automatically disable any API key that we've found has leaked publicly.

Enable tracking to see usage per API key on the [Usage page](#).

NAME	SECRET KEY	TRACKING	CREATED	LAST USED	PERMISSIONS
HelloKevinLee	sk-...mIA9	Enabled	Jun 22, 2023	Mar 15, 2024	All

186

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- add credit for API in <https://platform.openai.com/account/billing/overview>

Billing settings

Overview Payment methods Billing history Preferences

Pay as you go

Credit balance

\$19.99

Auto recharge is off

When your credit balance reaches \$0, your API requests will stop working. Enable automatic

186

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186

SAS Application using ChatGPT API

```

{ "model": "gpt-3.5-turbo",
  "message":
    [{ "role": "user",
        "content": <prompt>
      }]
}

{ "id": "xxx",
  "object": "chat.completion",
  "created": 1710533285,
  "model": "gpt-3.5-turbo-0125",
  "choices":
    [{ "index": 0,
        "message":
          { "role": "assistant",
            "content": <resp>
          },
        "logprobs": null,
        "finish_reason": "stop"
      }],
  "usage": {},
  "system_fingerprint": "xxx"
}

%let api_key = xxxxxx ;

filename prompt
"F:\biometrics\project\GenAI\OpenAI\prompt.txt";
filename resp
"F:\biometrics\project\GenAI\OpenAI\resp.json";

Proc http
  method = "POST"
  url="https://api.openai.com/v1/chat/completions"
  ct="application/json"
  in=prompt
  out=resp;
  Headers "Authorization" = "Bearer &api_key.";
Run;

libname response JSON fileref=resp;
data r_sas;
  set r_json.choices_message;
run;

```

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187

187

SAS Application using Gen AI (ChatGPT) API



```

{ "model": "gpt-3.5-turbo",
  "message":
    [{ "role": "user",
        "content": <prompt>
      }]
}

{ "id": "xxx",
  "object": "chat.completion",
  "created": 1710533285,
  "model": "gpt-3.5-turbo-0125",
  "choices":
    [{ "index": 0,
        "message":
          { "role": "assistant",
            "content": <resp>
          },
        "logprobs": null,
        "finish_reason": "stop"
      }],
  "usage": {},
  "system_fingerprint": "xxx"
}

%let api_key = xxxxxx ;

filename prompt
"F:\biometrics\project\GenAI\prompt.txt";
filename resp
"F:\biometrics\project\GenAI\resp.json";

Proc http
  method = "POST"
  url="https://api.openai.com/v1/chat/completions"
  ct="application/json"
  in=prompt
  out=resp;
  Headers "Authorization" = "Bearer &api_key.";
Run;

libname response JSON fileref=resp;
data r_sas;
  set response.choices_message;
run;

```

188

Python application using ChatGPT API

```
## import OpenAI ChatGPT library
import openai
```

```
## Create Prompt message
message = [{"role": "user", "content": "Tell me about CDISC in 200 words"}]
```

```
## Set up OpenAI ChatGPT API
openai.api_key = XXXXXXXX
```

```
## Send Prompt message to OpenAI
## Get the response
response = openai.chat.completion.create(
    model="gpt-3.5-turbo",
    messages=message,
    temperature=0
)
```

```
{
  "id": "chatcmpl-98x12kdpn8K1YUHFjddpKYSJwRY",
  "object": "chat.completion",
  "created": 1711920508,
  "model": "gpt-3.5-turbo-0125",
  "choices": [
    {
      "index": 0,
      "message": {
        "role": "assistant",
        "content": "CDISC, or the Clinical Data Interchange Standards Consortium, is a global non-profit organization that develops and promotes data standards for clinical research. Founded in 1997, CDISC aims to streamline the process of collecting, analyzing, and sharing clinical trial data to improve efficiency and facilitate data sharing among researchers, regulators, and industry stakeholders. CDISC standards cover a wide range of areas, including data collection, data submission, and data analysis. These standards help ensure that data is collected consistently across different studies, making it easier to compare and combine data from multiple sources. By using CDISC standards, researchers can save time and resources, reduce errors, and improve the quality and reliability of their data. In 2002, CDISC released its first set of standards, known as the Study Data Tabulation Model (SDTM). Since then, CDISC has developed several other standards, including the Analysis Data Model (ADaM) and the Define-XML standard for defining metadata in clinical trials. Overall, CDISC plays a crucial role in advancing the field of clinical research by promoting data standardization and interoperability, ultimately leading to better and more efficient healthcare outcomes for patients."
      },
      "logprobs": null,
      "finish_reason": "stop"
    }
  ],
  "usage": {
    "prompt_tokens": 18,
    "completion_tokens": 235,
    "total_tokens": 253
  },
  "system_fingerprint": "fp_b28b39ffa8"
}
```

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107

189

R application using ChatGPT API

```
## import OpenAI ChatGPT library
library(httr)
library(jsonlite)
```

```
# Set your OpenAI API key from .env file
api_key <- "xxxx"
```

```
# Define the request URL
api_url <- "https://api.openai.com/v1/chat/completions"
```

```
prompt <- "Tell me about CDISC in 200 words"
```

```
# Define the request body
body <- list(
  model = "gpt-3.5-turbo",
  messages = list(
    list(role = "system", content = "You are a helpful assistant."),
    list(role = "user", content = prompt)
  )
)
```

```
# Make the API request
response <- POST(
  api_url,
  add_headers('Authorization' = paste("Bearer", api_key),
    'Content-Type' = "application/json"),
  body = jsonlite::toJSON(body, auto_unbox = TRUE),
  encode = "json"
)
```

```
# Parse the response
result <- content(response, as = "parsed")
```

```
# Print the response
print(result$choices[[1]]$message$content)
```

```
result      List of 8
 $ id       : chr "chatcmpl-B3rIkdvKwMXRhLcd3k1p5rcyLSG1w"
 $ object   : chr "chat.completion"
 $ created  : int 1740259574
 $ model    : chr "gpt-3.5-turbo-0125"
 $ choices  :List of 1
 ..$ :List of 4
 .. ..$ index      : int 0
 .. ..$ message    :List of 3
 .. .. ..$ role    : chr "assistant"
 .. .. ..$ content : chr "CDISC, or the Clinical Data Interchange Standar..."
 .. .. ..$ refusal : NULL
 .. ..$ logprobs    : NULL
 .. ..$ finish_reason : chr "stop"
```

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LangChain

- Definition
- Functions
- Components

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191

LangChain

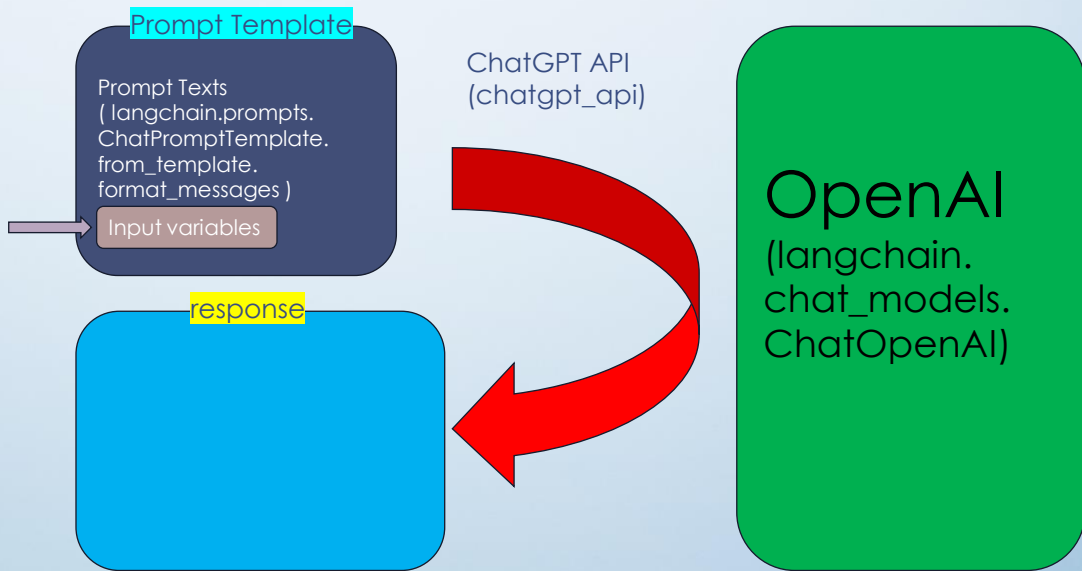
- Definition : Open-source development framework for developing LLM application.
- Functions
 - Modular components
 - Use cases : Common ways to combine components
- Components
 - Models
 - LLM : 20+ integrations
 - Chat models
 - Text Embedding models
 - Prompts
 - Prompt Templates
 - Output Parsers
 - Example Selections
 - Indexes
 - Document Loading
 - Text Splitting
 - Vector Stores
 - Retrievers
 - Chains
 - Prompt + LLM + Output Parsing
 - Can be used for building longer chains
 - Agents
 - Agent Types :
 - Algorithms for getting LLMs to use tools
 - Agent Toolkits : sort, sum

192

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LangChain Framework



193

193

Retrieval Augmented Generation (RAG)

- Definition
- Framework with LangChain



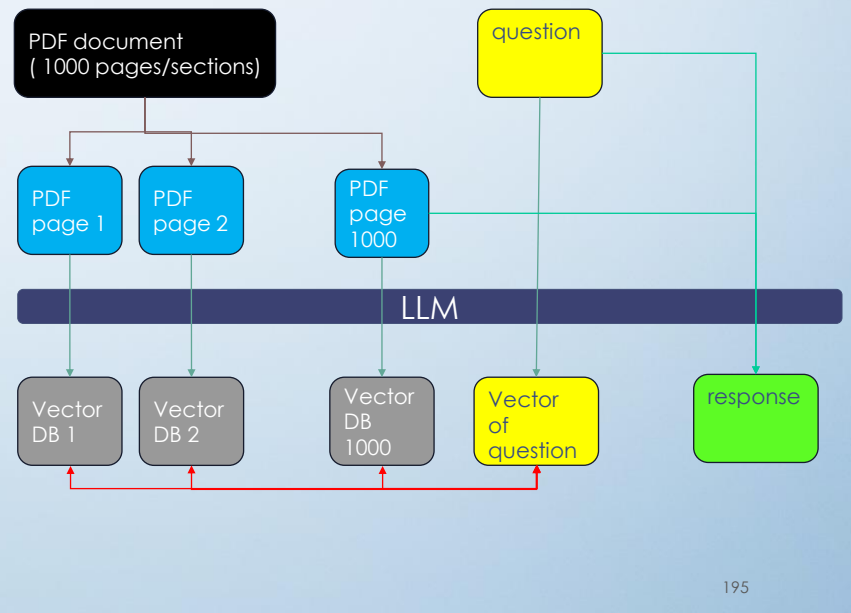
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194

RAG (Retrieval Augmented Generation)

- LLM driven vector DB
- Process
 1. Split document to smaller sections (1 to 1000)
 2. Create vector DB using LLM embedding (from PDF page 1 to 1000).
 3. Convert question to vector using same LLM embedding.
 4. Compare vector DB (1 to 1000) with

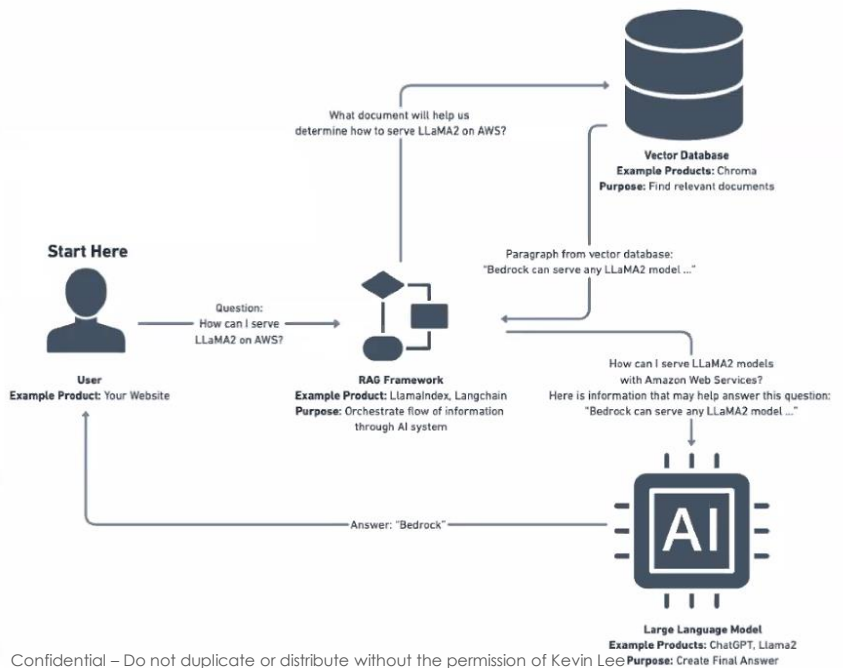


195

195

RAG Definition

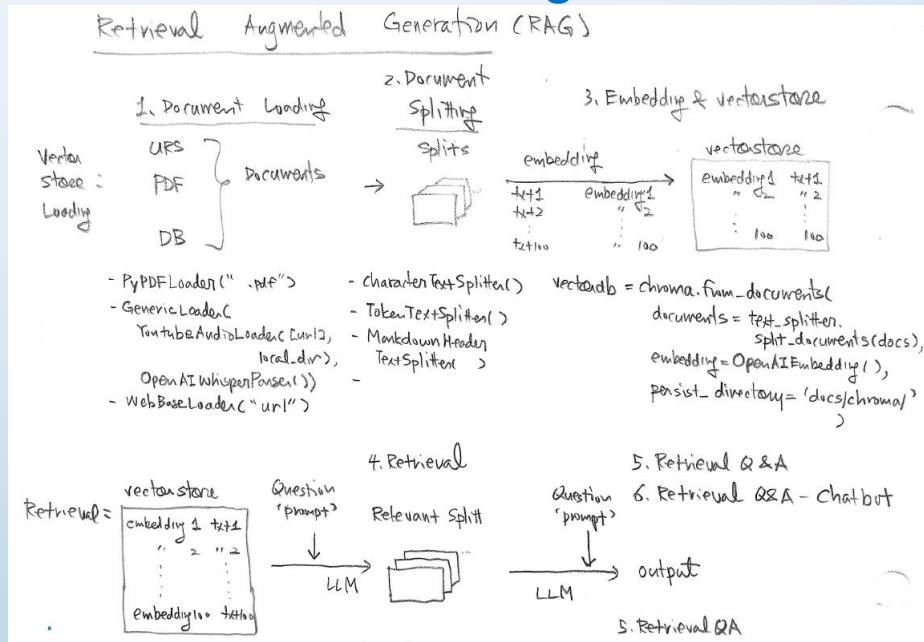
- Retrieval-Augmented Generation
- Process used to improve the accuracy and reliability of LLMs
- Process
 1. Create vector DB using LLM embedding.
 2. Retrieve the information relevant to questions from vector DB
 3. Feed the retrieved info to LLMs
 4. LLMs generates the contents from retrieved info.



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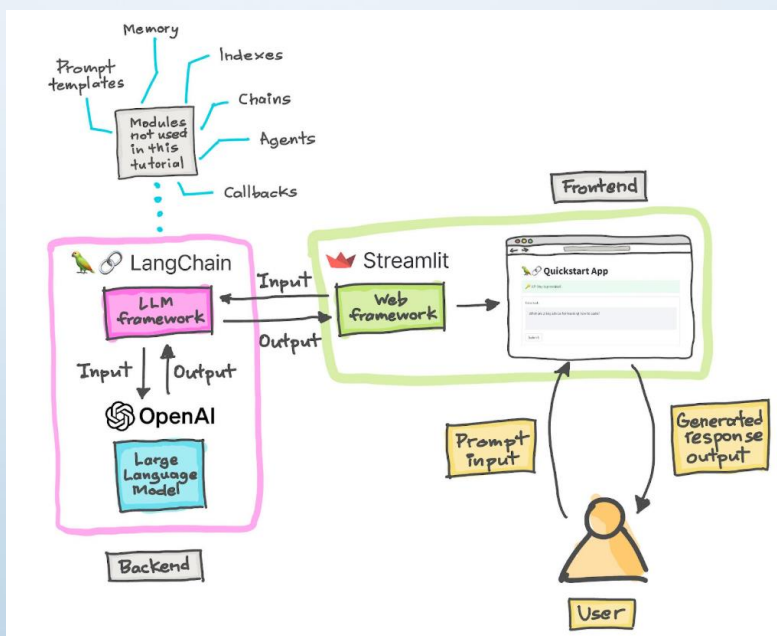
RAG framework in LangChain



197

197

Gen AI application architecture



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198

198

AI Agent

- Definition
- Framework with LangChain

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199

AI Agent

- System that can think and act on its own to complete a task for us.
- Requirements of AI Agent
 - Prompt
 - LLM
 - Tools
 - Memory

Examples

- I am visiting Detroit to attend the conference. My conference will be done in Detroit convention center. Find the hotel near by the convention center about 20 minute walking distance and with the reasonable prices around \$150.

1. Find the location of Detroit Convention Center. (tool : Google Map)
2. Find the hotels nearby within 20 min walk. (tool : Google Hotels)
3. Select the hotels around \$150. (tool : Google Hotels)

200

Example of AI Agent (Google Gemini)

I am visiting Detroit to attend the conference. My conferenc will be done in Detroit convention center. Find the hotel near by the convention center about 20 minu...

Google Maps & 1 more

Google Maps

✓ Query successful

Google Hotels

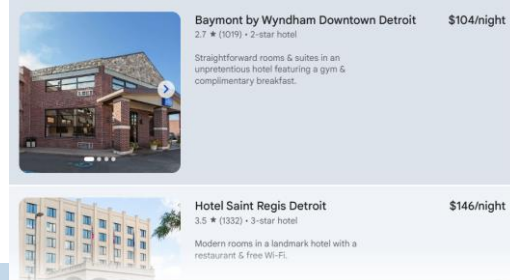
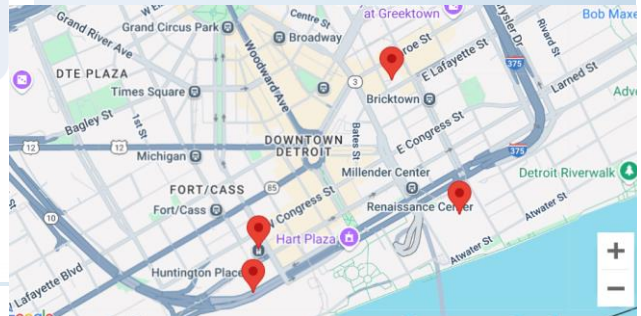
✓ Query successful

I found some hotels near the Detroit convention center. Now, I will use the hotel app to find options around \$150.

Here are some hotels near the Detroit convention center that fit your criteria:

- Baymont by Wyndham Downtown Detroit: \$104
- Hotel Saint Regis Detroit: \$146
- Viking Motel Detroit: \$94
- Hostel Detroit: \$110
- Stay Inn & Suites Detroit: \$141

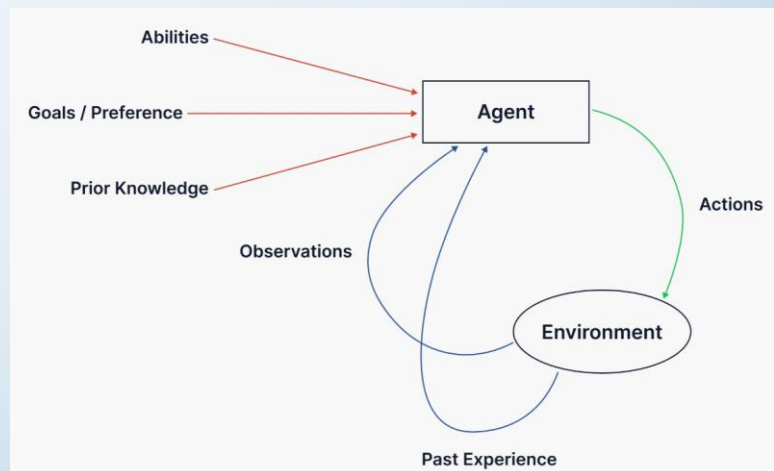
Would you like to explore other options or book a room?



201

AI Agent

- System that can interact with its environment, collect data, and autonomously perform tasks and make decisions to achieve the goals.
- Multiple prompts using different tools
- Main Components
 - LLM
 - Prompts
 - Tools
 - Memory (short-term /long-term memory, RAG)



202

Traditional GenAI vs AI Agent

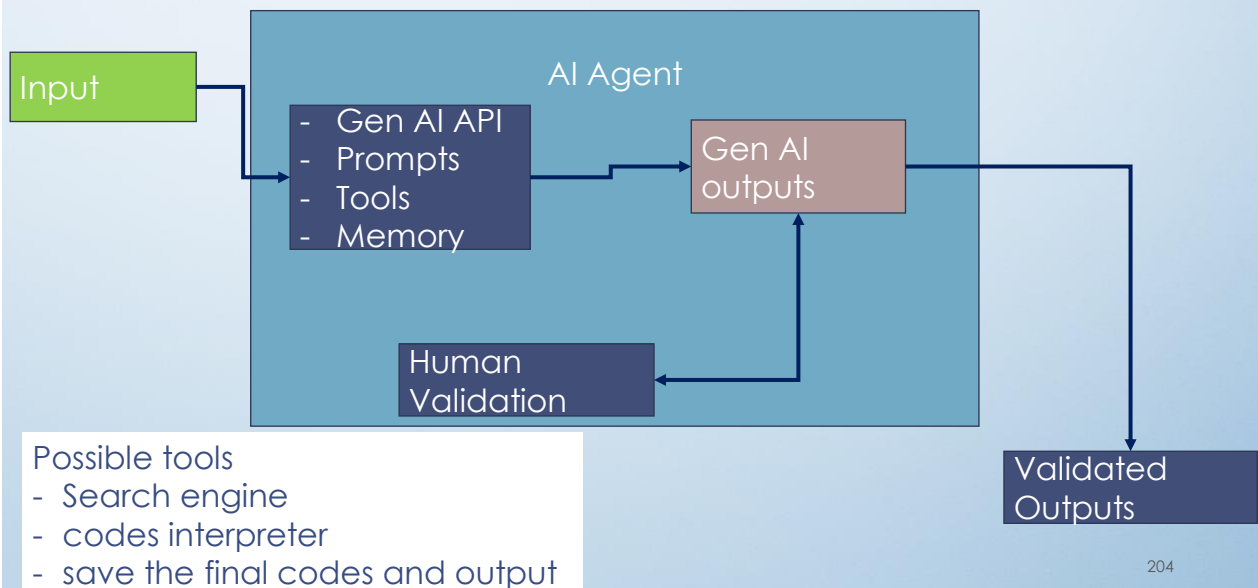
Traditional GenAI	AI Agent
One time direct answer generation – one prompt at a time	Multi-step problem processing – using Agentic workflow.
Dependent – require human prompt continuously	Autonomous – can start next actions without human prompt
Can answer	Can reason and decide
Stateless – No memory or very short-term memory	Stateful – can keep a context as long as the session is running
No tool use	Tool use – databases, RAG, APIs, MCP

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AI Agentic Artifacts Development



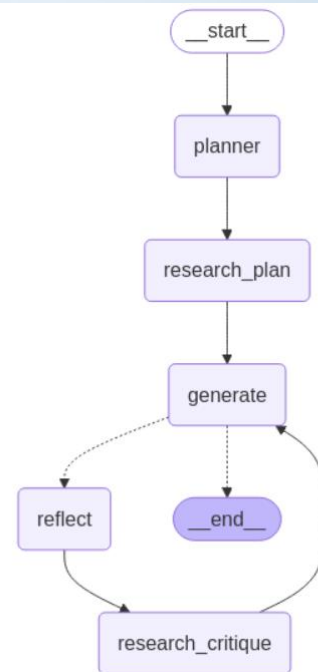
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204

How to build AI Agent workflow

1. Define Agent State
2. Define LLM model
3. Define Prompts
4. Define tools for AI Agents
5. Define functions for nodes
6. Define Memory
7. Define workflow
 - a) Start workflow with Agent State
 - b) Assign nodes to functions
 - c) Assign starting node
 - d) Assign conditional edges
 - e) Assign edges
 - f) Compile memory
8. Compile AI Agent
9. Prepare input (user prompts)
10. Run AI Agent

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Sample Codes for AI Agent

```

class AgentState(TypedDict):
    task: str
    plan: str
    draft: str
    ....

llm = ChatOpenAI(model="gpt-3.5-turbo")

prompt_system = "" "" "" "" ""
prompt_user = "" "" "" "" ""

tool_tavily = TavilyClient( )

def planner(state : AgentState ) :
    response = llm.invoke(prompt_user)
    return {"plan" : response.content }
def research_plan(state : AgentState ) :
    result = tool_tavily.search( )
    return {"draft" : result.content }
.....
mem_store = InMemoryStore( index={"embed":
"openai:text-embedding-3-small"})

```

```

workflow = StateGraph(AgentState)
workflow.add_node("planner", planner)
workflow.add_node("research_plan",
research_plan)
workflow.add_node("generate", generate)

workflow.set_entry_point("planner")

workflow.add_conditional_edges("generator"
, continue, {END: END, "reflect" : "reflect"} )

workflow.add_edge("planner",
"research_plan")
workflow.add_edge("research_plan",
"generate")

ai_agent =
workflow.compile(store=mem_store)

input = { }

final_response = ai_agent.invoke( input )

```

icate or di

206

6 different AI Agentic Design

1. ReAct Agent

- Thinks, takes action, looks at the result, repeats.
- Classic loop: "Should I Google this?" → Does it → Adjusts.
- Used in most AI products today (like basic chat assistants).

2. CodeAct Agent:

- Runs real code, not just JSON.
- So instead of saying "call API X," it writes and runs a Python script.
- More powerful. Used in research agents and dev assistants.

3. Modern Tool Use:

- Sends tasks to smart APIs (search, cloud, data), and lets them do the heavy lifting.
- The agent mostly routes and formats info.
- Think: a smart middleman between you and powerful services.

4. Self-Reflection:

- Agent checks its own work.
- Did it make a mistake? It catches it, critiques it, and tries again.
- Most AI errors happen **because this step is missing.**

5. Multi-Agent Workflow:

- One agent isn't doing everything.
- You have a planner, a researcher, and a writer — all working together.
- Like a mini team of AIs. More accurate. Less chaos.

6. Agentic RAG:

- This is what powers apps like Perplexity.
- The agent looks stuff up (retrieval), thinks about it, uses tools, and gives you a smarter answer.
- Works with real-time data, not just model memory.

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207

MCP

- Definition
- Structure



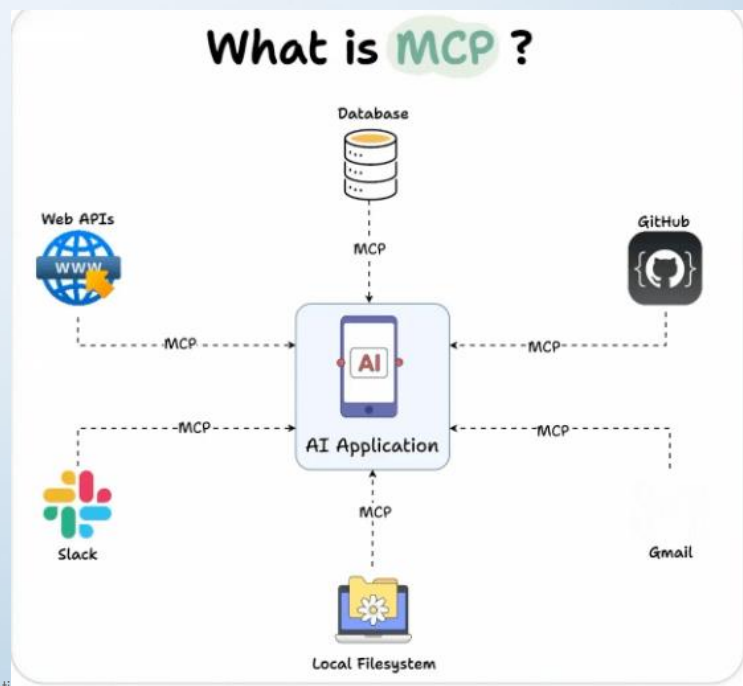
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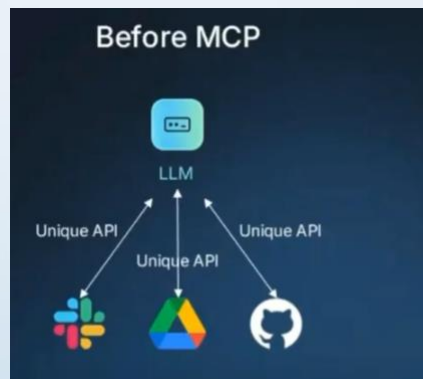
MCP

- Model Context Protocol (MCP) :
 - Open standard on how AI apps connect to different data sources and tools.
 - To simplify how AI models interact with external data and tools.
 - Initially developed by Anthropic in <https://modelcontextprotocol.io/introduction>



209

MCP



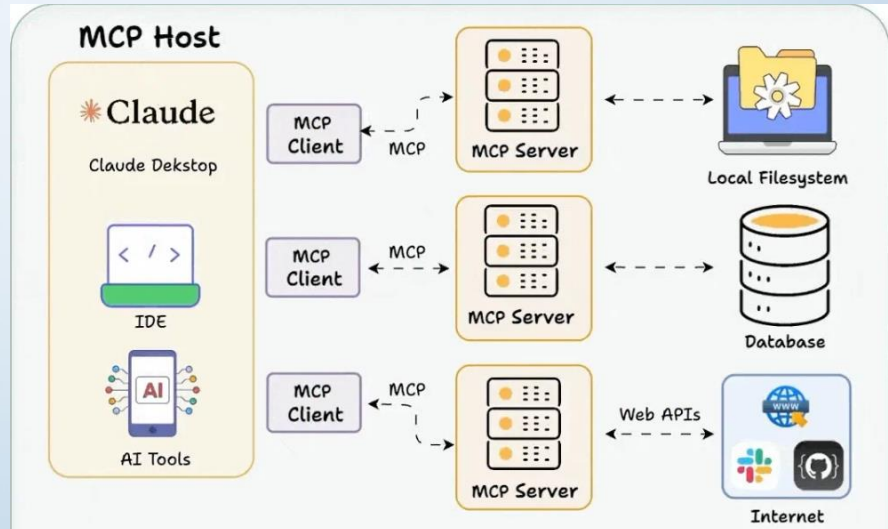
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MCP- high lever structure

- MCP host : AI app with LLM
- MCP client : operates within the host to enable communication with MCP servers
- MCP Server : expose specific capabilities and provide access to
 - Tools : enable LLM to perform actions thru server
 - Resources : Expose data and content from servers to LLMs
 - Prompts : Create reusable prompt template and workflows



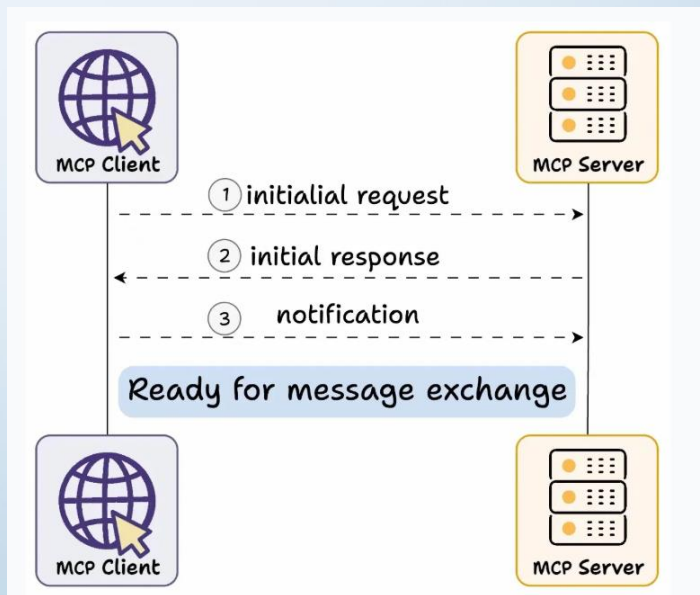
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MCP- how MCP client interact with MCP Servers

- MCP client :
- MCP Server



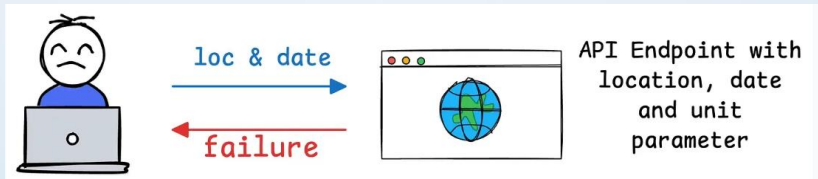
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MCP vs traditional APIs

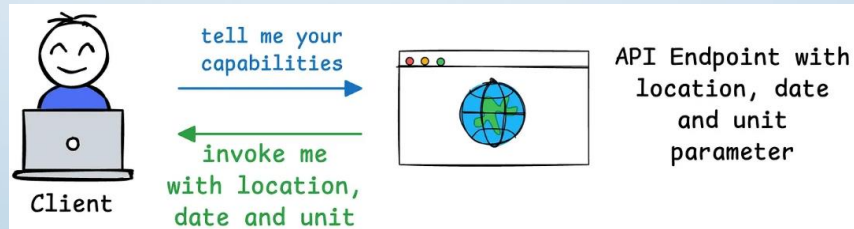
- Traditional APIs



Advantages of MCP

- Use LLMs to refine the inputs much easier
- Minimize the failures, but invoke the clients to adjust.
- Clients can adjust their request much easier.

- MCP



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213

AI Code Assistant

- Vibe-coding
- Github Copilot
- Gemini Code Assistant
- Tabnine
- Codeium



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Vibe Coding

- Andrej Karpathy (co-founder of openAI) – “There is a new kind of coding I call ‘vibe coding’, where you fully give in to the vibes, embrace exponentials, and forget that code even exists.”
- Coding with the help of AI
- MS CEO Satya Nadeli – 30% of the company’s code is now written by AI.

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215

Vibe Coding Rules

1. Start from templates: No need to reinvent
2. Use agent mode: Don’t code, compose
3. Chat in tasks: New Composer chat = new job
4. Voice-to-text is speed: Tools like Whisper Flow unlock flow
5. Run + test early: Don't wait for the deploy
6. Deploy fast: Use Vercel, catch errors upstream
7. Fix bugs with AI: Copy/paste errors into Composer
8. Save good prompts: They're reusable gold
9. Commit often: Cursor can push for you
10. Secure your secrets: Always .env
11. Iterate without fear: Polish comes later
12. Keep Composer lean: Short, scoped, clean
13. Use Perplexity to ideate: Let it find APIs & designs
14. Fork wisely: Use GitHub like a buffet
15. Just vibe: Build fast. Think less. Have fun.

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GitHub Copilot

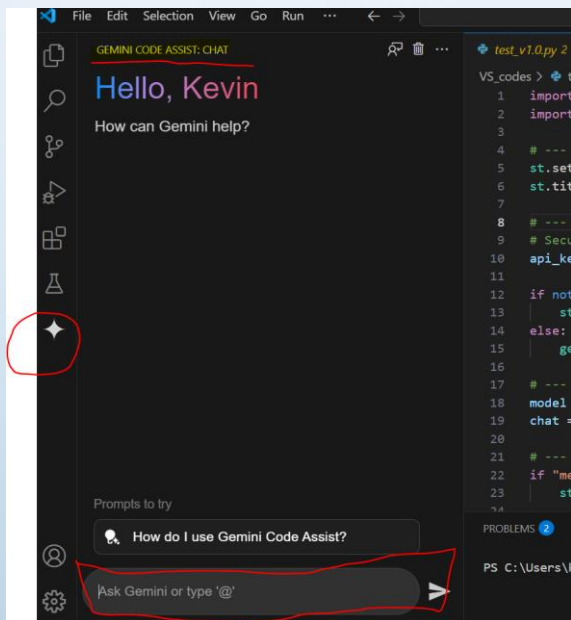
- Code suggestions,
- chat functionality,
- easy auto-complete navigation,
- multiple language and IDE support

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Gemini Code Assistant



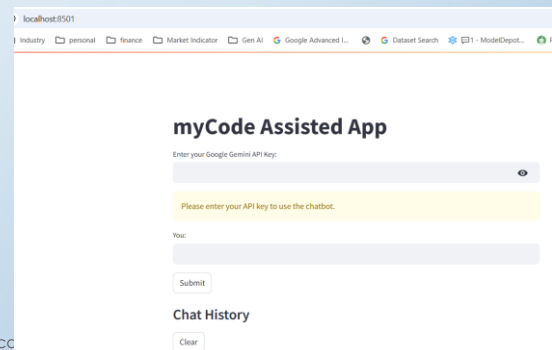
Prompt in Chat:

Create a streamlit script for a chatbot for google gemini.

Have a API kye placeholder, a input text, a submit button, and a output text area.

Have a clear button on the page.

Title of page is my code Assisted App.



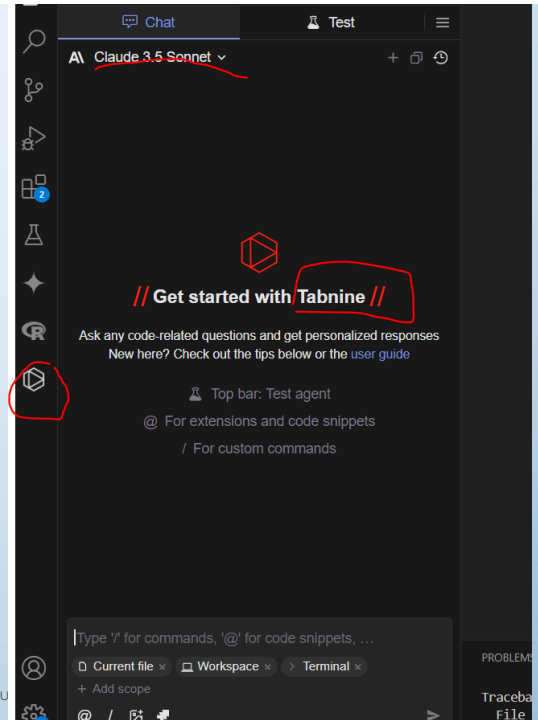
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Tabnine

- Code suggestions,
- chat functionality,
- easy auto-complete navigation,
- multiple language and IDE support

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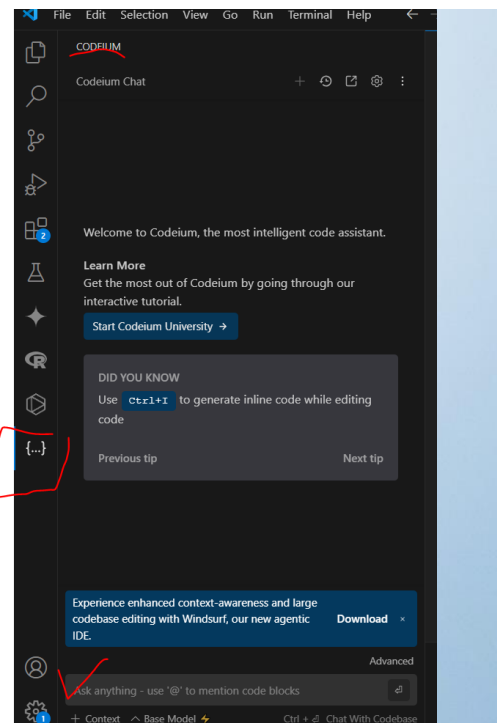


219

Codeium

- AI autocomplete,
- prompt-driven development,
- test generation,
- multiple programming languages support and IDE integration

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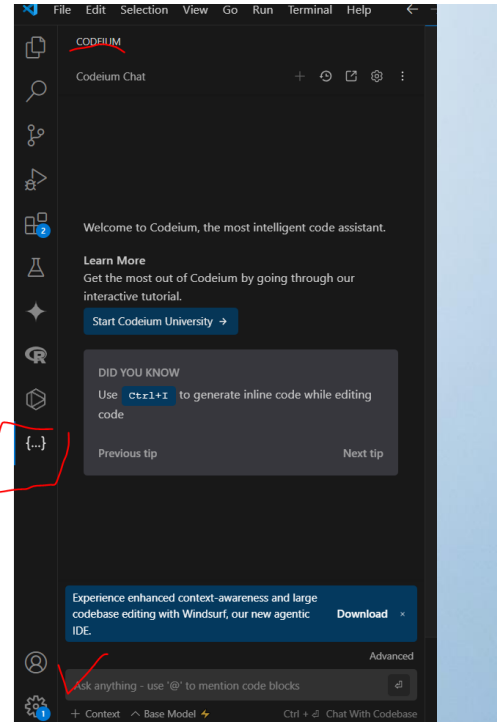


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Intellicode

- Real-world usage examples from GitHub,
- improved privacy

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Gen AI powered Application Development

- Pros
- Cons

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Gen AI Application Development - Pros

1. Rapid Prototyping and Development
 - Gen AI APIs enable programmers to quickly build sophisticated applications by leveraging pre-trained Gen AI models.
 - Significantly reduces development time compared to building AI capabilities from scratch
2. Advanced Functionality with Minimal Expertise
 - Provides access to cutting-edge Gen AI capabilities like natural language processing, image generation, and complex reasoning
 - Democratizes access to sophisticated Gen AI technologies across department
3. Scalability and Flexibility
 - Cloud-based AI APIs can easily scale to handle varying levels of computational demand
 - Flexible integration across different programming languages and frameworks
 - Can be quickly adapted to different use cases and business requirements
4. Cost-Effectiveness
 - Reduces infrastructure and computational costs associated with training and maintaining Gen AI models
 - Pay-as-you-go pricing models allow for more predictable budgeting
 - Eliminates the need for massive upfront investments in AI infrastructure
5. Continuous Improvement
 - Gen AI providers continuously update and improve their models
 - Access to the latest Gen AI advancements with minimal effort

223

223

Gen AI Application Development - Cons

1. Dependency on Gen AI company
 - Heavy reliance on third-party Gen AI API providers
 - Potential challenges in switching between different Gen AI service providers
 - Risk of service disruptions or significant pricing changes
2. Privacy and Data Security Concerns
 - Sending sensitive data through external APIs raises potential security risks
 - Compliance challenges with data protection regulations
 - Limited control over how data is processed and stored by API providers
3. Performance and Latency Issues
 - Network latency can impact real-time application responsiveness
 - Potential performance bottlenecks during high-traffic periods
 - Dependence on external service availability and response times
- Limited Customization
 - Pre-trained Gen AI models might not perfectly align with specific use cases
 - Restricted ability to fine-tune models compared to custom-built solutions
 - Generic responses that may lack domain-specific nuance
5. Cost Unpredictability
 - Usage-based pricing can lead to unexpected expenses (e.g., Gen AI application not suitable for record-based programming)
 - Complex pricing structures might make budgeting challenging
 - Potential for high costs with increased API usage
6. Ethical and Bias Considerations
 - Inherit potential biases from pre-trained models
 - Limited transparency in model decision-making processes
 - Potential for generating inappropriate or controversial content

224

224

Strategic Gen AI / LLM / ChatGPT Implementation at Enterprise

- Risks and Concerns
- Benefits and Use Cases
- Cross Functional Team
- Policy and Guidelines
- Training and Education
- PoC & Evaluation
- Deployment

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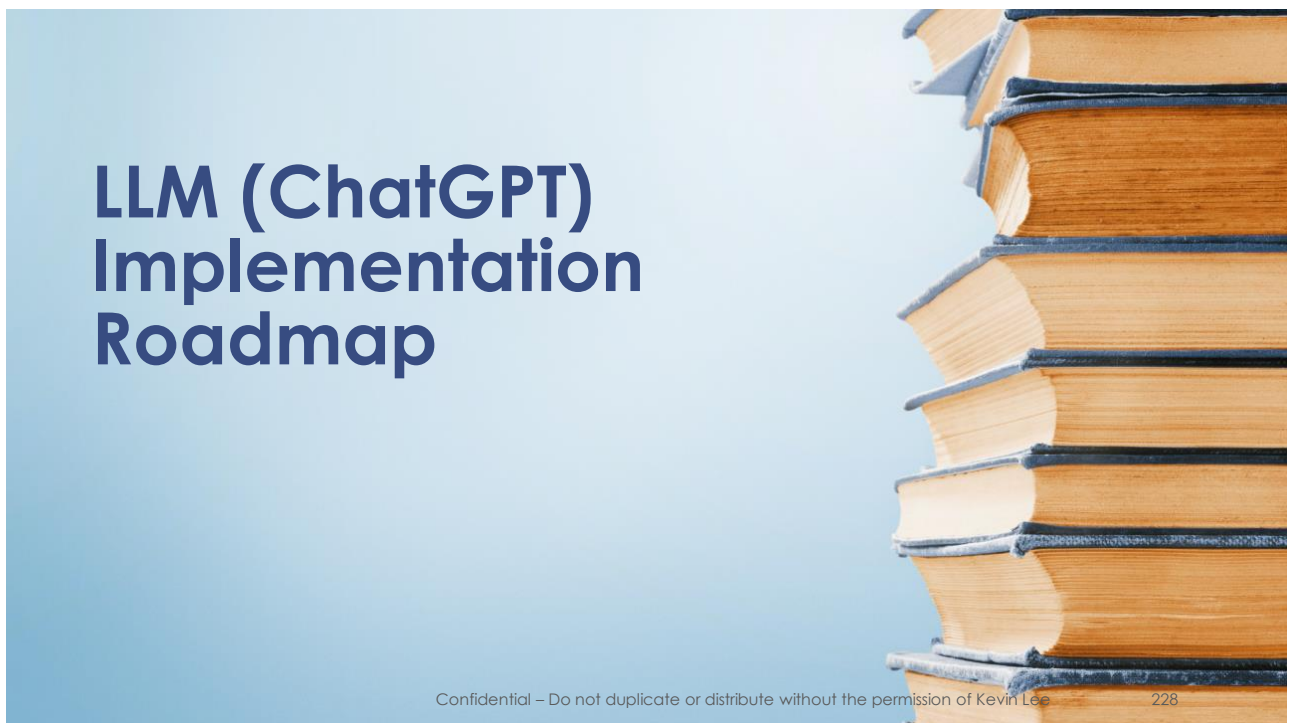
**Does your company allow ChatGPT
at Work?**

226

226

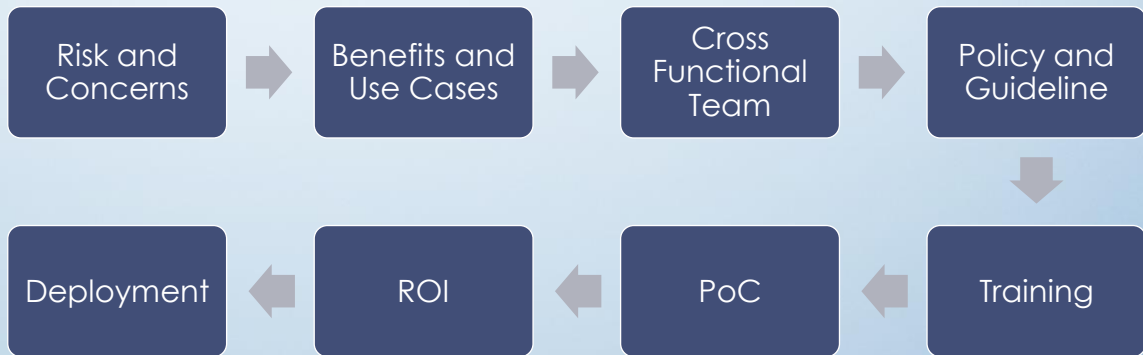


227



228

LLM (ChatGPT) Implementation Roadmap



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Concerns using AI (ChatGPT)

- Data Privacy & Security
- Bias
- Accuracy
- Regulatory Compliance
- Ethical Consideration
- Black Box Feature
- Resource Requirements
- Lack of Expertise
- Cost vs ROI

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Data Privacy

- Data Privacy – the right of a person to have control over how their personal information is collected and used. (e.g., clinical trial data)
- Since ML is built using data, ML algorithms can contain sensitive information even though it is one of a Big Data.
- The growth of ML has increased the possibility of using sensitive data in ways that may violate data privacy. (e.g., ChatGPT)



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Bias of Machine Learning

- Phenomenon that occurs when ML algorithms produces results that are prejudiced due to wrong assumptions
- Examples :
 - Amazon ML recruiting engine show a bias on women.
 - ML was trained to vet applicants using 10 years of resume
 - Most came from men, a reflection of male dominance across tech industry.



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Regulatory Compliance

- Regulatory Compliance – the process of adhering to laws, regulations, standards, and other rules set by governments and other regulatory bodies.
- For pre-market approval, AI-based software needs to follow regulatory guidelines (e.g., Good Machine Learning for Medical Device Development Guiding Principles)

**U.S. FOOD & DRUG
ADMINISTRATION**

Health Canada

**Medicines & Healthcare products
Regulatory Agency**

**Good Machine Learning Practice for Medical Device Development:
Guiding Principles**
October 2021

The U.S. Food and Drug Administration (FDA), Health Canada, and the United Kingdom's Medicines and Healthcare products Regulatory Agency (MHRA) have jointly identified 10 guiding principles that can inform the development of Good Machine Learning Practice (GMLP). These guiding principles will help promote safe, effective, and high-quality medical devices that use artificial intelligence and machine learning (AI/ML).

Artificial intelligence and machine learning technologies have the potential to transform health care by deriving new and important insights from the vast amount of data generated during the delivery of health care every day. They use software algorithms to learn from real-world use and in some situations may use this information to improve the product's performance. But they also present unique considerations due to their complexity and the iterative and data-driven nature of their development.

These 10 guiding principles are intended to lay the foundation for developing Good Machine Learning Practice that addresses the unique nature of these products. They will also help cultivate future growth in this rapidly progressing field.

Good Machine Learning Practice for Medical Device Development: Guiding Principles	
Multi-Disciplinary Expertise Is Leveraged Throughout The Total Product Life Cycle Clinical Study Participants and Data Sets Are Representative of the Intended Patient Population Selected Reference Datasets Are Based Upon Best Available Methods Focus Is Placed on the Performance of the Human-AI Team Users Are Provided Clear, Essential Information	Good Software Engineering and Security Practices Are Implemented Training Data Sets Are Independent of Test Sets Model Design Is Tailored to the Available Data and Reflects the Intended Use of the Device Testing Demonstrates Device Performance During Clinically Relevant Conditions Deployed Models Are Monitored for Performance and Re-training Risks are Managed

The 10 guiding principles identify areas where the International Medical Device Regulators Forum (IMDRF), international standards organizations, and other collaborative bodies could work to advance GMLP. Areas of collaboration include research, creating educational tools and resources, international harmonization, and consensus standards, which may help inform regulatory policies and regulatory guidelines.

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Ethical Consideration

- Who will make decision? AI or human?
- Can AI be responsible for their decision?
- Is AI transparent?



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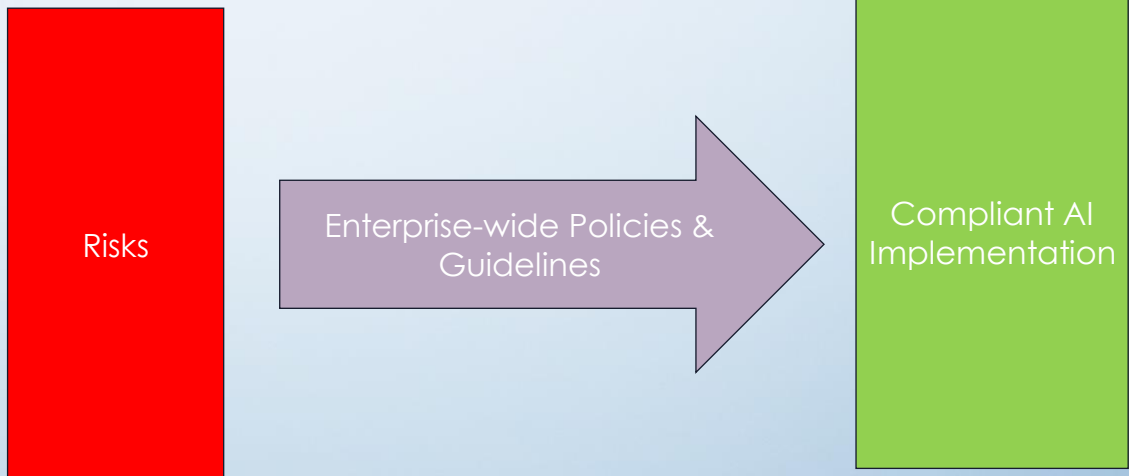
AI Implementation through Enterprise Policy and Guidelines

- Data Privacy
- Regulatory Compliance
- Secure Data Storage
- Secure Access in Gen AI



235

Risk mitigation through Policies and Guidelines



236

236

Data Privacy

- The protection and control of confidential company information collected, processed, and utilized by AI systems.
- Examples to protect data privacy in AI
 - Don't share sensitive data with AI
 - Be cautious of what you share online
 - Stay anonymous
 - Data Encryption
 - Secure Socket layer (SSL)
 - Transport Layer Security (TLS)

OpenAI Personal Data Removal Request

Under certain privacy or data protection laws, such as the GDPR, you may have the right to object to the processing of your personal data by OpenAI's models. You can submit that request using this form.

Please provide complete, accurate, and relevant answers on this form for evaluation. OpenAI may use additional sources to verify information, balancing privacy and free expression in accordance with applicable law. Submitting a request does not guarantee that information about you will be removed from ChatGPT outputs, and incomplete forms may not be processed.

Read this Help Center [article](#) for more about how we collect and use personal data to develop ChatGPT.

Your information

Please provide your own full legal name, even if you are making the request on behalf of someone else. If you are submitting the request on behalf of someone else, you must have the legal authority to act on their behalf.

First name *

Last name *

Email *

Please provide an email address we can use to contact you about your request.

Country where law applies *

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237

237

Regulatory Compliance

- Understand Regulatory Policy
 - FDA Comments: AI/ML will undoubtedly play a critical role in drug development, and FDA plans to develop and adopt a flexible **risk-based regulatory** framework that promotes innovation and protects patient safety.
- Policy Implementation
 - General Data Protection Regulation (GDPR)
 - Good Practice (GxP)
 - Code of Federal Regulation (CFR)
 - Health Insurance Portability and Accountability Act (HIPAA)

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Proposed Regulatory Framework for Modifications to Artificial Intelligence/Machine Learning (AI/ML)-Based Software as a Medical Device (SaMD)

Discussion Paper and Request for Feedback



238

Secure Data Storage

- Security of Data Storage and Data Processing
- Access Control of Data and ChatGPT
- Data Anonymization and Pseudonymization



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Secure Access to ChatGPT

- Access Control of Data and ChatGPT
- Strong User Authentication Mechanism (e.g., Multi-factor Authentication)
- Audit Trails of user interactions with ChatGPT
- Regular Security Audits on ChatGPT
- Data Privacy Impact Assessment by ChatGPT



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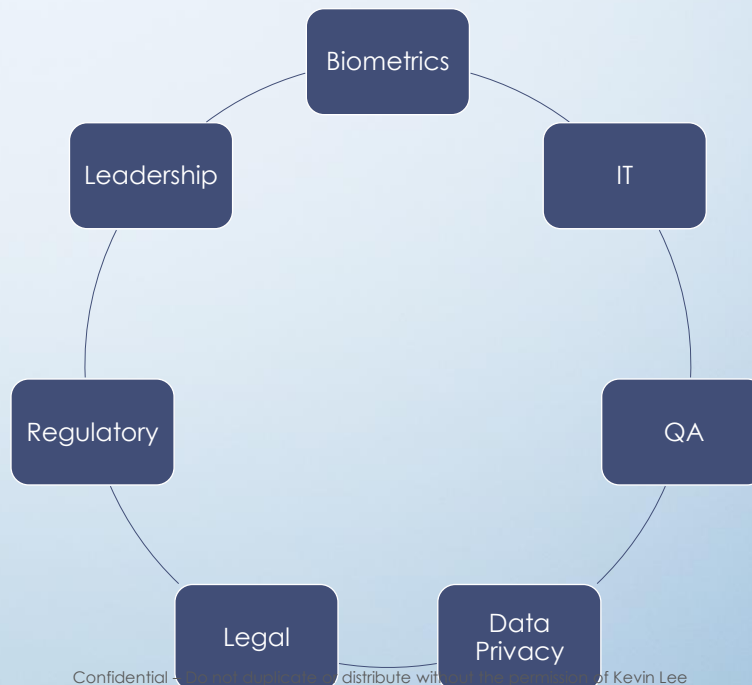
ChatGPT Implementation and Governance Cross-Functional Team

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Cross-Functional Team



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242

Employee Training

- Employees who will be using and managing ChatGPT
- Security best practices
- Data Privacy
- Data Handling Procedures with ChatGPT
- Ethics and Compliance



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243

PoC & Evaluation



44

244

Gen AI Deployment

- Enterprise License (CoPilot, ChatGPT Enterprise)
- Home-Grown Gen AI
- Customized Gen AI implementation at function



245

ChatGPT Enterprise

- Enterprise-grade security & privacy
- OpenAI does not train on enterprise business data.
- All conversations are encrypted.
- Compliance : CCPA, GDPR, SOC2, SOC3
- Enterprise Security Trust Portal



**ChatGPT
Enterprise**

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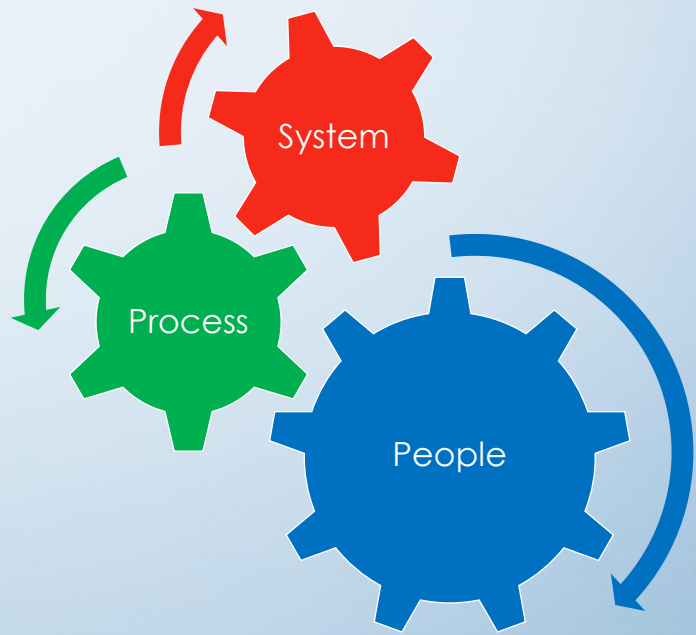
Gen AI implementation in Biometrics

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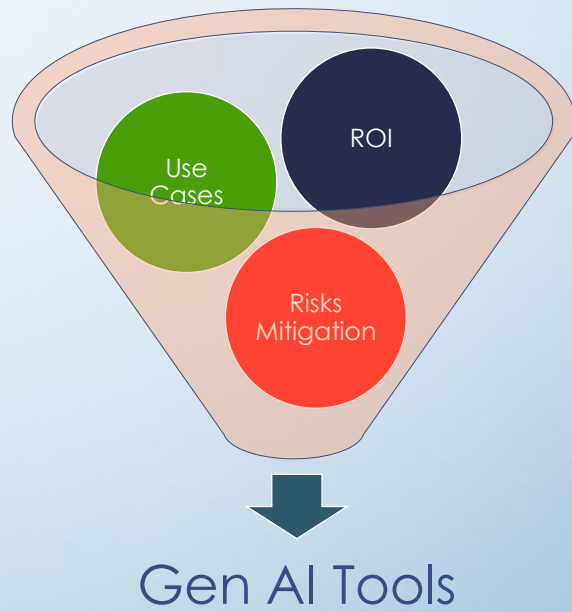
247

Three pillars to maximize Gen AI Capability in Biometrics



248

Use Cases driven Gen AI Tools



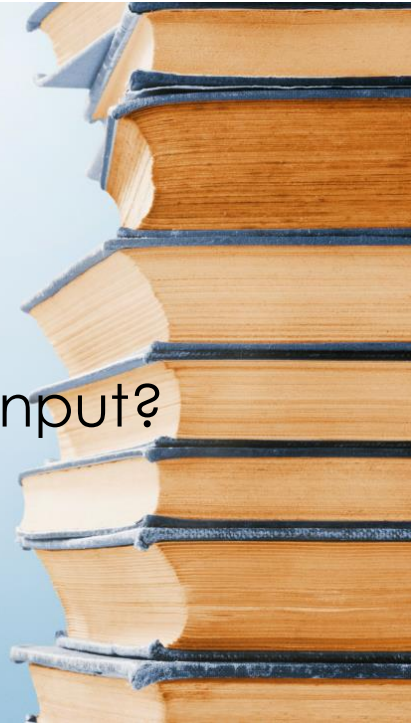
249



250

Gen AI Process

- Validation?
- Human intervention
- Final output or source of input?



251

ChatGPT is like a very smart, capable, recently graduated intern/new hire.



=



252

What will we do with the results from an intern?



253

**The output of Gen AI is
not the final result,
but the beginning of new process.**

254

Gen AI driven Process



255

People : Biometrics team powered with Gen AI

-Super Biometrics Team

256

Gen AI - Electricity



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257

257

ChatGPT Users vs Non-Users Comparison in Boston Consulting

- Results of ChatGPT Users
 - finished **12.2%** more tasks on average
 - completed tasks **25.1%** more quickly
 - produced **40%** higher quality results



258

258

Gen AI gives us a Head Start. We don't need to start from the blank page.



259

259

Future of Gen AI and US



260

ChatGPT Popularity

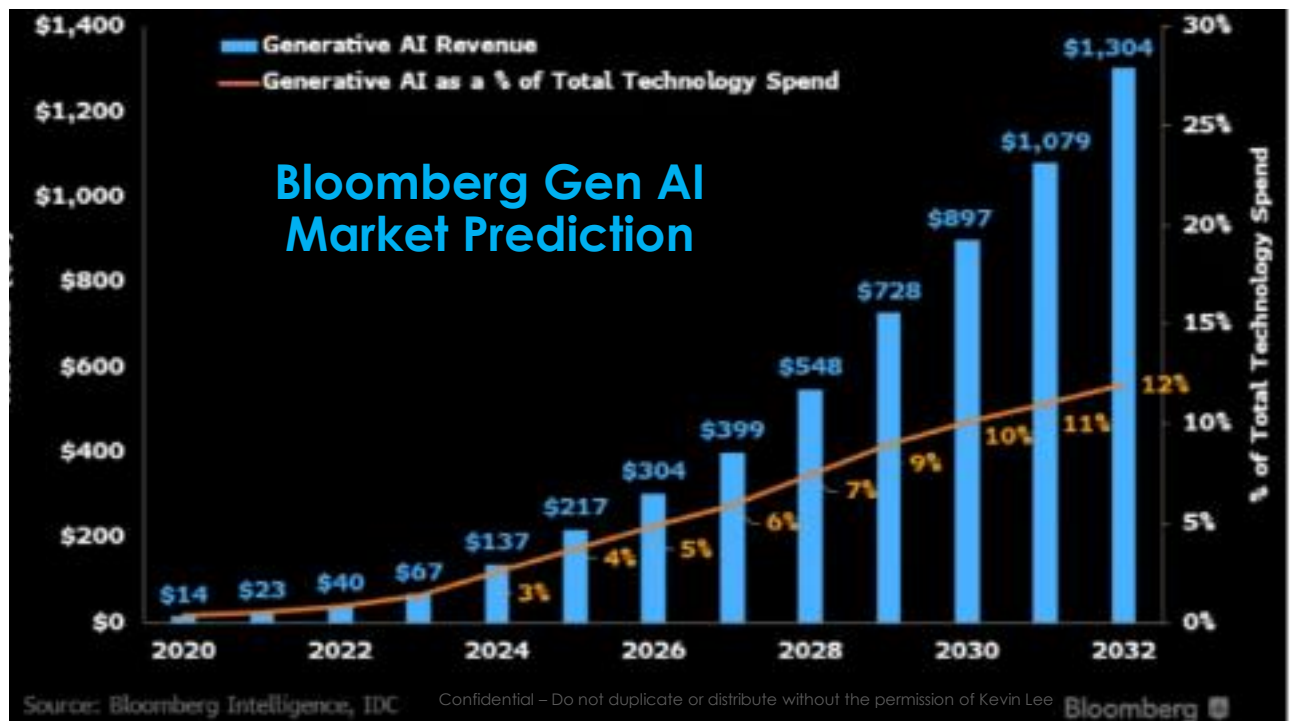
Road To 100 Million Users For Various Platforms



ChatGPT gained 100M users within 2 months since its release

261

Bloomberg Gen AI Market Prediction



262

Gen AI has arrived



263

263

Attacking vs Understanding



264

What is Gen AI for us?



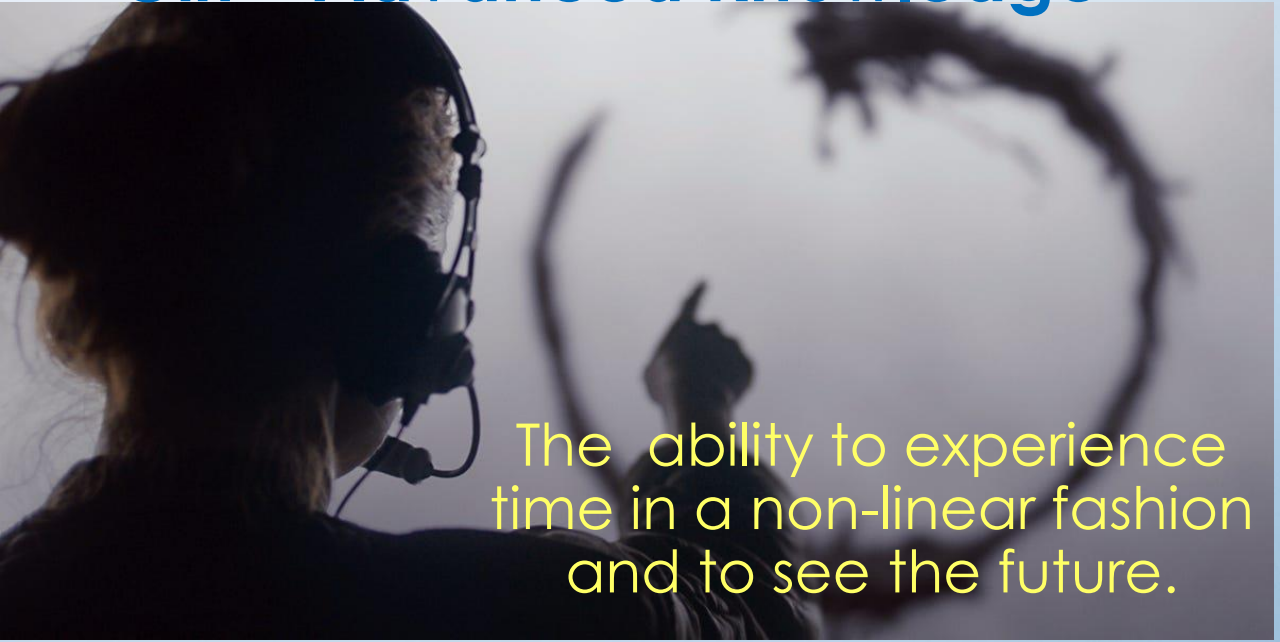
Replacing
us?

Augmenting
us?

265

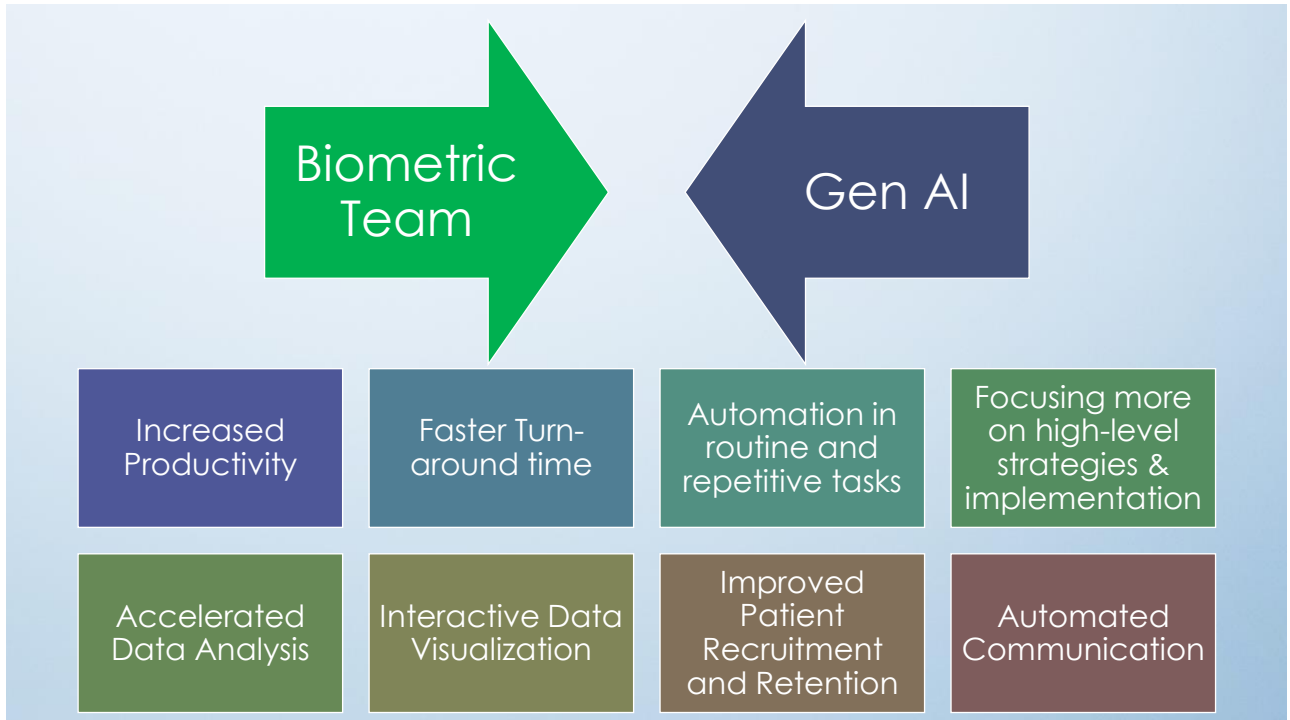
265

Gift – Advanced Knowledge



The ability to experience
time in a non-linear fashion
and to see the future.

266



267

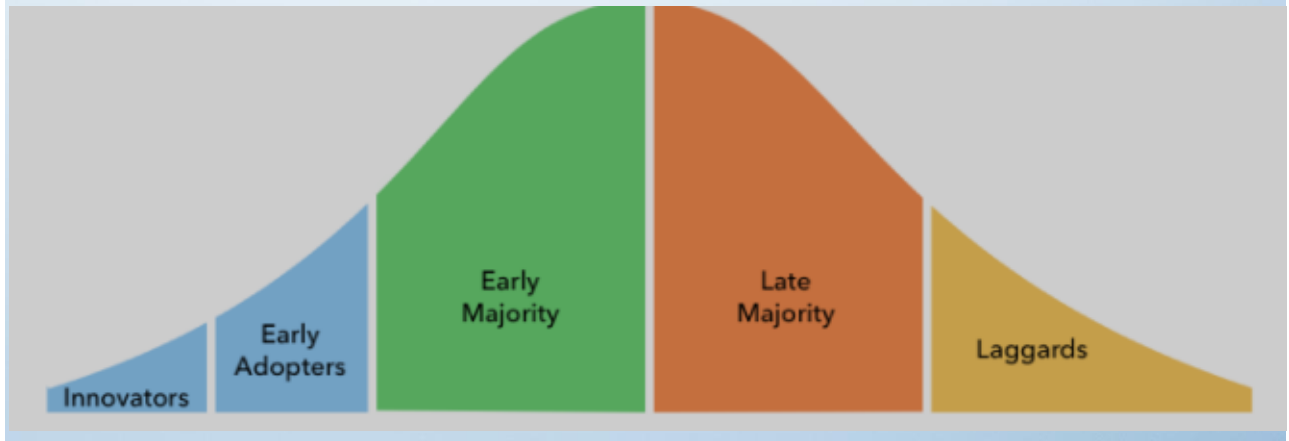
Future of Gen AI & Biometric team?



268

Discussion :

Where do we want our Biometric team to be in Gen AI Adopter?



269

Q & A (Discussion)



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270

270

